Mathematics 4024
Study Guide
GRADE 11



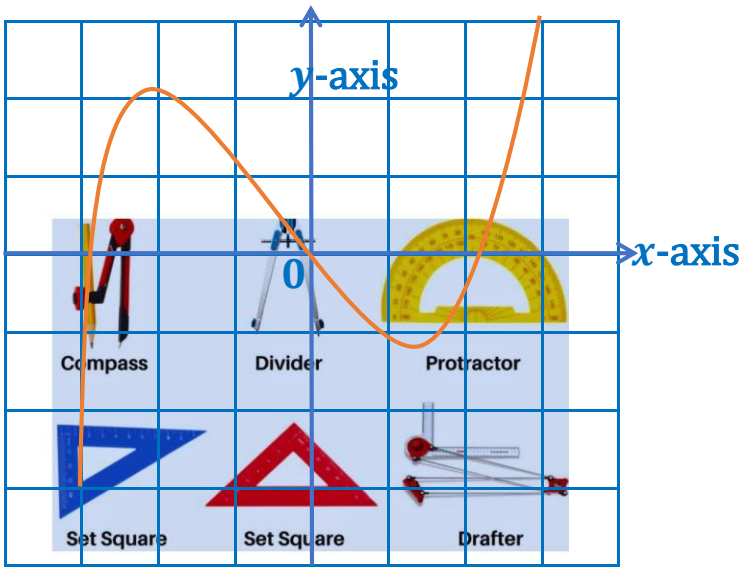
Complete Study Guide with Past Exam Questions and Teaching Notes.

Mastering Grade 11 Mathematics: A Comprehensive Guide for Grade 11

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$x = -\frac{b}{2a}$$

$$y = \frac{4ac-b^2}{4a}$$



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APPROXIMATIONS

- Approximation is a branch of mathematics which is used to simplify complex calculations and provide an estimate of the actual value.

Types of Approximations

1. Rounding: Rounding numbers to the nearest whole number, tenth, hundredth, etc.
2. Truncation: Truncating numbers to a specific number of decimal places.
3. Estimation: Estimating values based on mental calculations or rough estimates.

Rounding Rules

1. If the digit to be rounded is less than 5, round down.
2. If the digit to be rounded is 5 or greater, round up.

Absolute, Relative and Percentage Errors

- The absolute error of a dimension is the absolute difference between the true value and the recorded value.
- Absolute error = |recorded value – true value|.
- If the recorded value is not given, then the absolute error can be obtained by considering the least unit of measurement for the given dimension which is given by $\frac{n}{2}$.
- The relative error of a dimension is the ratio of the absolute error to the true value.
- Relative error $= \frac{\text{absolute error}}{\text{true value}}$.
- The percentage error states the relative error as a percentage.
- Percentage = relative error $\times$ 100%.

$$= \frac{\text{absolute error}}{\text{true value}} \times 100\%.$$

Tolerance

- Tolerance is the maximum allowable deviation from a specified value or dimension.
- Tolerance = absolute error × 2
- In engineering, tolerance is the limits of acceptable error in a measurement for a structure to serve its function properly.

Types of Tolerance

1. Unilateral Tolerance: A tolerance that allows for deviation in only one direction.
   Example: 10.0cm ± 0.1cm (deviation allowed only above 10.0cm)
2. Bilateral Tolerance: A tolerance that allows for deviation in both directions.
   Example: 10.0cm ± 0.1cm (deviation allowed both above and below 10.0cm)

Tolerance Limits

1. Upper Limit: The maximum allowable value.
2. Lower Limit: The minimum allowable value.

Importance of Tolerance

1. Quality Control: Tolerance ensures that products meet specified standards.
2. Interchangeability: Tolerance allows for interchangeability of parts.
3. Safety: Tolerance ensures safety by preventing deviations that could lead to accidents.

2018.P1.Oct

(a) The mass (m) of a bag of mealie meal is 25kg. Complete the statement in the answer space. ..... $\leq m <$ .....

(b) The length of a line is 8cm. If this is recorded as 8.2cm, calculate the percentage error.

Solution

(a) Absolute error $= 0.5$

Lower limit $= 25 - 0.5$

$= 24.5kg$

Upper limit $= 25 + 0.5$

$= 25.5kg$

$\therefore 24.5kg \leq m \leq 25.5kg$

(b) P.E $= R.E \times 100$

$= \frac{8.2-8}{8} \times 100$

$= \frac{0.2}{8} \times 100$

$= \frac{0.2\times10}{8\times10} \times 100$

$= \frac{2}{80} \times 100$

$= \frac{1}{2} \times 5$

$= \frac{5}{2}$

$= 2.5\%$

2019.P1.Nov

The length of piece of wire is measured as 4.5cm. Calculate

(a) the tolerance,

(b) the relative error.

Solution

(a) $\text{Tolerance} = 2 \times A.E$

$= 2 \times 0.05$

$= 0.1cm$

(b) $\text{R.E} = \frac{A.E}{TV}$

$= \frac{0.05}{4.5}$

$= \frac{0.05 \times 100}{4.5 \times 100}$

$= \frac{5}{450}$

$= \frac{1}{90}$

2020.P1.Nov

Chiyalwa estimated the length of a string as 15cm. If the true length of the string was 14.6cm, find the

(a) absolute error,

(b) relative error.

Solution

(a) $A.E = |R.V - T.V|$

$= |15 - 14.6|cm$

$= |0.4|cm$

$= 0.4cm$

(b) $R.E = \frac{A.E}{TV}$

$= \frac{0.4}{14.6}$

$= \frac{0.4 \times 10}{14.6 \times 10}$

$= \frac{4}{146}$

$= \frac{2}{73}$

2021.P1.Nov

A learner measured the length of a line to be 2.2cm. If the actual length is 2cm, find the

(a) absolute error,

(b) percentage error.

Solution

(a) $A.E = |R.V - T.V|$

$= |2.2 - 2|cm$

$= |0.2|cm$

$= 0.2cm$

(b) $P.E = R.E \times 100$

$P.E = \frac{0.2}{2} \times 100$

$= \frac{0.2 \times 10}{2 \times 10} \times 100$

$= \frac{2}{20} \times 100$

$= 1 \times 10$

$= 10\%$

2022.P1.Nov

The mass of a loaf of bread is 702.1g correct to 1 decimal place. Find the

(a) lower limit,

(b) relative error of the mass of the loaf of bread.

Solution

(a) Lower limit = T.V – A.E

= 702.1 – 0.05

= 702.05g

(b) R.E = $\frac{A.E}{TV}$

= $\frac{0.05}{702.1}$

= $\frac{0.05\times100}{702.1\times100}$

= $\frac{5}{70\ 210}$

= $\frac{1}{14042}$

2023.P1.Nov

(a) In a particular examination, Nellie recorded the volume, v, of water as 6.4 litres correct to one decimal place. Complete the statement in the answer space. ..... $\leq v \leq$ .....

(b) The actual length of the longest side of a rectangular playing field is 102m. A boy measured the same side as 99.8m. Calculate the relative error.

Solution

(a) Lower limit $= 6.4 - 0.05$

$= 6.35$ litres

Upper limit $= 6.4 + 0.05$

$= 6.45$ litres

$\therefore 6.35l \leq v \leq 6.45l$

(b) A.E $= |$R.V $-$ T.V$|$

$= |99.8 - 10\ 2|$m

$= |-2.2|$m

$= 2.2$m

$$\text{R.E} = \frac{AE}{TV}$$

$$\text{RE} = \frac{2.2}{102}$$

$$= \frac{2.2 \times 10}{102 \times 10}$$

$$= \frac{22}{1\,020}$$

$$= \frac{11}{510}$$

2024.P1.GCE

The length of a sugarcane is measured as 2.6m. If its true length is 2.5m, find the

(a) absolute error,

(b) percentage error.

Solution

(a) $AE = |RV - TV|$

$= |2.6m - 2.5m|$

$= 0.1m$

(b) $PE = RE \times 100\%$

$= \frac{0.1}{2.5} \times 100$

$= \frac{0.1 \times 10}{2.5 \times 10} \times 100$

$= \frac{1}{25} \times 100$

$= \frac{100}{25}$

$= 4\%$

SEQUENCES AND SERIES

- A sequence is an ordered set of numbers, where a pattern exists.
- A term is a number in a sequence.
- A series is the sum of the terms of a sequence.
- It is a mathematical expression that represents the sum of a sequence of numbers, often expressed using sigma notation.

Arithmetic Progression (AP)

- An arithmetic progression is a sequence in which each term is formed by adding a constant amount to the previous term.
- This constant amount is called a common difference, and it may be either positive or negative.
- If the common difference is positive, then the AP increases and it decreases if the common difference is negative.
- We letter $\boldsymbol{a}$ for the first term and the letter $\boldsymbol{d}$ for the common difference.
- Another name for an arithmetic progression is an arithmetic sequence.

The nth term of an arithmetic progression

- First term $\quad T_1 = a$
- Second term $\quad T_2 = a + d$
- Third term $\quad T_3 = a + 2d$
- Fourth term $\quad T_4 = a + 3d$
- Fifth term $\quad T_5 = a + 4d$
- n^{th} term $\quad T_n = a + (n-1)d$
- $d = T_2 - T_1 = T_3 - T_2$

The arithmetic mean of an arithmetic progression

- The arithmetic mean is the middle term of three consecutive terms in an AP.
- If a, b and c are three consecutive terms in an AP, then b is the arithmetic mean of a and c.
- Therefore, $b = \frac{a+c}{2}$

$$= \frac{1}{2}(a + c)$$

The sum of an arithmetic progression

- The sum of an arithmetic progression is called an arithmetic series.
- An arithmetic series is an AP with plus signs written between the terms.
- Therefore, $S_n = \frac{n}{2}[2a + (n-1)d]$.

2018.P1.Oct

For an arithmetic progression 16, 25, 34, 43, 52, ..., write down the

(a) next term,

(b) n^{th} term.

Solution

(b) $a = 16 \; and \; d = 9$

$T_n = a + (n-1)d$

$T_6 = 16 + (6-1)9$

$= 16 + (5)9$

$= 16 + 45$

$= 61$

(a) $T_n = a + (n-1)d$

$= 16 + (n - 1)9$

$= 16 + 9n - 9$

$= 9n + 16 - 9$

$= 9n + 7$

2019.P1.Nov

(a) Given that $17 + m + 27 + \ldots$ are consecutive terms of an arithmetic progression, find the arithmetic mean, m.

(b) For the sequence 11, 13, 15, ..., find the formula for the nth term.

Solution

(a) $m = \frac{17+27}{2}$

$= \frac{44}{2}$

$= 22$

(b) $a = 11 \; and \; d = 2$

$T_n = a + (n-1)d$

$= 11 + (n-1)2$

$= 11 + 2n - 2$

$= 2n + 11 - 2$

$= 2n + 9$

2020.P1.Nov

For the sequence 7, 11, 15, 19, ..., find the

(a) 9th term,
(b) sum of the first 16 terms.

Solution

(a) $a = 7 \; and \; d = 4$

$T_n = a + (n-1)d$

$T_9 = 7 + (9-1)4$

$= 7 + (8)4$

$= 7 + 32$

$= 39$

(b) $S_n = \frac{n}{2}[2a + (n-1)d]$

$= \frac{16}{2}[2(7) + (16 - 1)4]$

$= 8[14 + (15)4]$

$= 8(14 + 60)$

$= 8(74)$

$= 592$

2021.P1.Nov

The third and fourth terms of an arithmetic progression are 6 and 3 respectively. Find the

(a) first term,
(b) formula for the nth term.

Solution

(a) $a + 2d = 6$

$a + 3d = 3$

$d = -3$

$a + 2(-3) = 6$

$a - 6 = 6$

$a = 6 + 6$

$a = 12$

(b) $T_n = a + (n - 1)d$

$= 12 + (n - 1)(-3)$

$= 12 - 3n + 3$

$= 12 + 3 - 3n$

$= 15 - 3n$

2022.P1.Nov

Given that $2, -1, -4, \ldots$ are consecutive terms of an Arithmetic progression, find the

(a) common difference, d,

(b) formula for the nth term,

Solution

(a) $d = T_2 - T_1$

$= -1 - 2$

$= -3$

(b) $T_n = a + (n-1)d$

$= 2 + (n - 1)(-3)$

$= 2 - 3n + 3$

$= 2 + 3 - 3n$

$= 5 - 3n$

2023.P1.Nov

Given that 25 and 13 are the first and third terms of an arithmetic progression respectively, find the

(a) second term,

(b) formula for the nth term.

Solution

(a) $a = 25$

$a + 2d = 13$

$25 + 2d = 13$

$2d = 13 - 25$

$2d = -12$

$d = \frac{-12}{2}$

$d = -6$

$T_n = a + (n-1)d$

$= 25 + (2 - 1)(-6)$

$= 25 + (1)(-6)$

$= 25 - 6$

$= 19$

(b) $T_n = a + (n-1)d$

$= 25 + (n - 1)(-6)$

$= 25 - 6n + 6$

$= 25 + 6 - 6n$

$= 31 - 3n$

2024.P1.GCE

The sequence 20, 17, 14, …, is an Arithmetic progression. Find the

(a) 9th term,

(b) sum of the first 10 terms.

Solution

(a) $\boldsymbol{a = 20 \; and \; d = -3}$

$$\boldsymbol{T_n = a + (n-1)d}$$

$$\boldsymbol{T_9 = 20 + (9-1)(-3)}$$

$$\boldsymbol{= 20 + (8)(-3)}$$

$$\boldsymbol{= 20 - 24}$$

$$\boldsymbol{= -4}$$

(b) $\boldsymbol{S_n = \frac{n}{2}[2a + (n-1)d]}$

$$\boldsymbol{S_{10} = \frac{10}{2}[2(20) + (10-1)(-3)]}$$

$$\boldsymbol{= 5[40 + (9)(-3)]}$$

$$\boldsymbol{= 5(40 - 27)}$$

$$\boldsymbol{= 5(13)}$$

$$\boldsymbol{= 65}$$

GEOMETRIC PROGRESSION

Objectives

By the end of this topic, learners should be able

1 To define Geometric Progression (GP)
2 To understand the concept of Geometric Progression (GP)
3 To learn how to find the nth term and the sum of the first n terms of a GP
4 To apply GP to solve problems in mathematics and real-world applications

Definition

- A Geometric Progression is a sequence of numbers in which each term is obtained by multiplying the previous term by a fixed constant, called the common ratio.
- The common ratio is abbreviate by a small letter r.

The nth term of a GP

$T_1 = a$

$T_2 = ar$

$T_3 = ar^2$

$T_4 = ar^3$

$T_5 = ar^4$

$T_6 = ar^5$

$T_7 = ar^6$

$T_8 = ar^7$

$T_9 = ar^8$

$\therefore T_n = ar^{n-1}$

The sum of the first n terms of a GP

$$S_n = \frac{a(1-r^n)}{1-r}, \text{ for } |r| < 1$$

$$S_n = \frac{a(r^n-1)}{r-1}, \text{ for } |r| > 1$$

The sum to infinity of a GP

$$S_\infty = \frac{a}{1-r}, \text{ for } |r| < 1$$

The Geometric Mean of a GP

- If a, b and c are consecutive terms in a GP, b is the geometric mean of a and c.

$$\frac{b}{a} = \frac{c}{b}$$

$$b \times b = a \times c$$

$$b^2 = ac$$

$$b = \pm\sqrt{ac}$$

$$\therefore \text{G.M} = \pm\sqrt{ac}$$

2018.P2.GCE

In a geometric progression, the third term is $\frac{2}{9}$ and the fourth term is $\frac{2}{27}$. Find

(i) the first term and the common ratio, [3]
(ii) the sum of the first 5 terms of the progression, [2]
(iii) the sum to infinity. [2]

Solution

(i) $ar^2 = \frac{2}{9} \ldots\ldots\ldots (i)$

$ar^3 = \frac{2}{27} \ldots\ldots\ldots (ii)$

Divide equation (ii) by equation (i)

$\frac{ar^3}{ar^2} = \frac{2}{27} \div \frac{2}{9}$

$r = \frac{2}{27} \times \frac{9}{2}$

$r = \frac{1}{3}$

$a\left(\frac{1}{3}\right)^2 = \frac{2}{9}$

$\frac{a}{9} = \frac{2}{9}$

$a = 2$

∴ The first term is 2 and the common ratio is $\frac{1}{3}$.

(ii) $S_n = \frac{a(1-r^n)}{1-r}$

$S_2 = \frac{2\left[1-\left(\frac{1}{3}\right)^5\right]}{1-\frac{1}{3}}$

$= \frac{2\left(1-\frac{1}{243}\right)}{\frac{2}{3}}$

$= 2\left(\frac{242}{243}\right) \times \frac{3}{2}$

$= \frac{242}{81}$

$= 2\frac{80}{81}$

(iii) $S_\infty = \frac{a}{1-r}$

$= \frac{2}{1-\frac{1}{3}}$

$= \frac{2}{\frac{2}{3}}$

$= 2 \times \frac{3}{2}$

$= 3$

2019.P2.Nov

Given the geometric progression 4, 8, 16, ..., find

(i) the geometric mean of 256 and 1 024, [2]
(ii) the 11th term, [2]
(iii) the sum of the first 11 terms. [3]

Solution

(i) $\text{GM} = \pm\sqrt{ac}$

$= \pm\sqrt{256 \times 1\,024}$

$= \pm\sqrt{262\,144}$

$= \pm 512$

$= 512 \; or \; -512$

(ii) 4, 8, 16, ...,

$a = 4 \; and \; r = 2$

$T_n = ar^{n-1}$

$T_{11} = 4 \times 2^{11-1}$

$= 4 \times 2^{10}$

$= 4 \times 1\,024$

$= 4\,096$

(iii) $S_n = \frac{a(r^n-1)}{r-1}$

$S_{11} = \frac{4(2^{11}-1)}{2-1}$

$= \frac{4(2\,048-1)}{1}$

$= 4(2\,047)$

$= 8\,188$

2020.P2.Nov

The first three terms of a geometric progression are: $m-2, m+1$ and $m+7$. Find

(i) the value of m, [3]

(ii) the common ratio, [2]

(iii) the sum of the first 6 terms. [2]

Solution

(i) $m-2, m+1, m+7, \ldots,$

$$\frac{m+1}{m-2} = \frac{m+7}{m+1}$$
$$(m+1)(m+1) = (m+7)(m-2)$$
$$m^2 + m + m + 1 = m^2 - 2m + 7m - 14$$
$$m^2 + 2m + 1 = m^2 + 5m - 14$$
$$m^2 - m^2 + 2m - 5m = -14 - 1$$
$$-3m = -15$$
$$m = \frac{-15}{-3}$$
$$m = 5$$

(ii) $r = \frac{T_2}{T_1}$

$$= \frac{6}{3}$$
$$= 2$$

(iii) $S_n = \frac{a(r^n-1)}{r-1}$

$$S_4 = \frac{3(2^6-1)}{2-1}$$
$$= \frac{3(63)}{1}$$
$$= 189$$

2021.P2.Nov

The second and the fifth terms of a geometric progression are 16 and 2 respectively. Find

(i) the common ratio and the first term, [3]

(ii) the 12th term, [2]

(iii) the sum of the first 10 terms. [2]

Solution

(i) $ar = 16$ ..... (i)

$ar^4 = 2$ ..... (ii)

$\frac{ar^4}{ar} = \frac{2}{16}$

$r^3 = \frac{1}{8}$

$r = \sqrt[3]{\frac{1}{8}}$

$r = \frac{1}{2}$

$a\left(\frac{1}{2}\right) = 16$

$\frac{a}{2} = 16$

$a = 16 \times 2$

$a = 32$

$\therefore$ The common ratio is $\frac{1}{2}$ and the first term is 32.

(ii) $T_n = ar^{n-1}$

$T_{12} = 32 \times \left(\frac{1}{2}\right)^{12-1}$

$= 32 \times \left(\frac{1}{2}\right)^{11}$

$= 32 \times \frac{1}{2\,048}$

$= \frac{1}{64}$

(iii) $S_n = \frac{a(1-r^n)}{1-r}$

$S_{10} = \frac{32\left[1-\left(\frac{1}{2}\right)^{10}\right]}{1-\frac{1}{2}}$

$= \frac{32\left(\frac{1023}{1024}\right)}{\frac{1}{2}}$

$= \frac{1\,023}{32} \div \frac{1}{2}$

$= 63\frac{15}{16}$

2022.P2.Nov

Given that the third and sixth terms of a geometric progression are $\frac{3}{4}$ and $\frac{3}{32}$ respectively, find the

(i) first term and the common ratio, [3]

(ii) n^{th} term, [2]

(iii) sum to infinity of the progression. [2]

Solution

(i) $ar^2 = \frac{3}{4}$ ..... (i)

$ar^5 = \frac{3}{32}$ ..... (ii)

$\frac{ar^5}{ar^2} = \frac{3}{32} \times \frac{4}{3}$

$r^3 = \frac{1}{8}$

$r = \sqrt[3]{\frac{1}{8}}$

$r = \frac{1}{2}$

$a\left(\frac{1}{2}\right)^2 = \frac{3}{4}$

$\frac{a}{4} = \frac{3}{4}$

$4a = 3 \times 4$

$a = 3$

$\therefore$ The common ratio is $\frac{1}{2}$ and the first term is 3.

(ii) $T_n = ar^{n-1}$

$T_{12} = 3 \times \left(\frac{1}{2}\right)^{n-1}$

$= 3 \times \left(\frac{1}{2}\right)^{n} \times \left(\frac{1}{2}\right)^{-1}$

$= 3 \times \left(\frac{1}{2}\right)^{n} \times 2$

$= 6\left(\frac{1}{2}\right)^{n}$

$= \frac{6}{2^n}$

(iii) $S_\infty = \frac{a}{1-r}$

$= \frac{3}{1-\frac{1}{2}}$

$= \frac{3}{\frac{1}{2}}$

$= 3 \div \frac{1}{2}$

$= 3 \times 2$

$= 6$

2023.P2.Nov

The nth term of a geometric progression is given by $180\left(\frac{1}{2}\right)^n$. Find the

(i) first 3 terms, [2]

(ii) common ratio, [2]

(iii) sum to infinity of the progression. [3]

Solution

(i) $T_n = 180\left(\frac{1}{2}\right)^n$

$T_1 = 180\left(\frac{1}{2}\right)^1$

$= 90$

$T_2 = 180\left(\frac{1}{2}\right)^2$

$= 45$

$T_3 = 180\left(\frac{1}{2}\right)^3$

$= 22\frac{1}{2}$

$\therefore$ The first three terms are 90, 45 and $22\frac{1}{2}$

(ii) $r = T_2 \div T_1$

$= 45 \div 90$

$= \frac{1}{2}$

(iii) $S_\infty = \frac{a}{1-r}$

$= \frac{90}{1-\frac{1}{2}}$

$= \frac{90}{\frac{1}{2}}$

$= 90 \div \frac{1}{2}$

$= 90 \times 2$

$= 180$

2024.P2.GCE

The first three terms of a geometric progression are $3w-2, 3w+6$ and $3w+30$. Find the

(i) value of w, [3]

(ii) n^{th} term, [2]

(iii) sum of the first 4 terms. [2]

Solution

(i) $\frac{3w+6}{3w-2} = \frac{3w+30}{3w+6}$

$(3w+6)(3w+6) = (3w+30)(3w-2)$

$9w^2 + 18w + 18w + 36 = 9w^2 - 6w + 90w - 60$

$9w^2 + 36w + 36 = 9w^2 + 84w - 60$

$9w^2 - 9w^2 + 36w - 84w = -60 - 36$

$-48w = -96$

$w = \frac{-96}{-48}$

$w = 2$

(ii) $4, 12, 36, \ldots,$

$a = 4 \; and \; r = 3$

$T_n = ar^{n-1}$

$= 4 \times 3^{n-1}$

$= 4 \times 3^n \div 3^1$

$= 4 \times 3^n \times \frac{1}{3}$

$= 4 \times \frac{1}{3} \times 3^n$

$= \frac{4}{3} \times 3^n$

$= \frac{4(3^n)}{3}$

(iii) $S_n = \frac{a(r^n-1)}{r-1}$

$S_4 = \frac{4(3^4-1)}{3-1}$

$= \frac{4(80)}{2}$

$= 160$

COORDINATE GEOMETRY

Objectives:

By the end of this topic, learners should

1. Define Coordinate geometry.
2. Calculate distance between two points on the straight line.
3. Find the midpoint of two points on the line segment.
4. Calculate the gradient of a straight line.
5. Determine the gradients of parallel and perpendicular lines.
6. Find the equation of a straight line.

Definition

- ✓ Coordinate geometry is a branch of mathematics which uses the coordinate system to make sense of geometrical relationships.
- ✓ The coordinate system is a system which uses Cartesian coordinates to establish the position of a point or any geometric figure.

The length (distance) of a straight line

- ✓ The length of a straight line is the distance between the two end points.
- ✓ It is called the magnitude of two given points.
- ✓ The length of a line joining two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\left|\overrightarrow{PQ}\right| = \sqrt{x^2 + y^2}$$

Example 1

Find the length of a straight line joining points P(4, 1) and Q(8, 4).

Solution

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(8-4)^2 + (4-1)^2}$$
$$= \sqrt{4^2 + 3^2}$$
$$= \sqrt{16 + 9}$$
$$= \sqrt{25}$$
$$= 5\ units$$

Alternatively:

$$\overrightarrow{PQ} = \overrightarrow{PO} + \overrightarrow{OQ}$$
$$\overrightarrow{PQ} = \begin{pmatrix} -4 \\ -1 \end{pmatrix} + \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$
$$\overrightarrow{PQ} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$
$$\left|\overrightarrow{PQ}\right| = \sqrt{x^2 + y^2}$$
$$\left|\overrightarrow{PQ}\right| = \sqrt{4^2 + 3^2}$$
$$\left|\overrightarrow{PQ}\right| = \sqrt{16 + 9}$$
$$\left|\overrightarrow{PQ}\right| = \sqrt{25}$$
$$\left|\overrightarrow{PQ}\right| = 5\ units$$

The midpoint of a line segment

- ✓ The midpoint of a line segment is a point which lies exactly halfway between two points.
- ✓ It is given by the formula: $M.P = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

Example 2

Find the coordinates of the midpoint of a line segment AB where A is (5, 2) and B is (1, 2).

Solution

$$\boldsymbol{M.P} = \left(\frac{\boldsymbol{x_1+x_2}}{\boldsymbol{2}}, \frac{\boldsymbol{y_1+y_2}}{\boldsymbol{2}}\right)$$

$$= \left(\frac{\boldsymbol{5+1}}{\boldsymbol{2}}, \frac{\boldsymbol{2+2}}{\boldsymbol{2}}\right)$$

$$= \left(\frac{\boldsymbol{6}}{\boldsymbol{2}}, \frac{\boldsymbol{4}}{\boldsymbol{2}}\right)$$

$$= (\boldsymbol{3}, \boldsymbol{2})$$

The gradient of a straight

- ✓ The gradient of a straight line joining two points is a measure of steepness of the line of those two points.
- ✓ It is the ratio of the change in y-coordinates to the change in the x-coordinates.
- ✓ It is normally denoted by m.
- ✓ The gradient of a straight line is constant.

$$\therefore Gradient(m) = \frac{change\ in\ y}{change\ in\ x}$$

$$= \frac{y_2 - y_1}{x_2 - x_1}$$

Example 3

Find the gradient of the straight line joining the points (4, 5) and (9, 15).

Solution

$$\boldsymbol{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

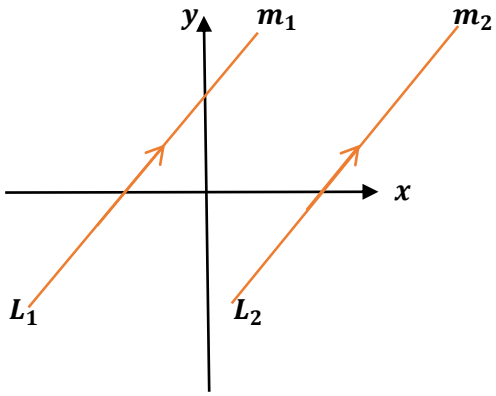
$$= \frac{\mathbf{15 - 5}}{\mathbf{9 - 4}}$$

$$= \frac{\mathbf{10}}{\mathbf{5}}$$

$$= \mathbf{2}$$

Gradient of parallel lines

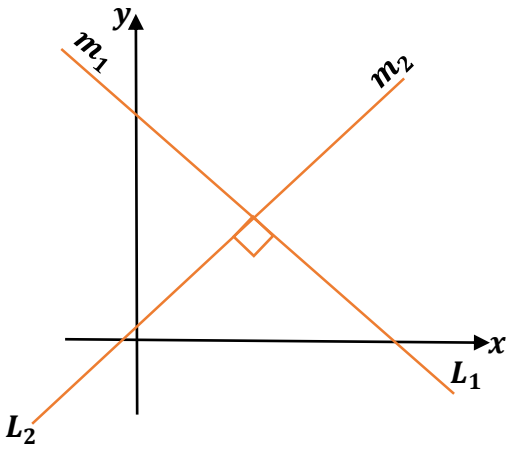
- ✓ If two lines are parallel, their gradients are equal.
- ✓ Consider two lines, L_1 with gradient m_1 and $L_2$with gradient m_2



$$\therefore \boldsymbol{m_1 = m_2}$$

Gradients of perpendicular lines

- ✓ If two lines are perpendicular, the product of their gradients is -1
- ✓ If L_1 has gradient m_1 and L_2 has gradient m_2, then $\boldsymbol{m_1 m_2 = -1}$.



- ✓ Multiplying gradients for perpendicular lines L_1 and L_2 shows that $\boldsymbol{m_1 \times m_2 = -1}$.

The equation of a straight line

✓ To find the equation of a straight line when given two points, we follow the following steps:

(i) Use the given points to find the gradient $m = \frac{y_2 - y_1}{x_2 - x_1}$.

(ii) Take the gradient together with one of the points and feed them into one of the general standard form for the equation of a straight line either $\boldsymbol{y = mx + c}$ or $\boldsymbol{y - y_1 = m(x - x_1)}$.

(iii) Write out the equation $\boldsymbol{y = mx + c}$ replacing $\boldsymbol{m}$ and $\boldsymbol{c}$ with their values.

Example 4

Find the equation of a straight line that passes through the points (2, 3) and (4, 4)

Solution

Method 1

$\boldsymbol{(2,3)\ and\ (4,4)}$

$$\boldsymbol{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

$$\boldsymbol{= \frac{4-3}{4-2}}$$

$$\boldsymbol{= \frac{1}{2}}$$

$$\boldsymbol{y - y_1 = m(x - x_1)}$$

$$\boldsymbol{y - 3 = \frac{1}{2}(x - 2)}$$

$$\boldsymbol{2y - 6 = x - 2}$$

$$\boldsymbol{2y = x - 2 + 6}$$

$$\boldsymbol{2y = x + 4}$$

Method 2

(2, 3) and (4, 4)

$$\boldsymbol{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

$$\boldsymbol{= \frac{4-3}{4-2}}$$

$$\boldsymbol{= \frac{1}{2}}$$

$$\boldsymbol{y = mx + c}$$

$$\boldsymbol{4 = \frac{1}{2}(4) + c}$$

$$\boldsymbol{4 = 2 + c}$$

$$\boldsymbol{4 - 2 = c}$$

$$\boldsymbol{c = 2}$$

$$\boldsymbol{\therefore y = \frac{1}{2}x + 2}$$

$$\boldsymbol{2y = x + 4}$$

2018.P1.Oct

(a) Find the equation of a straight line passing through (1, 5) and (2, 10).

(b) The point (5, 2) is the midpoint of a straight line joining A(x, 9) and B(3, y). Find the value of x and the value of y.

Solution

(a) (1, 5) and (2, 10)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{10-5}{2-1}$$

$$= \frac{5}{1}$$

$$= 5$$

$$y - y_1 = m(x - x_1)$$

$$y - 5 = 5(x - 1)$$

$$y - 5 = 5x - 5$$

$$y = 5x - 5 + 0$$

$$y = 5x$$

(b) $A(x, 9)\ and\ B(3, y)$

$$\text{M.P} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$(5, 2) = \left(\frac{x+3}{2}, \frac{9+y}{2}\right)$$

$$5 = \frac{x+3}{2}$$

$$x + 3 = 10$$

$$x = 10 - 3$$

$$x = 7$$

$$2 = \frac{9+y}{2}$$

$$9 + y = 4$$

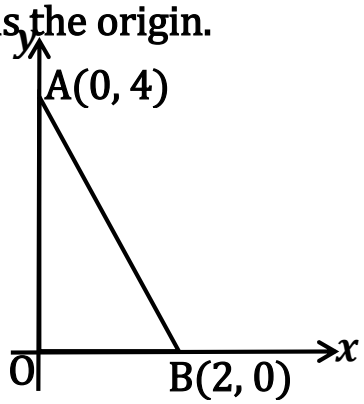
$$y = 4 - 9$$

$$y = -5$$

2019.P1.Nov

(a) A and B are points with coordinates (-3, 3) and (5, 9) respectively. Find the length of AB.

(b) In the diagram below, A is the point (0, 4) and B is the point (2, 0) and O is the origin.



Find the equation of the straight line through O which is parallel to the line AB.

Solution

(a) A(-3, 3) and B(5, 9)

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
$$= \sqrt{(5+3)^2 + (9-3)^2}$$
$$= \sqrt{8^2 + 6^2}$$
$$= \sqrt{64 + 36}$$
$$= \sqrt{100}$$
$$= 10\ units$$

(b) A(0, 4) and B(2, 0)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$
$$m = \frac{0-4}{2-0}$$
$$= \frac{-4}{2}$$
$$= -2$$
$$y - y_1 = m(x - x_1)$$
$$y - 0 = -2(x - 0)$$
$$y = -2x + 0$$
$$y + 2x = 0$$

2020.P1.Nov

(a) The points A and B are $(-5, 15)$ and $(3w, w)$ respectively. Given that the midpoint of the straight line joining A and B is (5, 10), find the value of w.

(b) P is the point (3, 1) and Q is another point such that the gradient of the line PQ is $-\frac{1}{2}$. Write down the equation of PQ.

Solution

(a) $A(-5, 15)\ and\ B(3w, w)$

$$M.P = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$(5, 10) = \left(\frac{3w-5}{2}, \frac{w+15}{2}\right)$$

$$5 = \frac{3w-5}{2}$$

$$3w - 5 = 10$$

$$3w = 10 + 5$$

$$3w = 15$$

$$w = \frac{15}{3}$$

$$w = 5$$

(b) $y - y_1 = m(x - x_1)$

$$y - 1 = -\frac{1}{2}(x - 3)$$

$$2y - 2 = -1(x - 3)$$

$$2y - 2 = -x + 3$$

$$2y + x = 3 + 2$$

$$2y + x = 5$$

2021.P1.Nov

(a) Find the equation of a straight line passing through the points (3, 4) and (7, 12).

(b) Find the equation of a straight line which is perpendicular to the line $2y - x = 3$ and passing through the point $(-4, 1)$.

Solution

(a) (3, 4) and (7, 12)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$
$$m = \frac{12-4}{7-3}$$
$$= \frac{8}{4}$$
$$= 2$$

$$y - y_1 = m(x - x_1)$$
$$y - 4 = 2(x - 3)$$
$$y - 4 = 2x - 6$$
$$y = 2x - 6 + 4$$
$$y = 2x - 2$$

(b) $2y - x = 3$

$$y = \frac{1}{2}x + \frac{3}{2}$$
$$m_1 = \frac{1}{2}$$
$$m_1 \times m_2 = -1$$
$$m_2 = -2$$
$$y - y_1 = m(x - x_1)$$
$$y - 1 = -2(x + 4)$$
$$y - 1 = -2x - 8$$
$$y + 2x = -8 + 1$$
$$y + 2x = -7$$

2022.P1.Nov

(a) Find the gradient of the line passing through the points $A(4, -6)$ and B(2, 4).

(b) ABC is a straight line. The coordinates of A and B are (2, 1) and $(-6, 5)$ respectively. Given that B is the midpoint of AC, find the coordinates of C.

Solution

(a) A(4, -6) and B(2, 4)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4+6}{2-4}$$

$$= \frac{10}{-2}$$

$$= -5$$

(b) $A(2, 1)\ and\ B(-6, 5)$

$$\text{M.P} = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$(-6, 5) = \left(\frac{x+2}{2}, \frac{y+1}{2}\right)$$

$$-6 = \frac{x+2}{2}$$

$$x + 2 = -12$$

$$x = -12 - 2$$

$$x = -14$$

$$5 = \frac{y+1}{2}$$

$$y + 1 = 10$$

$$y = 10 - 1$$

$$y = 9$$

$$\therefore C(-14, 9)$$

2023.P1.Nov

(a) The point A$(-5, 3)$ and the point B have a midpoint $(4, -2)$. Find the coordinates of B.

(b) Find the equation of a line parallel to the line $2x + y = 4$ passing through $(-5, 3)$.

Solution

(a) $A(-5, 3)\ and\ B(x, y)$

$M.P = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

$(4, -2) = \left(\frac{x-5}{2}, \frac{y+3}{2}\right)$

$4 = \frac{x-5}{2}$

$x - 5 = 8$

$x = 8 + 5$

$x = 13$

$-2 = \frac{y+3}{2}$

$y + 3 = -4$

$y = -4 - 3$

$y = -7$

$\therefore B(13, -7)$

(b) $y - y_1 = m(x - x_1)$

$y - 3 = -2(x + 5)$

$y - 3 = -2x - 10$

$y + 2x = -10 + 3$

$y + 2x = -7$

2024.P1.GCE

(a) Two points A and B have coordinates $(-5, -3)$ and $(-1, a)$ respectively. Given that the gradient of the line AB is 2, find the value of a.

(b) The straight line $y = \frac{1}{2}x + p$ passes through the point $(-2, 2)$. Find the coordinates of the y-intercept.

Solution

(a) A(-5, -3) and B(-1, a)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$2 = \frac{a+3}{-1+5}$$
$$a + 3 = 2(-1 + 5)$$
$$a + 3 = -2 + 10$$
$$a + 3 = 8$$
$$a = 8 - 3$$
$$a = 5$$

(b) $y = \frac{1}{2}x + p$

$$2 = \frac{1}{2}(-2) + p$$
$$2 = -1 + p$$
$$p = 2 + 1$$
$$p = 3$$
$$(0, 3)$$

RELATIONS AND FUNCTIONS

Objectives

By the end of this topic, learners should

1 Define relations and functions.
2 Find the inverses of one-to-one functions.
3 Simplify composite functions.
4 Solve problems involving linear functions.

Inverse functions

- The inverse of a function is a function that reverses the mapping of the original function.
- We use these notations for an inverse function $f^{-1}(x)$ and $f^{-1}: x \rightarrow$.
- The inverse of a function exists if, and only if, the original function is a one-to-one mapping.
- A mapping is the matching of the elements of one set to the elements of another set by a means of a rule.
- A function is a one-to-one or a many-to-one mapping.
- We use these notations for a function $f(x) =$ and $f: x \rightarrow . x \leq 0, or\ x \geq 0$.
- If a function is a many-to-one mapping, we can restrict the domain of the function in such a way that the inverse function exists.
- This restriction is often of the form
- The domain of a function is the set of x-values for which the function is defined.

- The range of a function is the set of y-values for which the function is defined.
- The graphs of $f(x)$ and $f^{-1}(x)$ are reflections of each other in the line $y = x$.
- A function that is identical to its inverse is called a self-inverse function.

Composite functions

- A composite function is a function that is a combination of two or more functions.
- We use these notations for a composite function $g \circ f$ or $gf(x)$.
- The combination of functions is not commutative, e.g $fg(x) \neq gf(x)$.

Applications

- A linear function is a function of the form $f(x) = mx + c$.
- A linear cost function is a function of the form $C(x) = ax + b$, where a is the cost of making or selling a single item, x is the number of items made or sold and b is the fixed cost of production or sales.

2018.P1.Oct

Given that $f(x) = 2x - 5$ and $g(x) = x + 3$, find

(a) $f^{-1}(x)$,
(b) $f^{-1}(-10)$,
(c) $gf(3)$.

Solution

(a) Let $f(x) = y$

$2x - 5 = y$

$2x = y + 5$

$x = \frac{y+5}{2}$

$y = \frac{x+5}{2}$

$\therefore f^{-1}(x) = \frac{x+5}{2}$

(b) $f^{-1}(-10) = \frac{-10+5}{2}$

$= -\frac{5}{2}$

$= -2\frac{1}{2}$

(c) $gf(3)$

$g[2(3) - 5]$

$g(1)$

$1 + 3$

2019.P1.Nov

The functions f and g are defined by $f(x) = 2x - 3$ and $g(x) = 3x$. Find

(a) $f^{-1}(x)$,
(b) $gf(x)$,
(c) $gf(2)$.

Solution

(a) Let $f(x) = y$

$2x - 3 = y$

$2x = y + 3$

$x = \frac{y+3}{2}$

$y = \frac{x+5}{2}$

$\therefore f^{-1}(x) = \frac{x+3}{2}$

(b) $gf(x)$

$g(2x - 3)$

$3(2x - 3)$

$6x - 9$

(c) $gf(2)$

$6(2) - 9$

$12 - 9$

3

2020.P1.Nov

The functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = 4x + 1$. Find

(a) $f^{-1}(x)$,
(b) $gf(x)$,
(c) $gf(-2)$.

Solution

(a) Let $f(x) = y$

$3x - 2 = y$

$3x = y + 2$

$x = \frac{y+2}{3}$

$y = \frac{x+2}{3}$

$\therefore f^{-1}(x) = \frac{x+2}{3}$

(b) $gf(x)$

$g(3x - 2)$

$4(3x - 2) + 1$

$12x - 8 + 1$

$12x - 7$

(c) $gf(-2)$

$12(-2) - 7$

$-24 - 7$

-31

2021.P1.Nov

The functions f and g are defined by $f(x) = \frac{x+9}{2}$ and $g(x) = x - 2$. Find

(a) $f^{-1}(x)$,
(b) $fg(x)$,
(c) $fg(-5)$.

Solution

(a) Let $f(x) = y$

$\frac{x+9}{2} = y$

$x + 9 = 2y$

$x = 2y - 9$

$y = 2x - 9$

$\therefore f^{-1}(x) = 2x - 9$

(b) $fg(x)$

$f(x - 2)$

$\frac{x-2+9}{2}$

$\frac{x+7}{2}$

(c) $fg(-5)$

$\frac{-5+7}{2}$

$\frac{2}{2}$

1

2022.P1.Nov

Given that $f(x) = 3x + 1$ and $g(x) = 4x - 1$, find

(a) $f^{-1}(x)$,
(b) $f^{-1}(-5)$,
(c) $fg(x)$.

Solution

(a) Let $f(x) = y$

$3x + 1 = y$

$3x = y - 1$

$x = \frac{y-1}{3}$

$y = \frac{x-1}{3}$

$\therefore f^{-1}(x) = \frac{x-1}{3}$

(b) $f^{-1}(-5)$

$\frac{-5-1}{3}$

$\frac{-6}{3}$

-2

(c) $fg(x)$

$f(4x - 1)$

$3(4x - 1) + 1$

$12x - 3 + 1$

$12x - 2$

$2(6x - 1)$

2023.P1.Nov

The functions g and h are defined by $g(x) = 3x + 1$ and $h(x) = \frac{x-1}{2}$.

Find

(a) $g^{-1}(x)$,
(b) $gh(x)$,
(c) x if $gg(x) = 22$.

Solution

(a) Let $g(x) = y$

$3x + 1 = y$

$3x = y - 1$

$x = \frac{y-1}{3}$

$y = \frac{x-1}{3}$

$\therefore g^{-1}(x) = \frac{x-1}{3}$

(b) $gh(x)$

$g\left(\frac{x-1}{2}\right)$

$3\left(\frac{x-1}{2}\right) + 1$

$\frac{3x-3}{2} + 1$

$\frac{3x-3+2}{2}$

$\frac{3x-1}{2}$

(c) $gg(x) = 22$

$g(3x + 1) = 22$

$3(3x + 1) + 1 = 22$

$9x + 3 + 1 = 22$

$9x + 4 = 22$

$9x = 22 - 4$

$9x = 18$

$x = \frac{18}{9}$

$x = 2$

2024.P1.GCE

The functions f and g are defined by $f(x) = 3x - 4$ and $g(x) = \frac{x+1}{3}$. Find

(a) $f^{-1}(x)$,
(b) $gf(x)$,
(c) $gf(-3)$.

Solution

(a) Let $f(x) = y$

$3x - 4 = y$

$3x = y + 4$

$x = \frac{y+4}{3}$

$y = \frac{x+4}{3}$

$\therefore f^{-1}(x) = \frac{x+4}{3}$

(b) $gf(x)$

$g(3x - 4)$

$\frac{3x-4+1}{3}$

$\frac{3x-3}{3}$

$\frac{3(x-1)}{3}$

$x - 1$

(c) $gf(-3)$

$-3 - 1$

-4

Objectives

By the end of this topic, learners should

1. Define a quadratic function.
2. Find the turning point of a parabola.
3. Determine the minimum, maximum and the axis of symmetry of the graph of quadratic function.
4. Calculate the x- and y-intercepts of a parabola
5. Sketch the graph of a quadratic function.
6. Apply the concepts for quadratic functions to solve real life problems.

The Quadratic function

- A quadratic function is a function of the form $f(x) = ax^2 + bx + c$, where a, b and c are constants and $a \neq 0$.
- The graph of a quadratic function is called a parabola.
- If a is positive, then the parabola has the shape of smile (∪) and the parabola has a minimum point.
- If a is negative, then the parabola has the shape of a frown (∩) and the parabola has a maximum point.

The Turning point of a parabola

- This is the point where a parabola turns, which is the minimum or maximum point of the parabola.
- Every parabola has a turning point and is symmetrical about the vertical line that passes through the turning point.
- This vertical line is called the axis of symmetry of the parabola.
- The x-coordinate of the turning point of $y = ax^2 + bx + c$ is given by the formula:

  $y = ax^2 + bx + c$

$\frac{dy}{dx} = 2ax + b$

$\frac{dy}{dx} = 0$

$2ax + b = 0$

$2ax = -b$

$x = -\frac{b}{2a}$

- The y-coordinate of the turning point of $y = ax^2 + bx + c$ can be found by substituting the x-coordinate of the turning point into the equation of the parabola which can give us the formula:

$y = ax^2 + bx + c$

$y = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c$

$y = a\left(\frac{b^2}{4a^2}\right) - \frac{b^2}{2a} + c$

$y = \frac{ab^2}{4a^2} - \frac{b^2}{2a} + c$

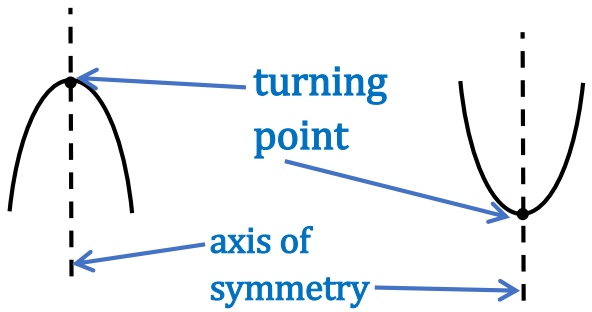
$y = \frac{b^2}{4a} - \frac{b^2}{2a} + c$

$y = \frac{b^2 - 2b^2 + 4ac}{4a}$

$y = \frac{-b^2 + 4ac}{4a}$

$y = \frac{4ac - b^2}{4a}$

- The x-coordinate of the turning point is also known as the equation of the axis of symmetry of the graph.
- The y-coodinate of the turning point is the minimum or maximum value of the graph.



The x- and y-intercepts of a parabola

- The parabola $y = ax^2 + bx + c$ intersects the y-axis at the point where $x = 0$ and the x-axis at the point where $y = 0$ because in the y-axis set $x = 0$ and in the x-axis set $y = 0$.
- To find the y-intercept, $y = f(0)$:

  $y = ax^2 + bx + c$

  $f(0) = a(0)^2 + b(0) + c$

  $f(0) = c$

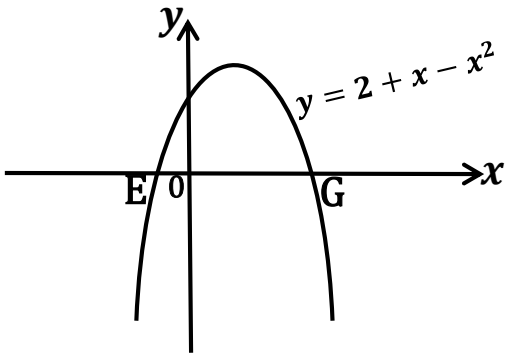
  $\therefore c$ is the y-intercept.
- The x-intercepts of the quadratic function are also called the roots of the corresponding quadratic equation for $y = 0$:

  $y = ax^2 + bx + c$

  $ax^2 + bx + c = 0$
- To find the x-intercepts, we use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ or factorization method if the corresponding equation has real roots.

2018.P1.Oct

The diagram below shows a sketch of the graph of $y = 2 + x - x^2$ passing through the points E and G.



Find the

(a) coordinates of E and G,

(b) maximum value of y.

Solution

(a) $-x^2 + x + 2 = 0$

$x^2 - x - 2 = 0$

$x^2 + x - 2x - 2 = 0$

$x(x+1) - 2(x+1) = 0$

$(x-2)(x+1) = 0$

$x - 2 = 0 \; or \; x + 1 = 0$

$x = 2 \; or \; x = -1$

$\therefore E(-1, 0) \; and \; G(2, 0)$

(b) $y = \frac{4ac - b^2}{4a}$

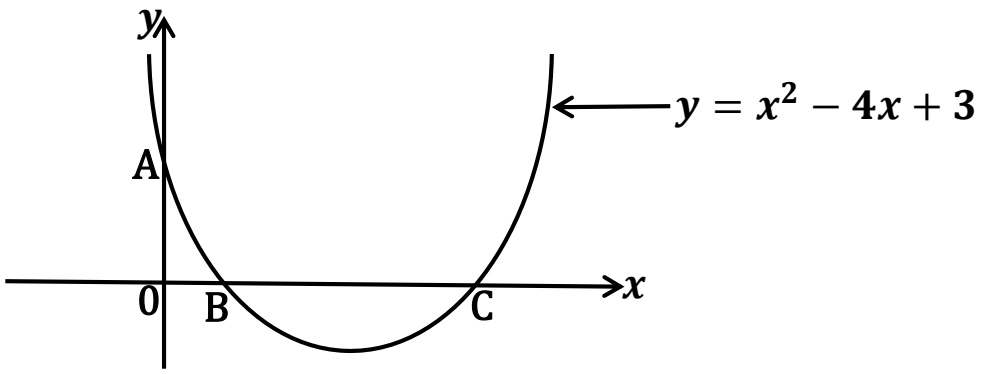
$y = \frac{4(-1)(2) - 1^2}{4(-1)}$

$y = \frac{-9}{-4}$

$y = 2\frac{1}{4}$

2019.P1.Nov

The sketch shown below represents the graph of the curve $y = x^2 - 4x + 3$, passing through the points A, B and C.



Find the

(a) the coordinates of the points B and C,

(b) minimum value of y.

Solution

(a) $x^2 - 4x + 3 = 0$

$x^2 - 4x + 3 = 0$

$x^2 - x - 3x + 3 = 0$

$x(x-1) - 3(x-1) = 0$

$(x-3)(x-1) = 0$

$x - 3 = 0 \; or \; x - 1 = 0$

$x = 3 \; or \; x = 1$

$\therefore B(1,0) \; and \; C(3,0)$

(b) $y = \frac{4ac - b^2}{4a}$

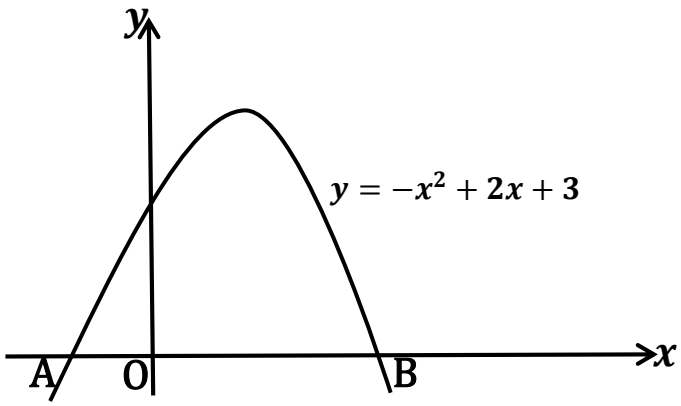
$y = \frac{4(1)(3) - (-4)^2}{4(1)}$

$y = \frac{-4}{4}$

$y = -1$

2020.P1.Nov

The diagram below shows the sketch of the graph of $y = -x^2 + 2x + 3$ cutting the x-axis at A and B.



Find the

(a) coordinates of A,

(b) maximum value of the graph.

Solution

(a) $-x^2 + 2x + 3 = 0$

$-x^2 - x + 3x + 3 = 0$

$-x(x + 1) + 3(x + 1) = 0$

$(3 - x)(x + 1) = 0$

$3 - x = 0 \; or \; x + 1 = 0$

$x = 3 \; or \; x = -1$

$\therefore A(-1, 0)$

(b) $y = \frac{4ac - b^2}{4a}$

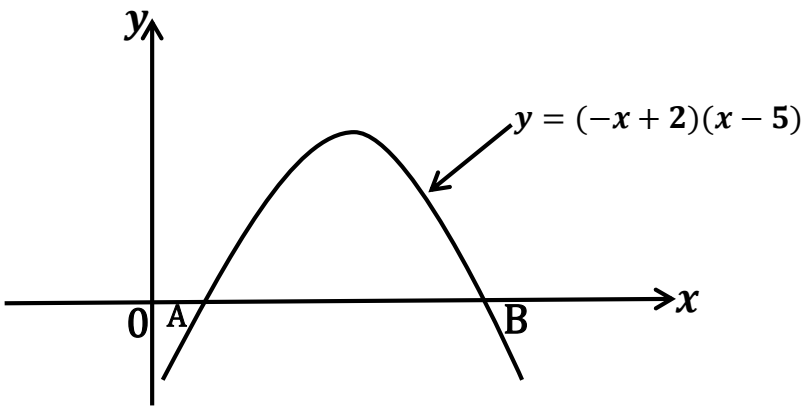
$y = \frac{4(-1)(3) - (2)^2}{4(-1)}$

$y = \frac{-16}{-4}$

$y = 4$

2021.P1.Nov

The diagram below shows the graph of $y = (-x + 2)(x - 5)$. The graph cuts the x-axis at A and B.



Find the

(a) coordinates of the points A and B,

(b) maximum value of y.

Solution

(a) $(-x + 2)(x - 5) = 0$

$-x + 2 = 0 \; or \; x - 5 = 0$

$x = 2 \; or \; x = 5$

$\therefore A(2, 0) \; and \; B(5, 0)$

(b) $y = -x^2 + 7x - 10$

$y = \frac{4ac - b^2}{4a}$

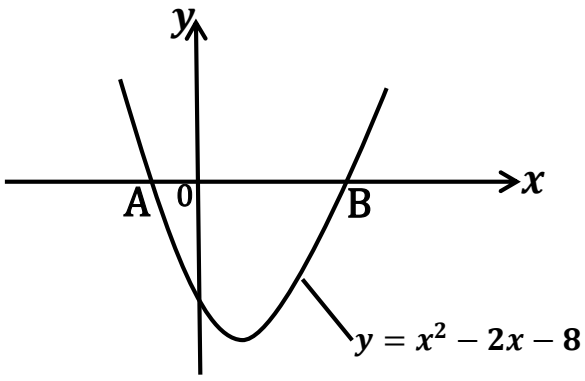
$y = \frac{4(-1)(-10) - (7)^2}{4(-1)}$

$y = \frac{-9}{-4}$

$y = 2\frac{1}{4}$

2022.P1.Nov

The sketch below shows the graph of $y = x^2 - 2x - 8$.



Find the coordinates of

(a) A and B,

(b) the minimum point on the graph.

Solution

(b) $x^2 - 2x - 8 = 0$

$x^2 - 4x + 2x - 8 = 0$

$x(x - 4) + 2(x - 4) = 0$

$(x + 2)(x - 4) = 0$

$x + 2 = 0 \; or \; x - 4 = 0$

$x = -2 \; or \; x = 4$

$\therefore A(-2, 0) \; and \; B(4, 0)$

(a) $x = -\frac{b}{2a}$

$x = \frac{2}{2(1)}$

$x = 1$

$y = \frac{4ac - b^2}{4a}$

$y = \frac{4(1)(-8) - (-2)^2}{4(1)}$

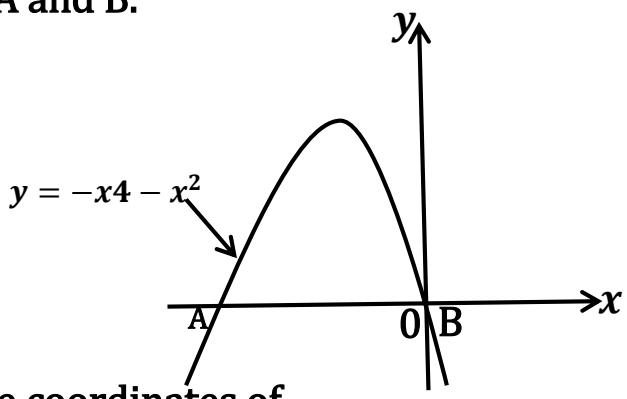
$y = \frac{-36}{4}$

$y = -9$

$\therefore$ *The minimum turning point on the graph is* $(1, -9)$

2023.P1.Nov

The diagram below show the sketch of the graph of $y = -4x - x^2$ cutting the x-axis at A and B.



Find the coordinates of

(a) A,

(b) the turning point of the graph.

Solution

(a) $-4x - x^2 = 0$

$-x(4 + x) = 0$

$-x = 0 \; or \; 4 + x = 0$

$x = 0 \; or \; x = -4$

$\therefore A(-4, 0)$

(b) $\frac{dy}{dx} = -4 - 2x$

$-4 - 2x = 0$

$-2x = 4$

$x = \frac{4}{-2}$

$x = -2$

$y = -4(-2) - (-2)^2$

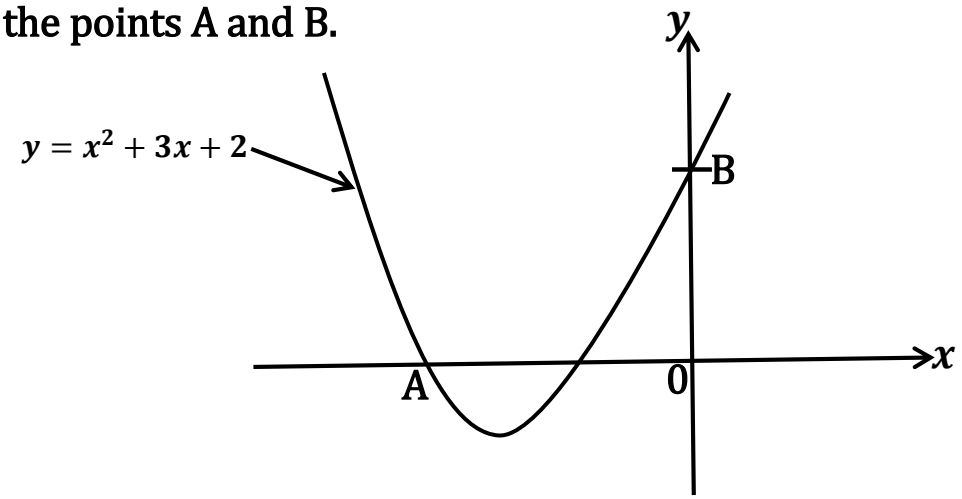
$y = 8 - 4$

$y = 4$

$\therefore (-2, 4)$

2024.P1.GCE

The diagram below shows a sketch of the graph of $y = x^2 + 3x + 2$, passing through the points A and B.



Find the

(a) coordinates of A and B,

(b) minimum turning point.

Solution

(b) $x^2 + 3x + 2 = 0$

$x^2 + x + 2x + 2 = 0$

$x(x + 1) + 2(x + 1) = 0$

$(x + 2)(x + 1) = 0$

$x + 2 = 0 \; or \; x + 1 = 0$

$x = -2 \; or \; x = -1$

∴A(-2, 0) and B(0, 2)

(a) $x = -\frac{b}{2a}$

$= -\frac{3}{2(1)}$

$= -\frac{3}{2}$

$= -1\frac{1}{2}$

$y = \frac{4ac - b^2}{4a}$

$y = \frac{4(1)(2) - (3)^2}{4(1)}$

$y = \frac{8-9}{4}$

$y = -\frac{1}{4}$

$\therefore \left(-1\frac{1}{2}, -\frac{1}{4}\right)$

QUADRATIC EQUATIONS

Objectives

By the end of this topic, learners should

1. Define a quadratic equation.
2. Solve quadratic equations by graphical method.
3. Solve quadratic equations using the factorization method.
4. Solve quadratic equations using the completing the square method.
5. Solve quadratic equations using the quadratic formula method.
6. Apply quadratic equations to solve real-life problems.

Definition

- A quadratic equation is an equation of the form $ax^2 + bx + c$, where a, b and c are constants and $a \neq 0$, and x is the unknown quantity.
- The values of x which satisfy the equation are known as the roots of the quadratic equation.

Graphical Method

- The graph of any function of the form $y = ax^2 + bx + c$ where a, b and c are constants and $a \neq 0$, is a parabola.
- If $y = f(x)$ is any function, the equation $f(x) = 0$ is satisfied by the values of x at the points where the graph of $f(x)$ crosses or touches the x-axis (where $y = 0$).

Factorization Method

- When we use factorization to solve a quadratic equation, we use the zero product rule, which states that: if $A \times B = 0$, then either A must be zero or B must be zero.
- We have to look for two numbers which gives the product of a and c as well as the sum of b.

Completing the square method

- Completing the Square is a mathematical technique used to solve quadratic equations of the form $ax^2 + bx + c = 0$.
- The method of completing the square of a quadratic equation is the method that was used to derive the quadratic formula.
- You will only use the method of completing the square if you want, or if you are a genius or if you are specifically asked to do so.

Step-by-Step Guide

1. Start with a quadratic equation: Begin with a quadratic equation in the form $ax^2 + bx + c = 0$.
2. Move the constant term: Move the constant term (c) to the right-hand side of the equation: $ax^2 + bx = -c$.
3. Make the coefficient of x^2 equal to 1: Divide both sides of the equation by the coefficient of x^2 (a), so that the coefficient of x^2 becomes 1:
   $x^2 + \left(\frac{b}{a}\right)x = -\frac{c}{a}$.
4. Add and subtract $\left(\frac{b}{2a}\right)^2$: Add $\left(\frac{b}{2a}\right)^2$ to both sides of the equation to complete the square: $x^2 + \left(\frac{b}{a}\right)x + \left(\frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$.

5 Factor the left-hand side: Factor the left-hand side of the equation into a perfect square: $\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \left(\frac{b}{2a}\right)^2$.

6 Take the square root: Take the square root of both sides of the equation: $x + \frac{b}{2a} = \pm\sqrt{-\frac{c}{a} + \left(\frac{b}{2a}\right)^2}$.

7 Solve for x: Solve for x by subtracting $\frac{b}{2a}$ from both sides of the equation: $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$.

Example

Solve the quadratic equation $x^2 + 5x + 6 = 0$ using the Completing the Square method.

Solution

$$x^2 + 5x + 6 = 0.$$

$$x^2 + 5x = -6.$$

$$x^2 + 5x + \left(\frac{5}{2}\right)^2 = -6 + \left(\frac{5}{2}\right)^2.$$

$$\left(x + \frac{5}{2}\right)^2 = -6 + \left(\frac{5}{2}\right)^2.$$

$$x + \frac{5}{2} = \pm\sqrt{-6 + \left(\frac{5}{2}\right)^2}.$$

$$x = \frac{-5 \pm \sqrt{25-24}}{2} = \frac{-5 \pm \sqrt{1}}{2}.$$

$$x = \frac{-5+1}{2} \; or \; x = \frac{-5-1}{2}.$$

$$x = -2 \; or \; x = -3.$$

2018.P1.Oct

Solve the equation $2x^2 = 8$

Solution

$x^2 = \frac{8}{2}$

$x^2 = 4$

$x^2 - 4 = 0$

$x^2 - 2^2 = 0$

$(x + 2)(x - 2) = 0$

$x + 2 = 0$ or $x - 2 = 0$

$x = -2$ or $x = 2$

2019.P1.Nov

Solve the equation $(x - 2)(2x + 5) = 0$.

Solution

$(x - 2)(2x + 5) = 0$

$x - 2 = 0$ or $2x + 5 = 0$

$x = 2$ or $2x = -5$

$x = 2$ or $x = -\frac{5}{2}$

$x = 2$ or $x = -2\frac{1}{2}$

2020.P1.Nov.

Solve the equation $5x^2 = x$.

Solution

$5x^2 = x$

$5x^2 - x = 0$

$x(5x - 1) = 0$

$x = 0$ or $5x - 1 = 0$

$x = 0$ or $5x = 1$

$x = 0$ or $x = \frac{1}{5}$

2021.P1.Nov

Solve the equation $(2x - 5)(x + 3) = 0$.

Solution

$(2x - 5)(x + 3) = 0$

$2x - 5 = 0$ or $x + 3 = 0$

$2x = 5$ or $x = -3$

$x = \frac{5}{2}$ or $x = -3$

$x = 2\frac{1}{2}$ or $x = -3$

2022.P1.Nov

Solve the equation $4y^2 - 8y = 0$.

Solution

$$4y^2 - 8y = 0$$

$$4y(y - 2) = 0$$

$$4y = 0 \text{ or } y - 2 = 0$$

$$y = \frac{0}{4} \text{ or } y = 2$$

$$y = 0 \text{ or } y =$$

2023.P1.Nov

Solve the equation $x^2 = 4(x - 3)^2$

Solution

$$x^2 = 4(x - 3)^2$$

$$x^2 - 4(x - 3)^2 = 0$$

$$[x + 4(x - 3)][x - 4(x - 3)] = 0$$

$$(x + 4x - 12)(x - 4x + 12) = 0$$

$$(5x - 12)(-3x + 12) = 0$$

$$5x - 12 = 0 \text{ or } -3x + 12 = 0$$

$$5x = 12 \text{ or } 3x = 12$$

$$x = \frac{12}{5} \text{ or } x = \frac{12}{3}$$

$$x = 2\frac{2}{5} \text{ or } x = 4$$

2024.P1.GCE

Solve the equation $(3x-5)(x+2)=0$.

Solution

$$(3x-5)(x+2)=0$$

$$3x-5=0 \text{ or } x+2=0$$

$$3x=5 \text{ or } x=-2$$

$$x=\frac{5}{3} \text{ or } x=-2$$

$$x=1\frac{2}{3} \text{or } x=-2$$

2018.P2.GCE

Solve the equation $x^2-4x-2=0$, giving your answers correct to 2 decimal places. [5]

Solution

$$x^2-4x-2=0$$

$$a=1, b=-4 \text{ and } c=-2$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$x=\frac{-(-4)\pm\sqrt{(-4)^2-4(1)(-2)}}{2(1)}$$

$$x=\frac{4\pm\sqrt{16+8}}{2}$$

$$x=\frac{4\pm\sqrt{24}}{2}$$

$$x=\frac{4+\sqrt{24}}{2} \quad \text{or} \quad x=\frac{4-\sqrt{24}}{2}$$

$$x=\left(4+\sqrt{24}\right)\div 2 \quad \text{or} \quad x=\left(4-\sqrt{24}\right)\div 2$$

$$x=4.449\,489\,743 \quad \text{or} \quad x=-0.449\,489\,742$$

$$x=4.45 \quad \text{or} \quad x=-0.45$$

By Completing the Square Method

$x^2-4x-2=0$
$x^2-4x=2$
$x^2-4x+(-2)^2=2+(-2)^2$
$(x-2)^2=2+4$
$(x-2)^2=6$
$x-2=\pm\sqrt{6}$
$x=2\pm\sqrt{6}$
$x=2+\sqrt{6}$ *or* $x=2-\sqrt{6}$
$x=4.449\,742\,783\,18$ *or* $x=-0.449\,489\,742\,783\,18$
$\therefore x=4.45$ *or* $x=-0.45$

2019.P2.Nov

Solve the equation $p^2 - p = 4$, giving your answers correct to 2 decimal places. [5]

Solution

$$p^2 - p = 4$$

$$p^2 - p + \left(-\frac{1}{2}\right)^2 = 4 + \left(-\frac{1}{2}\right)^2$$

$$\left(p - \frac{1}{2}\right)^2 = 4 + \frac{1}{4}$$

$$\left(p - \frac{1}{2}\right)^2 = \frac{17}{4}$$

$$p - \frac{1}{2} = \pm\sqrt{\frac{17}{4}}$$

$$p - \frac{1}{2} = \pm\frac{\sqrt{17}}{2}$$

$$p = \frac{1}{2} \pm \frac{\sqrt{17}}{2}$$

$$p = \frac{1 \pm \sqrt{17}}{2}$$

$$p = \frac{1+\sqrt{17}}{2} \text{ or } p = \frac{1-\sqrt{17}}{2}$$

$$p = 2.561\,552\,812\,808\,83 \; or \; p = -1.561\,552\,812\,808\,83$$

$$\therefore p = 2.56 \quad or \quad p = -1.56$$

2020.P2.Nov

Solve the equation $x^2 - x - 19 = 0$, giving your answers correct to 2 decimal places. [5]

Solution

$$x^2 - x - 19 = 0$$

$$x^2 - x = 19$$

$$x^2 - x + \left(-\frac{1}{2}\right)^2 = 19 + \left(-\frac{1}{2}\right)^2$$

$$\left(x - \frac{1}{2}\right)^2 = 19 + \frac{1}{4}$$

$$\left(x - \frac{1}{2}\right)^2 = \frac{77}{4}$$

$$x - \frac{1}{2} = \pm\sqrt{\frac{77}{4}}$$

$$x - \frac{1}{2} = \pm\frac{\sqrt{77}}{2}$$

$$x = \frac{1}{2} \pm \frac{\sqrt{77}}{2}$$

$$x = \frac{1 \pm \sqrt{77}}{2}$$

$$x = \frac{1+\sqrt{77}}{2} \text{ or } x = \frac{1-\sqrt{17}}{2}$$

$$x = 4.887\,482\,193\,696\,06 \text{ or } x = -3.887\,482\,193\,696\,06$$

$$\therefore x = 4.89 \text{ or } x = -3.89$$

2021.P2.Nov

Solve the equation $2x^2 - 13x + 14 = 0$, giving your answers correct to 2 decimal places. [5]

Solution

$$2x^2 - 13x + 14 = 0$$

$$2x^2 - 13x = -14$$

$$x^2 - \frac{13}{2}x = -7$$

$$x^2 - \frac{13}{2}x + \left(-\frac{13}{4}\right)^2 = -7 + \left(-\frac{13}{4}\right)^2$$

$$\left(x - \frac{13}{4}\right)^2 = -7 + \frac{169}{16}$$

$$\left(x - \frac{13}{4}\right)^2 = \frac{57}{16}$$

$$x - \frac{13}{4} = \pm\sqrt{\frac{57}{16}}$$

$$x - \frac{13}{4} = \pm\frac{\sqrt{57}}{4}$$

$$x = \frac{13}{4} \pm \frac{\sqrt{57}}{4}$$

$$x = \frac{13 \pm \sqrt{57}}{4}$$

$$x = \frac{13+\sqrt{57}}{4} \text{ or } x = \frac{13-\sqrt{57}}{4}$$

$$x = 5.137\,458\,609 \text{ or } x = 1.362\,541\,391$$

$$\therefore x = 5.14 \text{ or } x = 1.36$$

2022.P2.Nov

Solve the equation $3x^2 - 5x - 7 = 0$, giving your answers correct to 2 decimal places. [5]

Solution

$$3x^2 - 5x - 7 = 0$$

$$3x^2 - 5x = 7$$

$$x^2 - \frac{5}{3}x = \frac{7}{3}$$

$$x^2 - \frac{5}{3}x + \left(-\frac{5}{6}\right)^2 = \frac{7}{3} + \left(-\frac{5}{6}\right)^2$$

$$\left(x - \frac{5}{6}\right)^2 = \frac{7}{3} + \frac{25}{36}$$

$$\left(x - \frac{5}{6}\right)^2 = \frac{109}{36}$$

$$x - \frac{5}{6} = \pm\sqrt{\frac{109}{36}}$$

$$x - \frac{5}{6} = \pm\frac{\sqrt{109}}{6}$$

$$x = \frac{5}{6} \pm \frac{\sqrt{109}}{6}$$

$$x = \frac{5 \pm \sqrt{109}}{6}$$

$$x = \frac{5+\sqrt{109}}{6} \text{ or } x = \frac{5-\sqrt{109}}{6}$$

$$x = 2.573\,384\,418\,151\,76 \text{ or } x = -0.906\,717\,751\,485\,09$$

$$\therefore x = 2.57 \text{ or } x = -0.91$$

2023.P2.Nov

Solve the equation $3x^2 + 13x - 7 = 0$, giving your answers correct to 2 decimal places. [5]

Solution

$$3x^2 + 13x - 7 = 0$$

$$x^2 + 13x = 7$$

$$x^2 + \frac{13}{3}x = \frac{7}{3}$$

$$x^2 + \frac{13}{3}x + \left(\frac{13}{6}\right)^2 = \frac{7}{3} + \left(\frac{13}{6}\right)^2$$

$$\left(x + \frac{13}{6}\right)^2 = \frac{7}{3} + \frac{169}{36}$$

$$\left(x + \frac{13}{6}\right)^2 = \frac{253}{36}$$

$$x + \frac{13}{6} = \pm\sqrt{\frac{253}{36}}$$

$$x + \frac{13}{6} = \pm\frac{\sqrt{253}}{6}$$

$$x = -\frac{13}{6} \pm \frac{\sqrt{253}}{6}$$

$$x = \frac{-13 \pm \sqrt{253}}{6}$$

$$x = \frac{-13+\sqrt{253}}{6} \text{ or } x = \frac{-13-\sqrt{253}}{6}$$

$$x = 0.484328953 \quad \text{or} \quad x = -4.817662287$$

$$\therefore x = 0.48 \quad \text{or} \quad x = -4.82$$

2024.P2.GCE

Solve the equation $7 - 2x^2 = x$, giving your answers correct to 2 decimal places. [5]

Solution

$$7 - 2x^2 = x$$

$$-2x^2 - x + 7 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(-2)(7)}}{2(-2)}$$

$$x = \frac{1 \pm \sqrt{1 + 56}}{-4}$$

$$x = \frac{1 \pm \sqrt{57}}{-4}$$

$$x = \frac{1 + \sqrt{57}}{-4} \text{ or } x = \frac{1 - \sqrt{57}}{-4}$$

$$x = -2.137\,458\,608\,817\,69 \quad \text{or} \quad x = 1.637\,458\,608\,817\,69$$

$$x = -2.14 \quad \text{or} \quad x = 1.64$$

VARIATION

Objectives

By the end of this topic, learners should

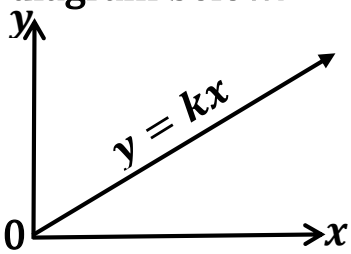
1. Define variation.
2. Describe various types of variation.
3. Distinguish between direct and inverse variation.
4. Distinguish between joint and partial variation.
5. Draw and interpret graphs of variation.
6. Apply the concepts for variation to solve real-life problems.

Definition

- In everyday language, a variation is a change to something.
- In mathematics, we use the term variation to describe specific relationships between two or more variables.

Direct variation

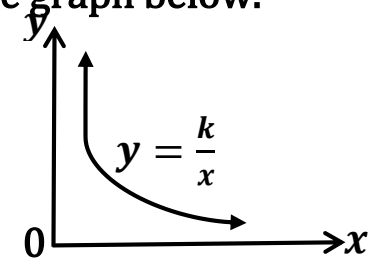
- In a direct variation, as one variable increases, the other variable increases as well.
- Look at the diagram below.



- As the value of x increases, the value of y increases as well.
- We say that y varies directly with x and we write this as $y \propto x$, where the symbol $\propto$ means "varies directly".
- This can be written as an equation as $y = kx$, where $k > 0$ and k is called the constant of variation.

Inverse variation

- In an inverse variation, as one variable increases, the other variable decreases.
- Look at the graph below.



- As the value of x increases, the value of y decreases.
- We say that y varies inversely with x and we write this as $y \propto \frac{1}{x}$, where the symbol $\propto$ means "varies inversely".
- This can be written as an equation as $y = k\left(\frac{1}{x}\right) = \frac{k}{x}$, where $k > 0$ and k is called the constant of variation.

Joint variation

- A joint variation is a variation where a quantity varies directly with a product of two or more other quantities.
- For example, if y varies directly with x and also with z, we say that y varies jointly with x and z.
- This is written as $y \propto xz$, and in equation form it becomes $y = kxz$, where $k > 0$ and k is the constant of variation.

Partial variation

- A partial variation is a variation between two variables, such that its graph is a straight line that does not pass through the origin.
- If y varies partially with x, then the relationship between y and x can be expressed as $y = mx + b$, where b is the fixed part of the partial variation and m is the variable part of the partial variation.

Graphs of Variations

- The equation of a direct variation is $y = kx$ and its graph is a straight line that passes through the origin, and the value of k is equal to the gradient of the graph, m.
- The graph of an inverse variation is a curve that does not touch either of the axes.
- The equation of a partially direct variation is $y = kx + b$ and its graph is a straight line that does not pass through the origin, and the value of k is equal to the gradient of the graph, m and the value of b is the y-intercept of the graph.

2018.P1.Oct

y varies directly as the square of x and $y = 96$ when $x = 4$. Find the

(a) value of the constant k,

(b) value of y when $x = 5$,

(c) values of x when $y = 24$.

Solution

(c) $y \propto x^2$

$y = kx^2$

$96 = k(4)^2$

$96 = 16k$

$k = \frac{96}{16}$

$k = 6$

(a) $y = 6x^2$

$y = 6(5)^2$

$y = 6 \times 25$

$y = 150$

(b) $y = 6x^2$

$24 = 6x^2$

$x^2 = \frac{24}{6}$

$x^2 = 4$

$x = \pm\sqrt{4}$

$x = \pm 2$

$x = 2 \text{ or } x = -2$

2019.P1.Nov

a varies directly as b and as the square of c and $a = 30$ when $b = 2.5$ and $c = 2$. Find the

(a) value of k, the constant of variation,

(b) value of a when $b = 2$ and $c = 3$,

(c) values of c when $a = 300$ and $b = 4$.

Solution

(a) $a \propto bc^2$

$a = kbc^2$

$30 = k(2.5)(2)^2$

$30 = 10k$

$k = \frac{30}{10}$

$k = 3$

(b) $a = 3bc^2$

$a = 3(2)(3)^2$

$a = 6 \times 9$

$a = 54$

(c) $a = 3bc^2$

$300 = 3(4)c^2$

$c^2 = \frac{300}{12}$

$c^2 = 25$

$c = \pm\sqrt{25}$

$c = \pm 5$

$c = 5 \; or \; x = -5$

2020.P1.Nov

It is given that y varies directly as the cube root of x and inversely as the square of z and $y = 2$ when $x = 8$ and $z = 4$. Find

(a) k, the constant of variation,

(b) the equation connecting x, y and z,

(c) the values of z when $y = 12$ and $x = 27$.

Solution

(a) $y \propto \frac{\sqrt[3]{x}}{z^2}$

$y = \frac{k\sqrt[3]{x}}{z^2}$

$2 = \frac{k\left(\sqrt[3]{8}\right)}{4^2}$

$2k = 2 \times 16$

$k = \frac{32}{2}$

$k = 16$

(b) $y = \frac{16\sqrt[3]{x}}{z^2}$

(c) $y = \frac{16\sqrt[3]{x}}{z^2}$

$12 = \frac{16 \times \sqrt[3]{27}}{z^2}$

$12z^2 = 16 \times 3$

$12z^2 = 48$

$z^2 = \frac{48}{12}$

$z^2 = 4$

$z = \pm\sqrt{4}$

$z = \pm 2$

$z = 2 \; or \; x = -2$

2021.P1.Nov

It is given that y varies directly as $2x$ and inversely as z^2 and $y = 4$ when $x = 8$ and $z = 2$. Find the

(a) value of k, the constant of variation,

(b) value of y when $x = 27$ and $z = 3$,

(c) values of z when $x = 24$ and $y = 3$.

Solution

(a) $y \propto \frac{2x}{z^2}$

$y = \frac{2kx}{z^2}$

$4 = \frac{2k(8)}{2^2}$

$16k = 4 \times 4$

$k = \frac{16}{16}$

$k = 1$

(b) $y = \frac{2x}{z^2}$

$y = \frac{2 \times 27}{3^2}$

$y = \frac{54}{9}$

$y = 6$

(c) $y = \frac{2x}{z^2}$

$3 = \frac{2 \times 24}{z^2}$

$3z^2 = 48$

$z^2 = \frac{48}{3}$

$z^2 = 16$

$z = \pm\sqrt{16}$

$z = \pm 4$

$z = 4 \text{ or } x = -4$

2022.P1.Nov

y varies directly as the square of x and inversely as z and $y = 2$ when $x = 4$ and $z = 24$. Find the

(a) value of k, the constant of variation,

(b) value of y when $x = 9$ and $z = 27$,

(c) values of x when $y = 8$ and $z = 6$.

Solution

(a) $y \propto \frac{x^2}{z}$

$y = \frac{kx^2}{z}$

$2 = \frac{k(4)^2}{24}$

$16k = 48$

$k = \frac{48}{16}$

$k = 3$

(b) $y = \frac{3x^2}{z}$

$y = \frac{3(9)^2}{27}$

$y = \frac{3\times3\times3\times3\times3}{3\times3\times3}$

$y = 9$

(c) $y = \frac{3x^2}{z}$

$8 = \frac{3x^2}{6}$

$x^2 = 16$

$x = \pm\sqrt{16}$

$x = \pm4$

$x = 4 \text{ or } x = -4$

2023.P1.Nov

If a varies jointly as b and the square root of c and $a = 21$ when $b = 5$ and $c = 36$, find the value of

(a) k, the constant of variation,

(b) a when $b = 9$ and $c = 100$,

(c) c when $a = 70$ and $b = 25$.

Solution

(a) $a \propto b\sqrt{c}$

$a = kb\sqrt{c}$

$21 = k(5)(\sqrt{36})$

$21 = 30k$

$k = \frac{21}{30}$

$k = \frac{7}{10}$

(b) $a = \frac{7b\sqrt{c}}{10}$

$a = \frac{7\times9\times\sqrt{100}}{10}$

$a = \frac{360}{10}$

$a = 63$

(c) $a = \frac{7b\sqrt{c}}{10}$

$70 = \frac{7\times25\sqrt{c}}{10}$

$175\sqrt{c} = 700$

$\sqrt{c} = \frac{700}{175}$

$\sqrt{c} = 4$

$(\sqrt{c})^2 = 4^2$

$c = 16$

2024.P1.GCE

Given that y varies directly as x and inversely as the square of z and that $x = 8$ when $y = 4$ and $z = 2$, find the

(a) constant of variation, k,

(b) value of y when $x = 18$ and $z = 3$,

(c) values of z when $x = 25$ and $y = 2$.

Solution

(a) $y \alpha \frac{x}{z^2}$

$y = \frac{kx}{z^2}$

$4 = \frac{8k}{2^2}$

$8k = 16$

$k = 2$

(b) $y = \frac{2x}{z^2}$

$= \frac{2 \times 18}{3^2}$

$= 4$

(c) $2 = \frac{2 \times 25}{z^2}$

$2z^2 = 50$

$z^2 = \frac{50}{2}$

$z^2 = 25$

$z = \pm\sqrt{25}$

$z = \pm 5$

$z = 5 \; or \; z = -5$

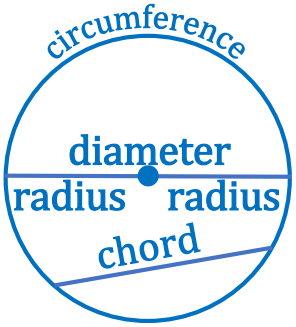
CIRCLE THEOREMS

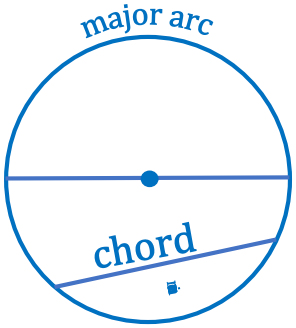
Objectives

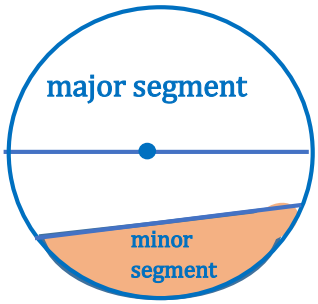
By the end of this topic, learners should

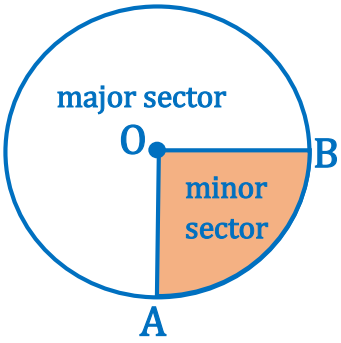
1 Understand the concept of a circle and its components.
2 Apply the circle theorems, including:
   (a) Angle in a semicircle theorem.
   (b) Angle at the center theorem.
   (c) Angle in a segment theorem.
   (d) Cyclic quadrilateral theorem.
3 Recognize and solve problems involving circle theorems.
4 Develop problem-solving skills and logical reasoning.

Properties and Parts of a circle









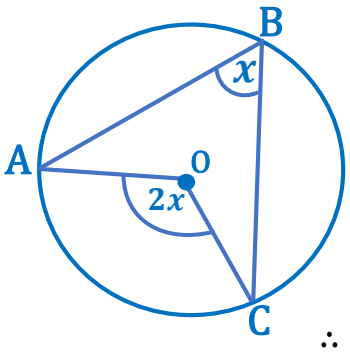
- A chord is a straight line drawn across a circle that joins any two points on its circumference.
- A chord that passes through the centre of a circle is called a diameter.
- A line drawn from the centre of a circle to its circumference is called a radius.

- A diameter is twice a radius and a radius is a half of a diameter, however, a diameter consists of two radii.
- A chord which is not a diameter divides the circumference into a major arc and a minor arc.
- A chord divides a circle into a major segment and a minor segment.
- A diameter divides a circle into two equal halves which are called semi-circles.
- A sector of a circle is an area bounded by two radii and an arc, while the two radii divide the circle into a major sector and a minor sector.
- A diameter (or radius) drawn perpendicular to a chord bisects the chord.
- The perpendicular bisector of a chord passes through the centre of a circle.
- Equal chords are equidistant from the centre of the circle.
- A line that cuts a circle in two points is called a secant.
- A line that touches a circle at one point only is called a tangent.

Angle Properties of a Circle

Theorem 1:

The angle which an arc subtends at the centre of a circle is twice that which it subtends at any point on the circumference.

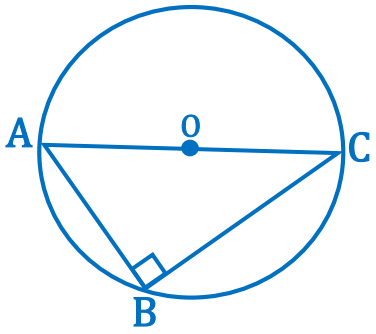


$\therefore A\hat{O}C = 2 \times A\hat{B}C$ and $A\hat{B}C = \frac{1}{2} \times A\hat{B}C$

$= \frac{A\hat{B}C}{2}$

Theorem 2:

An angle in a semi-circle is a right angle.



$$\therefore A\hat{O}C = 2 \times A\hat{B}C \text{ and } A\hat{B}C = \frac{1}{2} \times A\hat{B}C$$

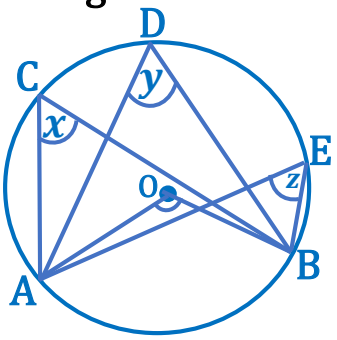
$$= \frac{A\hat{B}C}{2}$$

$$A\hat{O}C = 2 \times 90° \text{ and } A\hat{B}C = \frac{1}{2} \times 180°$$

$$A\hat{O}C = 180° \text{ and } A\hat{B}C = 90°$$

Theorem 3:

Angles in the same segment of a circle are equal.

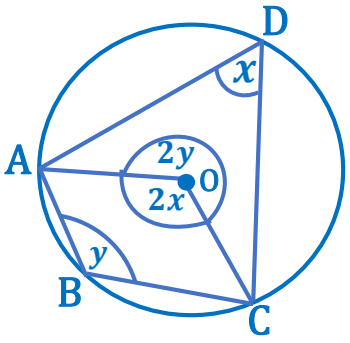


$$\therefore x = y = z = \frac{1}{2} \times A\hat{O}B$$

$$A\hat{C}B = A\hat{D}B = A\hat{E}B$$

Theorem 4:

The opposite angles of a cyclic quadrilateral are supplementary.

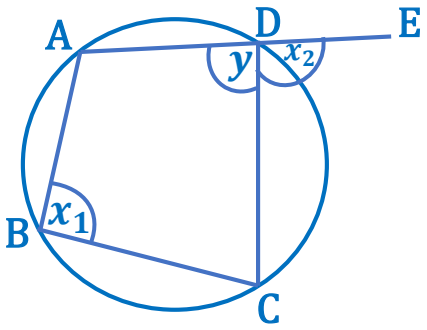


$\therefore 2x + 2y = 360°$ (divide throughout by 2)
$x + y = 180°$

$A\hat{D}C + A\hat{B}C = 180°$
$B\hat{A}D + B\hat{C}B = 180°$

Theorem 5:

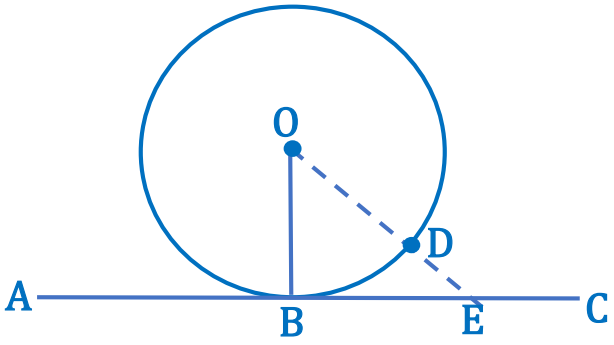
The exterior angle of a cyclic quadrilateral is equal to the interior opposite angle.



$\therefore x_1 + y = 180°$
$x_1 = 180° - y$
$x_2 + y = 180°$
$x_2 = 180° - y$
$x_1 = x_2$
$A\hat{B}C = E\hat{D}C$
$B\hat{A}D + B\hat{C}B = 180°$

Theorem 6:

A tangent to a circle is perpendicular to the radius at the point of contact.

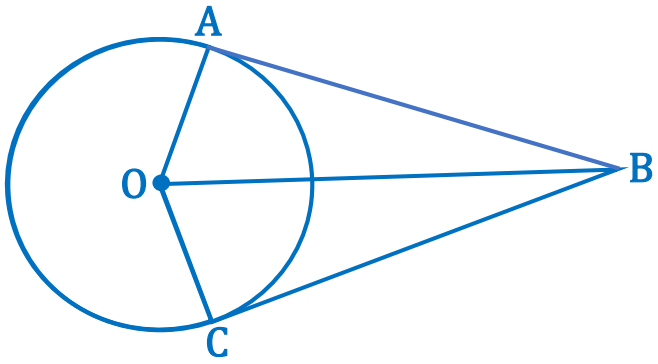


$\therefore \mathbf{OB} < \mathbf{OE}$

$\mathbf{O\hat{B}C} = \mathbf{90°}$

Theorem 7:

The tangents to a circle from an external point are equal.



$\mathbf{O\hat{A}B} = \mathbf{O\hat{C}B} = \mathbf{90°}$ (A tangent to a circle is perpendicular to the radius at the point of contact)

OA = OC (two radii)

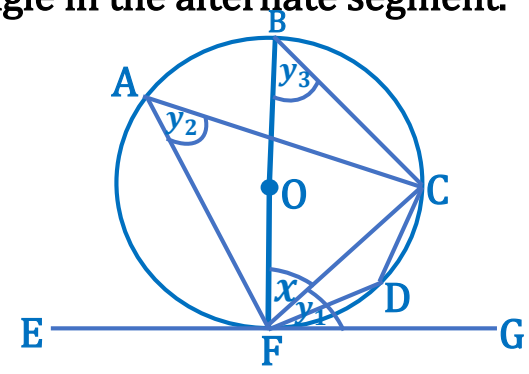
OB =OB (the common side also known as the hypotenuse side of the two triangles formed)

$\therefore \Delta\mathbf{AOB} \equiv \Delta\mathbf{OBC}$

$\therefore \mathbf{AB} = \mathbf{BC}$

Theorem 8:

If a chord to a circle is drawn from the point of contact at which the tangent touches the circle, then the angle which the tangent makes with the chord is equal to any angle in the alternate segment.



$\therefore x + y_1 = 90°$ (A tangent to a circle is perpendicular to the radius at the point of contact)
$y_1 = 90° - x$

$B\hat{C}F = 90°$ (an angle in a semi-circle is a right angle)

$x + y_3 = 180° - 90°$ (sum of interior angles of a triangle)
$x + y_3 = 90°$
$y_3 = 90° - x$

$\therefore y_1 = y_3$

$y_2 = y_3$ (angles in the same segment)

$\therefore C\hat{F}G = C\hat{B}F = C\hat{A}F$

REMEDIAL TEST

Complete each of the following statements by writing down the missing word.

1 A chord that passes through the centre of a circle is called a _____. [1]

2 A line drawn from the centre of a circle to its circumference is called a _____. [1]

3 A diameter divides a circle into two equal halves. These are called _____ [1]

4 A diameter (or radius) drawn to a chord _____ the chord. [1]

5 The angle which an arc subtends at the centre of a circle is twice that which it subtends at any point on the _____ [1]

6 An angle in a semi-circle is a _____ angle. [1]

7 Angles in the same segment of a circle are _____. [1]

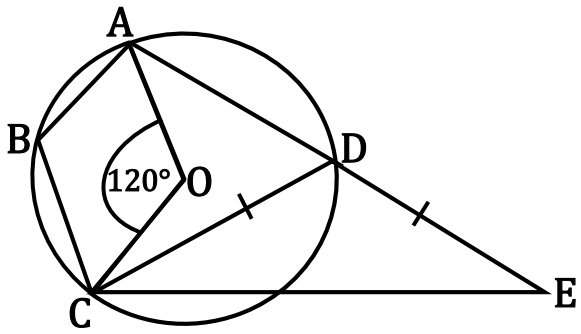
8 A cyclic quadrilateral is a quadrilateral of which all four _____ lie on the circumference of the same circle. [1]

9 The exterior angle of a cyclic quadrilateral is equal to the _____ angle. [1]

10 A line that touches a circle at one point only is called a _____. [1]

2018.P1.Oct

In the diagram below, O is the centre of the circle. A, B, C and D are points on the circumference. AD produced meets the line from C at E such that CD = DE and angle AOC = 120°.



Calculate angle

(a) ADC,

(b) ABC,

(c) DEC.

Solution

(a) $A\hat{D}C = \frac{120}{2}$ (The angle which an arc subtends at the centre of a circle is twice that which it subtends at any point on the circumference.)

$= 60°$

(b) $A\hat{B}C = 180 - 60$ (The opposite angles of a cyclic quadrilateral are supplementary)

$= 120°$

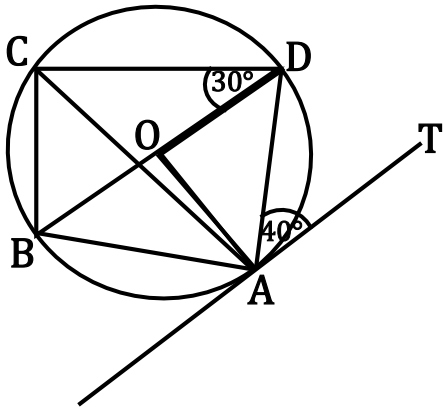
(c) $D\hat{E}C = \frac{180-120}{2}$ (Angles in a straight line & Sum of interior angles of a triangle)

$= \frac{60}{2}$

$= 30°$

2019.P1.Nov

In the diagram below, A, B, C and D are points on the circumference of a circle, centre O. $\hat{DAT} = 40°$, $\hat{BDC} = 30°$ and AT is a tangent to the circle at A.



Calculate

(a) $\hat{CBD}$,

(b) $\hat{BAC}$,

(c) $\hat{AOB}$.

Solution

(a) $\hat{CBD} = 180 - 90 - 30$

$= 60°$

(b) $\hat{BAC} = \hat{BDC}$

$= 30°$

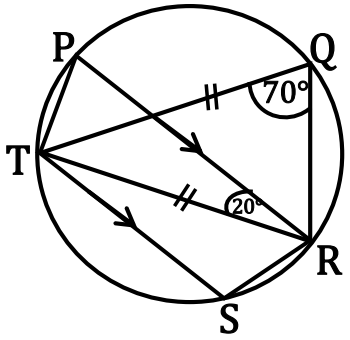
(c) $\hat{AOB} = 180 - (180 - 100)$

$= 180 - 80$

$= 100°$

2020.P1.Nov

In the diagram below, P, Q, R, S and T lie on a circle and TQ = TR. PR and TS are parallel. Angle TQR = 70° and angle TRP = 20°.



Find

(a) angle RTQ,

(b) angle RST,

(c) angle TRS.

Solution

(a) $\hat{RTQ} = 180 - 70 - 70$

$= 40°$

(b) $\hat{RST} = 180 - 70$

$= 110°$

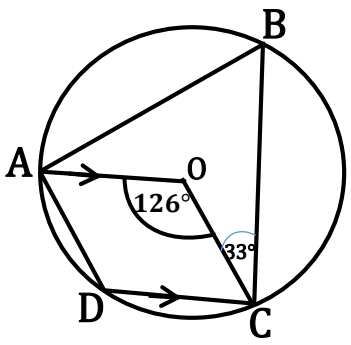
(c) $\hat{TRS} = 180 - (110 + 20)$

$= 180 - 130$

$= 50°$

2021.P1.Nov

The points A, B, C and D are on the circumference of a circle with centre O, such that angle AOC = 126°. AO is parallel to DC and angle OCB = 33°.



Find

(a) $A\hat{D}C$,

(b) $O\hat{C}D$,

(c) $B\hat{A}O$.

Solution

(a) $A\hat{D}C = 180 - (126 \div 2)$

$= 180 - 63$

$= 117°$

(b) $O\hat{C}D = \frac{180-126}{2} \times 2$

$= \frac{54}{2} \times 2$

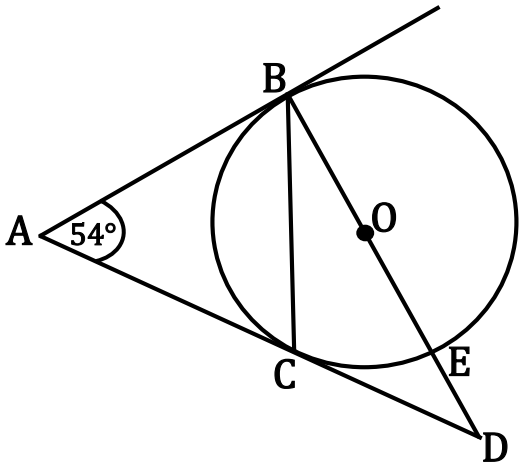
$= 54°$

(c) $B\hat{A}O = O\hat{C}B$

$= 33°$

2022.P1.Nov

In the following diagram, AB and AC are tangents to the circle, centre O. AC and BC produced, meet at D and angle BAC = 54°.



Calculate angle

(a) ACB,

(b) CBD,

(c) CDB.

Solution

(a) $A\hat{C}B = \frac{180-54}{2}$

$= \frac{126}{2}$

$= 63°$

(b) $C\hat{B}D = 90 - 63$

$= 27°$

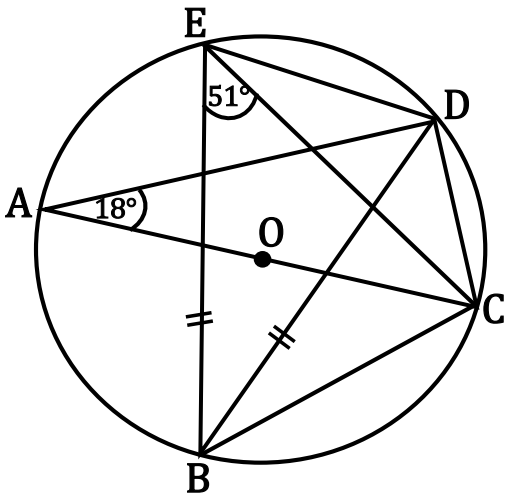
(c) $C\hat{D}B = 180 - (90 + 54)$

$= 180 - 144$

$= 36°$

2023.P1.Nov

In the following diagram, A, B, C, D, and E lie on the circumference of a circle centre O. BD = BE, angle CAD = 18° and angle BEC = 51°.



Find

(a) angle ADC,

(b) angle BED,

(c) angle EBC.

Solution

(a) $A\hat{D}C = 90°$ (An angle in semi-circle is a right angle)

(b) $B\hat{E}D = 51 + 18$ (Angles in the same segment of circle are equal)

$= 69°$

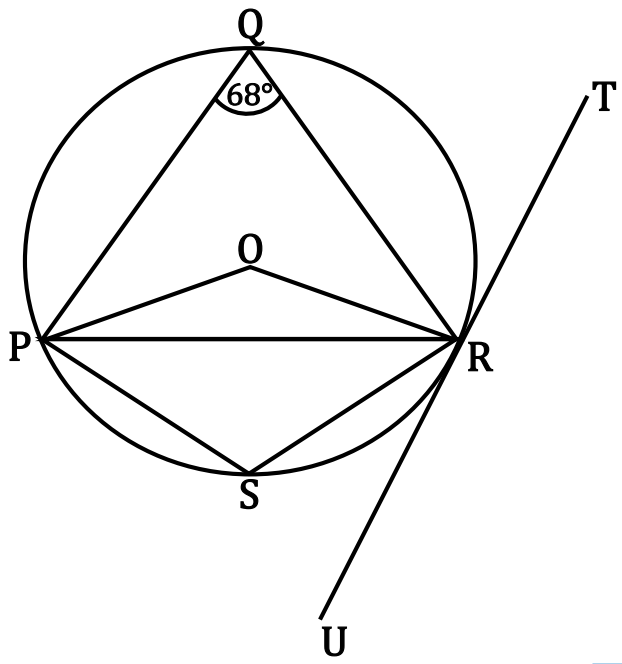
(c) $E\hat{B}C = 180 - (69 + 69) + 18$ (An isosceles triangle)

$= 180 - 138 + 18$

$= 60°$

2024.P1.GCE

In the diagram below, P, Q, R and S are points on the circumference of a circle with centre O and angle PQR = 68°. TRU is a tangent at R.



Find

(a) $\hat{POR}$,

(b) $\hat{PSR}$,

(c) $\hat{PRO}$.

Solution

(a) $\hat{POR} = 68 \times 2$

$= 136°$

(b) $\hat{PSR} = 180 - 68$

$= 112°$

(c) $\hat{PRO} = \frac{180-136}{2}$

$= \frac{44}{2}$

$= 22°$

CONSTRUCTION AND LOCI

Objectives

By the end of this topic, learners should

1 Define construction.
2 Construct line and angle bisectors.
3 Explain the meaning of a locus.
4 Describe the locus of a point in two dimensions.
5 Construct the locus of a point in two dimensions.
6 Describe the locus of a point in three dimensions.
7 Use a compass and straightedge to construct geometric figures.
8 Construct perpendicular bisectors, angle bisectors, and parallel lines.
9 Apply construction techniques to solve problems.

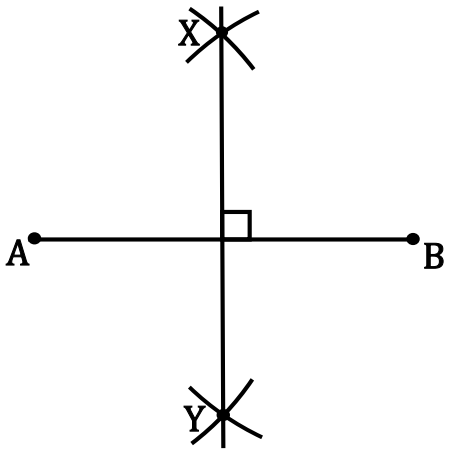
Definition

- A geometrical construction is an accurate drawing, using geometrical instruments.
- In real life, the word "construction" refers to something that is built, for example, a building, a road or a bridge.

Bisecting a line

- To bisect is to divide into two equal halves.
- Bisecting a line means dividing the line into two equal parts.
- The point where the line is divided is called the midpoint.
- Bisecting a line creates two line segments of equal length.
- A line that bisects another line at an angle of 90° is called a perpendicular bisector.
- To bisect a line:
  (i) Draw any line AB.
  (ii) Set the pairs of compasses to any length greater than (more than) half of AB.

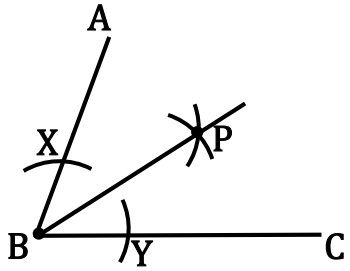
(iii) With centre A, construct arcs above and below line AB.
(iv) With centre B, construct arcs to cut the arcs in the previous step at points X and Y respectively.
(v) The line through X and Y is the perpendicular bisector of AB.
(vi) The results are shown on the diagram below:



Bisecting an angle

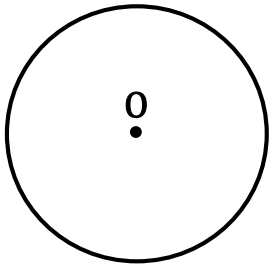
- Bisecting an angle means dividing the angle into two equal parts.
- The line that divides the angle is called the angle bisector.
- Bisecting an angle creates two angles of equal measure.
- To bisect an angle:

(i) Draw any $A\hat{B}C$.
(ii) With ecntre B and any radius, construct an arc to cut AB at X and BC at Y.
(iii) With cntres X and Y, and using a large enough radius, construct arcs to intersect at P.
(iv) The straight line through BP bisects $A\hat{B}C$.
(v) The results are shown in the diagram below:

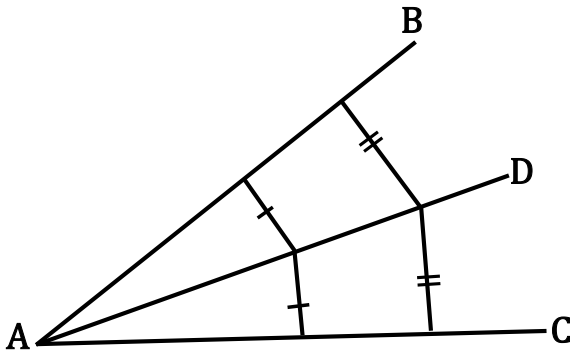


The locus of a point

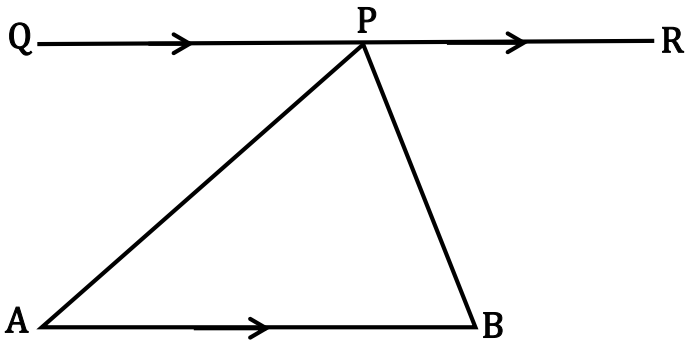
- The locus of a point is the set of all points that satisfy certain given conditions or properties.
- It is the path along which the point may move in order to satisfy one or more conditions.
- In other words, it is the path or shape that a point traces out as it moves according to certain rules or constraints.
- The plural of locus is Loci.
- The locus of a point which moves in such a way that it is always equidistant from a fixed point its locus is the circumference of the circle with centre O as shown in the diagram below:



- Equidistant (from) means equally far from, the same distance from.
- The locus of a point which moves in such a way that it is equidistant from two intersecting straight lines, AB and AC its locus is the bisector AD of $B\hat{A}C$ as shown in the diagram below:



- The locus of a point P which moves in such a way that the area of ΔAPB is constant its locus is an infinitely long straight line through the points Q, P and R, parallel to AB as shown in the diagram below.



2018.P2.GCE

(a) (i) Construct triangle PQR in which PQ = 10cm, QR = 8cm and $\mathbf{P\hat{Q}R = 50°}$.

(ii) Measure and write the length of PR.

(b) On your diagram, within triangle PQR, construct the locus of points which are

(i) equidistant from P and Q,

(ii) equidistant from PR and PQ,

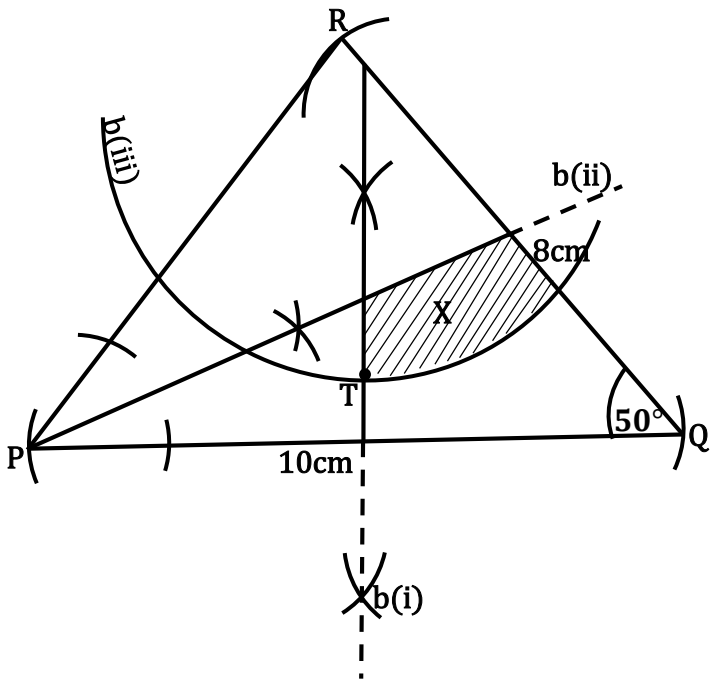
(iii) 5cm from R.

(c) A point T within triangle PQR is such that it is 5cm from R and equidistant from P and Q. Label the point T.

(d) Another point X is such that it is less than or equal to 5cm from R, nearer to Q than P and nearer to PQ than PR. Indicate clearly, by shading, the region in which X must lie.

Solution

(a)(ii) PQ = 7.7cm



2019.P2.Nov

(a) (i) Construct triangle ABC in which AB = 8cm, $\widehat{BAC} = 110°$ and $\widehat{ABC} = 35°$.

(ii) Measure and write the length of BC.

(b) Within the triangle ABC, construct the locus of points which are

(i) 3cm from AB,

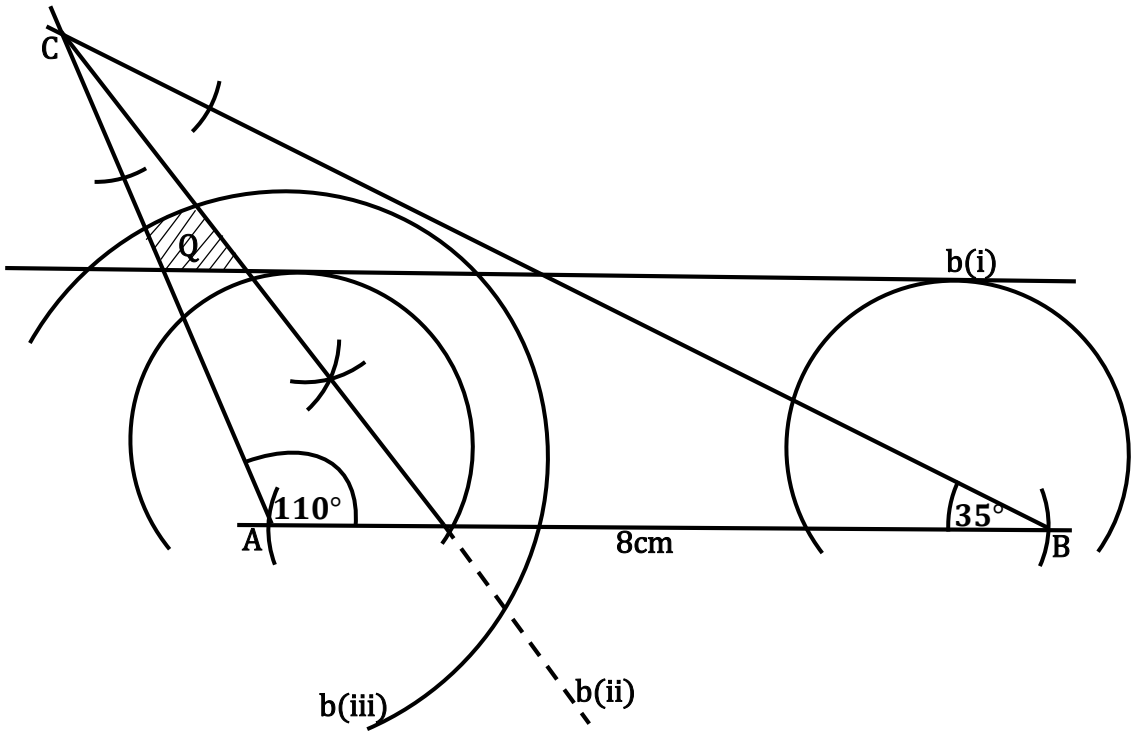
(ii) equidistant from AC and BC,

(iii) 4cm from A.

(c) A point Q inside triangle ABC is greater than or equal to 3cm from AB, less than or equal to 4cm from A and nearer to AC than BC. Indicate clearly, by shading, the region in which Q must lie.

Solution

(a) (ii) BC = 13cm



2020.P2.Nov

(a) Construct a quadrilateral PQRS in which PQ = 8cm, angle QPS = angle PQR = 120°, PS = 6cm and QR = 5cm.

(b) Measure and write the length of RS.

(c) Within the quadrilateral PQRS, draw the locus of points which are

(i) 4.5cm from Q,

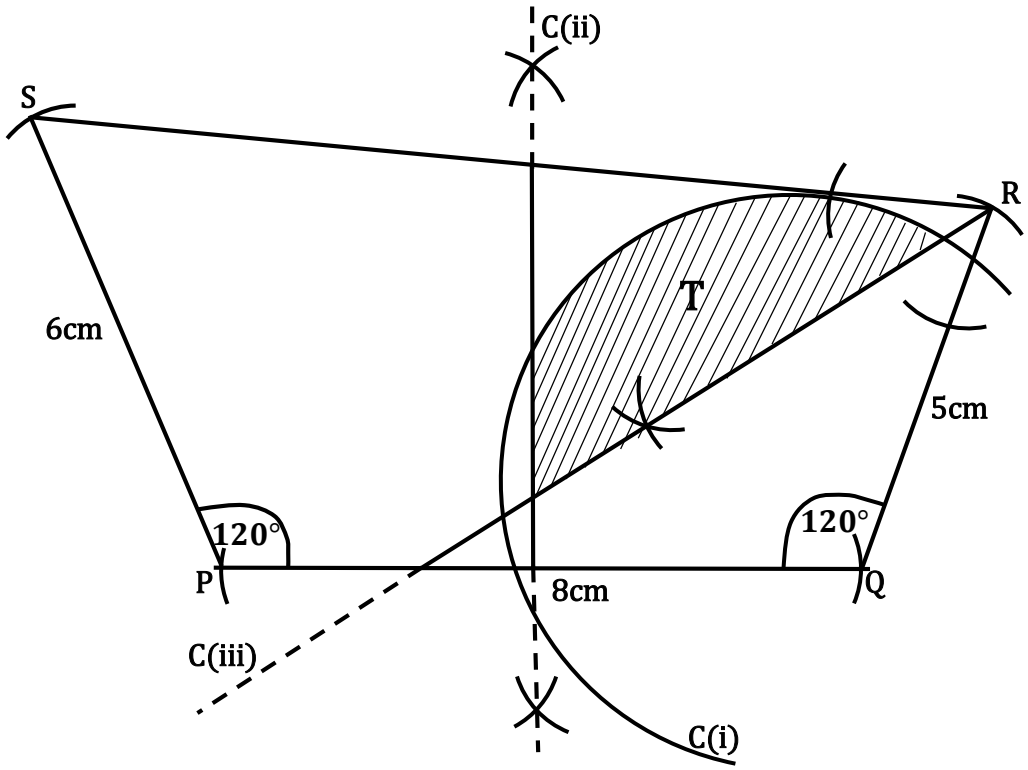
(ii) equidistant from P and Q,

(iii) equidistant from QR and RS.

(d) A point T, within the quadrilateral PQRS, is such that it is less than or equal to 4.5cm from Q, nearer to Q than P and nearer to RS than QR. Indicate, by shading, the region in which T must lie.

Solution

(b) RS = 13.4cm



2021.P2.Nov

(a) (i) Construct triangle ABC in which AB = 6cm, BC = 9cm and AC = 5cm.

(ii) Measure and write the size of angle ABC.

(b) Within the triangle ABC, draw the locus of points which are

(i) equidistant from A and C,

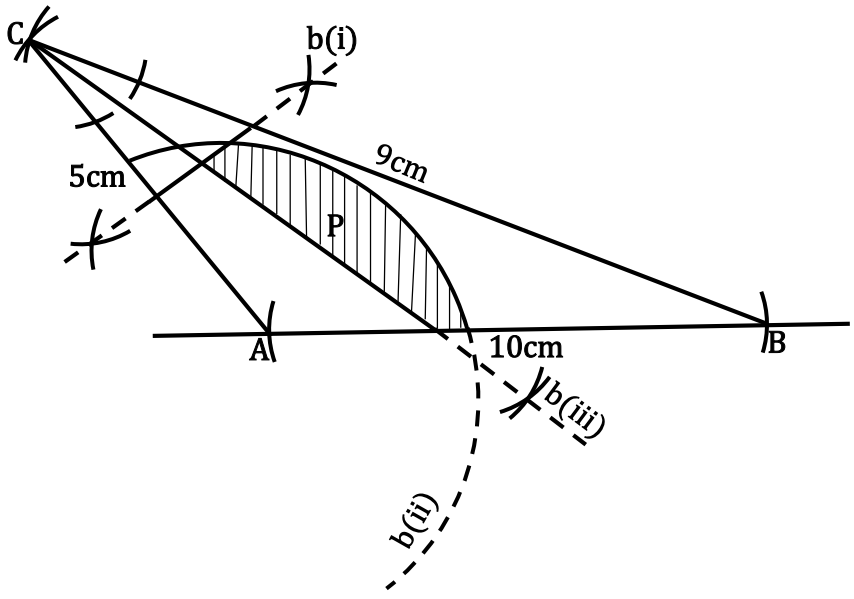
(ii) 3cm from A,

(iii) Equidistant from AC and BC.

(c) A point P, within the triangle ABC, is such that it is less than or equal to 3cm from A, nearer to BC than AC and closer to A than to C. Indicate clearly by shading, the region in which P must lie.

Solution

(a) (ii) $A\hat{B}C = 32°$



2022.P2.Nov

(a) Construct a parallelogram ABCD in which AB = 10cm, AD = BC = 6.5cm, angle BAD = 110 ° and angle ABC = 70°.

(b) Measure and write the length of AC.

(c) Within parallelogram ABCD, construct the locus of points which are

(i) 3cm from AB,

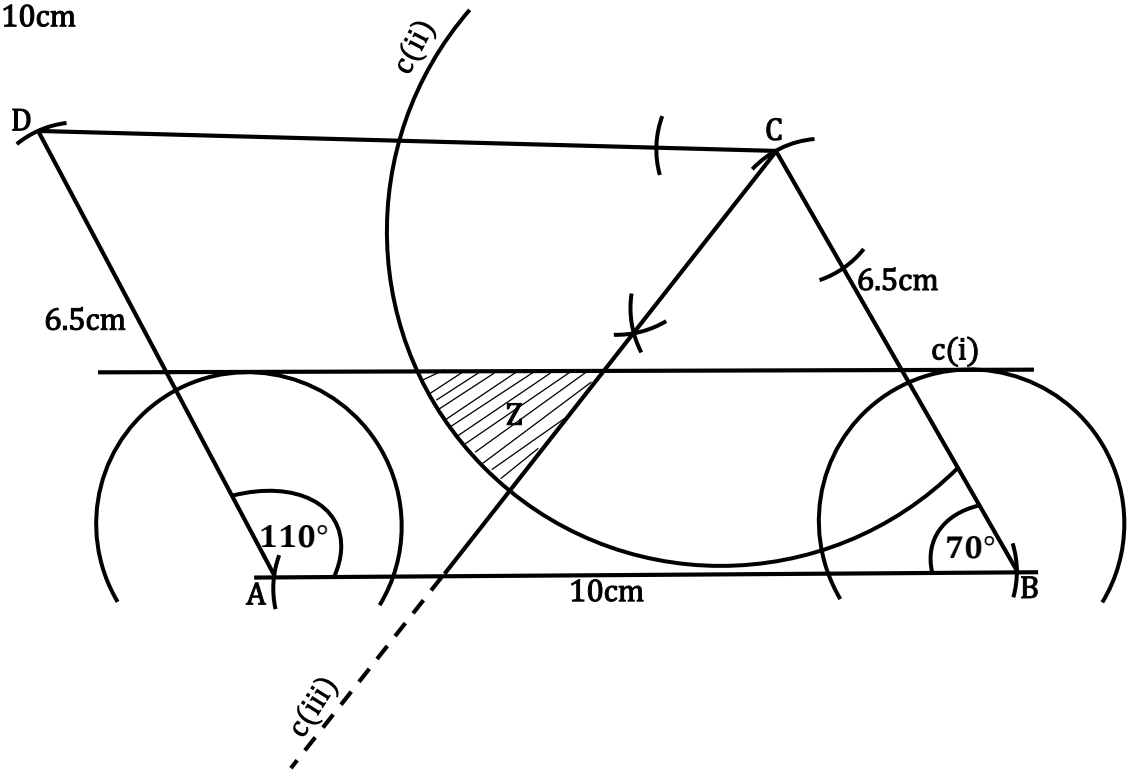
(ii) 6cm from C,

(iii) equidistant from BC and CD.

(d) A point Z, within the parallelogram ABCD, is such that it is less than or equal to 3cm from AB, less than or equal to 6cm from C and nearer to CD than BC. Indicate clearly, by shading, the region in which Z must lie.

Solution

(b) AC =10cm



2023.P2.Nov

(a) Construct a trapezium ABCD in which AB = 10cm, BC = 8cm, AD = 7cm, angle ABC =60° and angle BAD is 90°. [1]

(b) Measure and write the length of CD. [1]

(c) Within the trapezium ABCD, draw the locus of points which are

(i) 2cm from CD, [1]

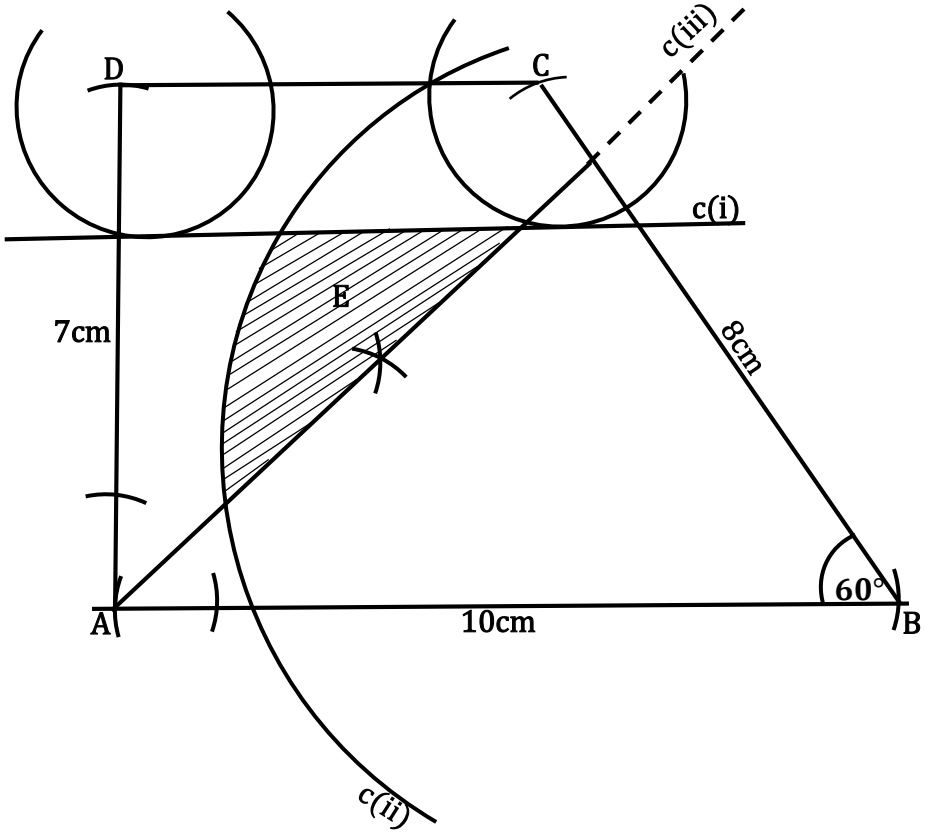
(ii) 9cm from B, [1]

(iii) equidistant from AB and AD. [2]

(d) A point E, within the trapezium ABCD, is such that it is greater than or equal to 2cm from CD, less than or equal to 9cm from B and nearer to AD than AB. Indicate, by shading, the region in which E must lie. [2]

Solution

(b) CD = 6cm



2024.P2.GCE

(a) Construct a quadrilateral PQRS in which PQ = 10cm, angle SPQ = 70°, QR = 6.5cm and PS = 5cm and SR = 7cm.

(b) Measure and write the angle PQR.

(c) On your diagram, construct the locus of points within the quadrilateral PQRS which are

(i) 3cm from PQ,

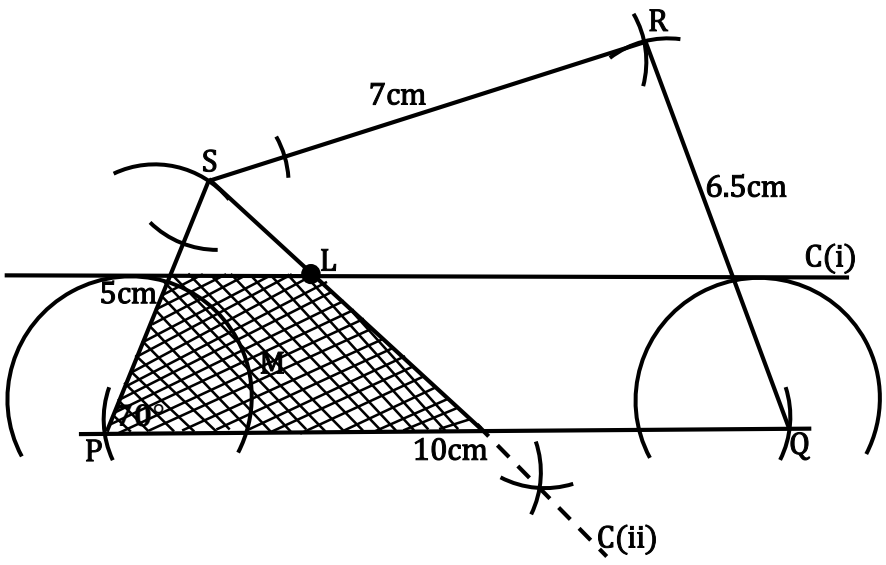
(ii) equidistant from PS and SR.

(d) L is a point inside quadrilateral PQRS such that it is 3cm from PQ and equidistant from PS and SR. Label the point L.

(e) M is another point inside quadrilateral PQRS such that it is less than or equal to 3cm from PQ and nearer to PS than SR. Indicate, by shading, the region in which M must lie.

Solution

(b) $P\hat{Q}R = 77°$



Objectives

By the end of this topic, learners should

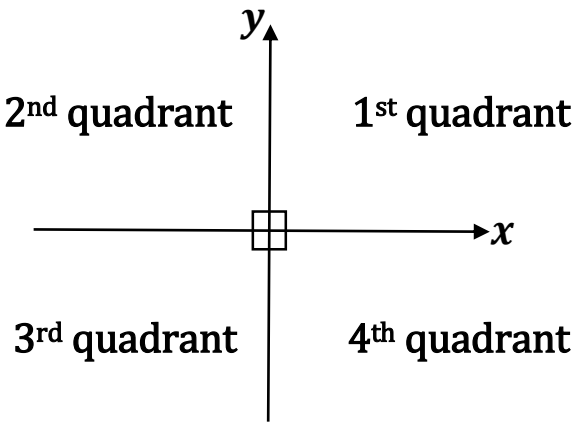
1 Define trigonometry.
2 Relate right-angled triangles to the three trigonometric ratios.
3 Describe the three trigonometric ratios on a right-angled triangle.
4 Calculate sides and angles of a right-angled triangle.
5 Work with special angles (30°, 45°, 60°).
6 Find sides and angles of non right-angled triangles.
7 Calculate the area of non right-angled triangles.
8 Determine the signs of three trigonometric ratios in the quadrants.
9 Draw graphs for sine, cosine and tangent curves.
10 Solve trigonometric equations.
11 Use trigonometry to solve practical problems.

Definition

- ✓ Trigonometry means the measurement of angles.
- ✓ From the word trigonometry, tri means 3, gon means angle and metric means measure.
- ✓ Therefore, trigonometry is an area of mathematics relating the study of trigonometric functions to the measurement of angles, and the behavior of wave functions.

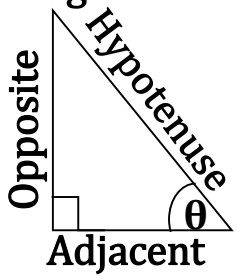
The General Angle

- ✓ Angles measured from the x-axis in an anti-clockwise direction are positive, and those measured in a clockwise direction are negative.
- ✓ The general angle has 4 quadrants.



Trigonometric Ratios and Trigonometric Functions

- ➢ Generally, there are three trigonometric ratios (functions) namely:
  - (a) Cosine
  - (b) Sine
  - (c) Tangent
- ➢ The following are inverses with respect to the trigonometric ratios
  - (a) Secant
  - (b) Cosecant
  - (c) Cotangent
- ➢ Consider a right angled triangle below:



- From the triangle above, the three trigonometric ratios can be defined as:

$$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}} \text{ and } \sec\theta = \frac{1}{\cos\theta} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$$

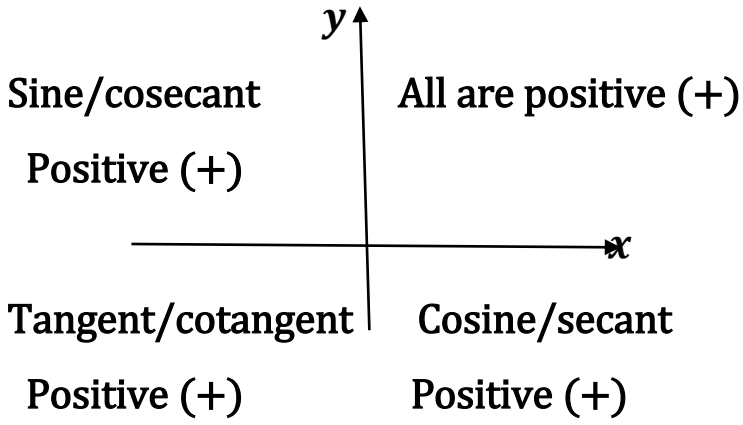
$$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}} \text{ and } \operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{\text{Hypotenuse}}{\text{Opposite}}$$

$$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}} \text{ and } \cot\theta = \frac{1}{\tan\theta} = \frac{\text{Adjacent}}{\text{Opposite}}$$

- Note that $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\cot\theta = \frac{\cos\theta}{\sin\theta}$.
- The above results can be summarized in one sentence for easier memorizing as *SOH-CAH-TOA.*

The Sign Conversion of Trigonometric Ratios

- Consider the diagram below:



- The results in the diagram above can easily be memorized as All Science Teachers are Crazy.
- Angles in the first quadrant are angles in the range $0° \leq \theta \leq 90°$ and $\theta = \alpha$

- Angles in the second quadrant are angles in the range $\mathbf{90° < \theta \leq 180°}$ and $\boldsymbol{\theta = 180° - \alpha}$.
- Angles in the third quadrant are angles in the range $\mathbf{180° < \theta < 270°}$ and $\boldsymbol{\theta = 180° + \alpha}$.
- Angles in the fourth quadrant are angles in the range $\mathbf{270° \leq \theta \leq 360°}$ and $\boldsymbol{\theta = 360° - \alpha}$.
- Therefore, the ratios of $\mathbf{150°, 210°}$ $\boldsymbol{and}$ $\mathbf{330°}$ are numerically the same as the ratios of $\mathbf{30°}$. So we can say that:

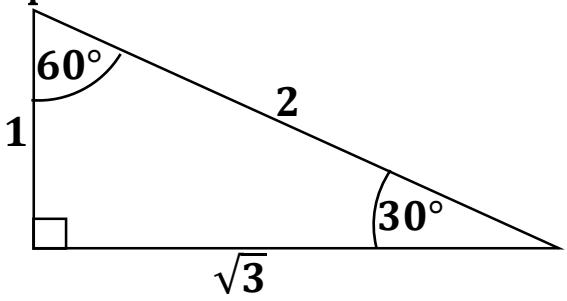
$\mathbf{\sin 150° = +\sin 30°}$
$\mathbf{\cos 150° = -\cos 30}$
$\mathbf{\tan 150° = -\tan 30}$

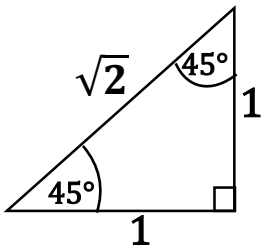
$\sin 210° = -\sin 30°$
$\cos 210° = -\cos 30°$
$\tan 210° = +\tan 30°$

$\sin 330° = -\sin 30°$
$\cos 330° = +\cos 30°$
$\tan 330° = -\tan 30°$

Special angles

- These are angles which can be evaluated without using a scientific calculator.
- Consider the two right-angled triangles below of sides 2 units and 1 unit base and perpendicular side.



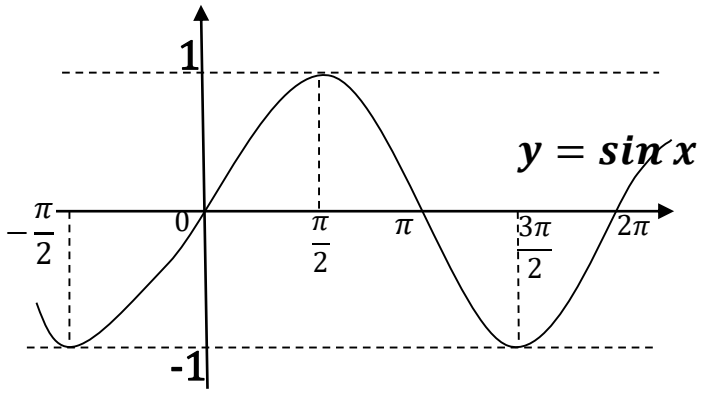


- Taking trigonometric ratios for $\mathbf{< 30°, < 45°}$ $\boldsymbol{and}$ $\mathbf{< 60°}$ as well as those for $\mathbf{< 0°}$ $\boldsymbol{and}$ $\mathbf{< 90°}$ gives:

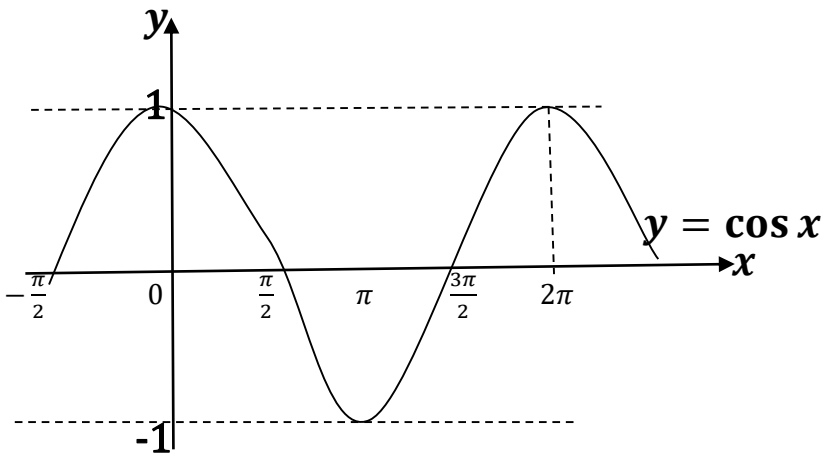
θ	$0°$	$30°$	$45°$	$60°$	$90°$
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	$Not\ defined$

Graphs of Trigonometric Functions

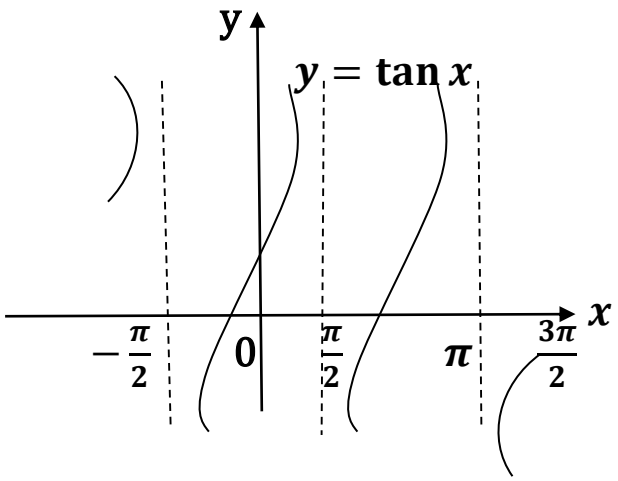
(a) The sine graph ($y = sin\ x$)



(b) The cosine graph ($y = cos\ x$)



(c) The tangent graph ($y = tan\ x$)

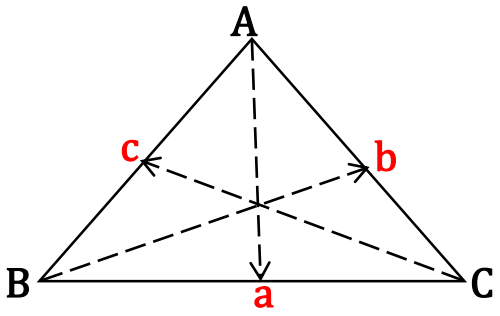


Note the following important points about the sine, cosine and tangent graphs:

- ✓ The sine and cosine graphs have a wave shape.
- ✓ The sine and cosine graphs have a minimum value of -1 and maximum value of 1. This implies that the sine and cosine graphs have amplitude of 1. However, amplitude is half the distance between the highest and lowest points of a wave.
- ✓ The sine and cosine graphs complete a full cycle in $360°$. This implies that the sine and cosine graphs have a period of $360°$. A period is the distance on the horizontal axis of one full cycle of a graph.
- ✓ The tangent graph does have neither minimum nor maximum value.
- ✓ The tangent graph is undefined where $x = 90°$ and $270°$. This implies that the tangent graph has asymptotes at $90°\ and\ 270°$. Asymptote is a straight line to which a graph draws closer and closer, without ever touching it.
- ✓ The tangent graph completes a full cycle in $180°$. However, the tangent graph has a period of $180°$.

The Sine and Cosine rules

- ✓ These are useful rules applied to find missing sides or angles for triangles that do not have right angles (non right-angled triangles).
- ✓ The sine rule is a rule to solve non right-angled triangles, when two angles and any side are given, or two sides and the non-included angle are given.
- ✓ Under sine rule, the angles are denoted by capital letters and the sides opposite to these angles are denoted by small letters respectively.
- ✓ Consider the non right-angled triangle below:



- ✓ In ΔABC above, the sine rule states that:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \text{ or } \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

- ✓ The cosine rule is a rule to solve non right-angled triangles, when two sides and the included angles are given, or three sides are given.
- ✓ Under cosine rule, the angles are denoted by capital letters and the sides opposite to these angles are denoted by small letters respectively.
- ✓ In ΔABC above, the cosine rule states that:

$$a^2 = b^2 + c^2 - 2bc\cos A$$
$$b^2 = a^2 + c^2 - 2ac\cos B$$
$$c^2 = a^2 + b^2 - 2ab\cos C$$

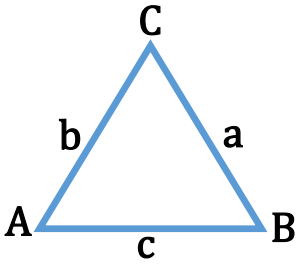
$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$\cos B = \frac{a^2+c^2-b^2}{2ac}$$

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

Area rule

- ✓ It is a rule to calculate the area of any triangle especially for non right-angled triangles, when two sides and the included angle are given.
- ✓ It is given by the general formula $A = \frac{1}{2}bh\sin\theta$, where b and h are the given two sides while θ is the included angle.
- ✓ Consider ΔABC below:



- ✓ In triangle ABC above, the Area rule states that:

$$Area = \frac{1}{2}bc\sin A$$

$$Area = \frac{1}{2}ac\sin B$$

$$Area = \frac{1}{2}ab\sin C$$

Trigonometric Identities

- ✓ A trigonometric identity is a relationship that is true for all values of the unknown variable.
- ✓ The following are the common trigonometric identities to be considered:

1 $c^2 = a^2 + b^2$

2 $sin^2\theta + \cos^2\theta = 1$

3 $1 + \cot^2\theta = \operatorname{cosec}^2\theta$

4 $\tan^2\theta + 1 = \sec^2\theta$

5 $\cos(A + B) = \cos A \cos B - \sin A \sin B$

6 $\cos(A - B) = \cos A \cos B + \sin A \sin B$

7 $\sin(A + B) = \sin A \cos B + \cos A \sin B$

8 $\sin(A - B) = \sin A \cos B - \cos A \sin B$

9 $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

10 $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

11 $\sin 2\theta = 2 \sin\theta \cos\theta$

12 $\cos 2\theta = \cos^2\theta - \sin^2\theta$

13 $\cos 2\theta = 1 - 2\sin^2\theta$

14 $\cos 2\theta = 2\cos^2\theta - 1$

15 $\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$

Harmonic form

- ✓ Changing an expression of the form $\alpha \sin\theta + \beta \cos\theta$ into the form $R \sin(\theta + a)$.
- ✓ We can express $\alpha \sin\theta + \beta \cos\theta$ into the form $R \sin(\theta + \alpha)$ by following the procedures outlined below:

STEP 1

Expand $R \sin(\theta + a)$

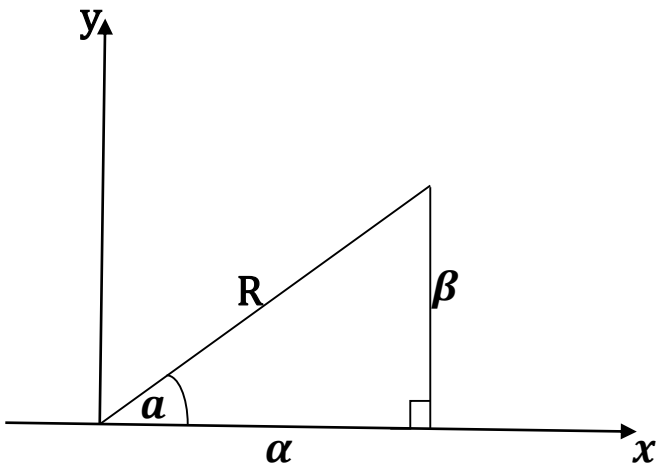
$$R \sin(\theta + a) = R\,[\sin(\theta + a)]$$
$$R \sin(\theta + a) = R(\sin\theta \cos a + \cos\theta \sin a)$$
$$= R \sin\theta \cos a + R \cos\theta \sin a$$

STEP 2

Form an equation connecting $\alpha \sin\theta + \beta \cos\theta$ and $R \sin(\theta + a)$.

$$R \sin(\theta + a)\; R \sin(\theta + a) = R \sin\theta \cos a + R \cos\theta \sin a$$
$$= \alpha \sin\theta + \beta \cos\theta$$
$$\therefore \alpha = R \cos a \text{ and } \beta = R \sin a$$
$$\cos a = \frac{\alpha}{R} \text{ and } \sin a = \frac{\beta}{R}$$

- ✓ Consider the diagram below:



✓ From the diagram above:

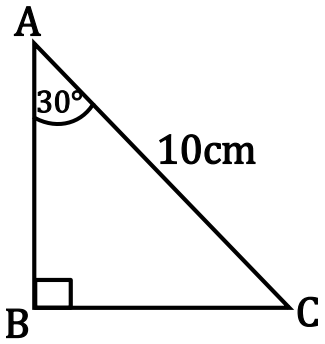
$$\tan a = \frac{\beta}{\alpha}$$

$$a = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$$

$$R = \sqrt{\alpha^2 + \beta^2}$$

2018.P1.Oct

In the diagram below, AC =10cm, angle ABC =90° and angle BAC = 30°.



Calculate BC

Solution

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\sin 30 = \frac{BC}{10}$$

$$\frac{1}{2} = \frac{BC}{10}$$

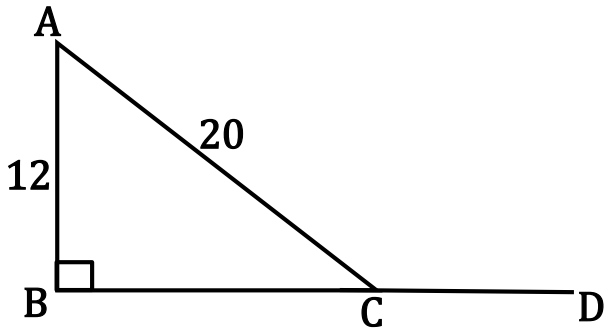
$$2BC = 10$$

$$BC = \frac{10}{2}$$

$$BC = 5cm$$

2019.P1.Nov

In the diagram below, BCD is a straight line, AB = 12cm, AC = 20cm and angle ABC = 90°.



Find the value of cos $A\hat{C}D$.

Solution

$$BC = \sqrt{AC^2 - AB^2}$$

$$= \sqrt{20^2 - 12^2}$$

$$= \sqrt{400 - 144}$$

$$= \sqrt{256}$$

$$= 16\text{cm}$$

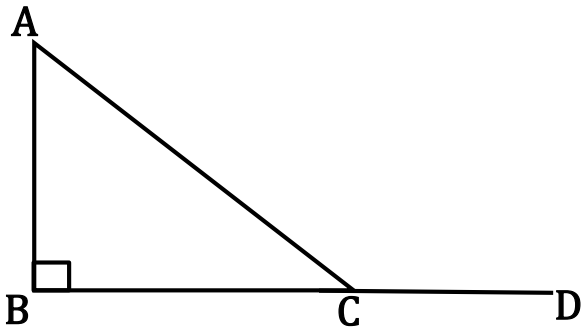
$$\cos A\hat{C}B = \frac{16}{20}$$

$$= \frac{4}{5}$$

$$\therefore \cos A\hat{C}D = -\frac{4}{5}$$

2020.P1.Nov

The diagram below shows ΔABC in which $A\hat{B}C = 90°$. BC is produced to D.



Given that $\tan B\hat{A}C = \frac{4}{3}$, find the value of $\cos A\hat{C}D$.

Solution

$$AC = \sqrt{BC^2 + AB^2}$$

$$= \sqrt{4^2 + 3^2}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25}$$

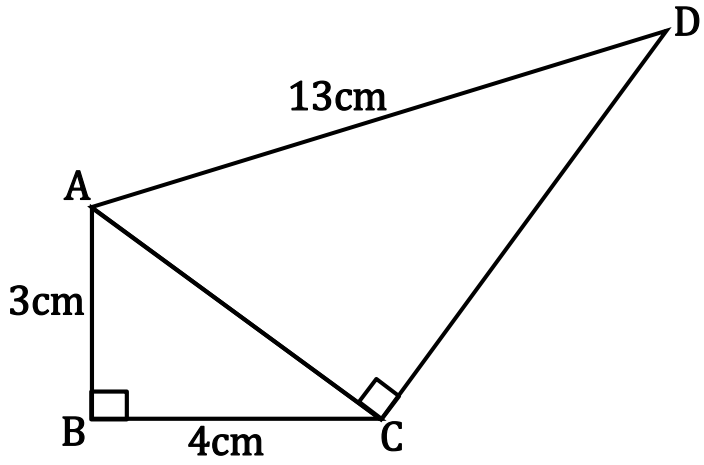
$$= 5$$

$$\cos A\hat{C}B = \frac{4}{5}$$

$$\therefore \cos A\hat{C}D = -\frac{4}{5}$$

2021.P1.Nov

In the diagram below, AB = 3cm, BC = 4cm, AD = 13cm and $A\hat{B}C = A\hat{C}D = 90°$.



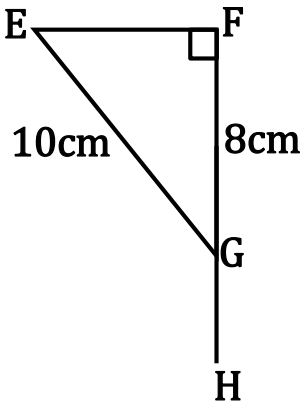
Calculate the value of tan $D\hat{A}C$.

Solution

$$AC = \sqrt{BC^2 + AB^2}$$
$$= \sqrt{4^2 + 3^2}$$
$$= \sqrt{16 + 9}$$
$$= \sqrt{25}$$
$$= 5\text{cm}$$
$$DC = \sqrt{AD^2 - AC^2}$$
$$= \sqrt{13^2 - 5^2}$$
$$= \sqrt{169 - 25}$$
$$= \sqrt{144}$$
$$= 12\text{cm}$$
$$\tan D\hat{A}C = \frac{12}{5}$$
$$= 2\frac{2}{5}$$

2022.P1.Nov

In the diagram below, FGH is a straight line, FG = 8cm, EG = 10cm and angle EFG = 90°.



Find the value of sin $\hat{EGH}$.

Solution

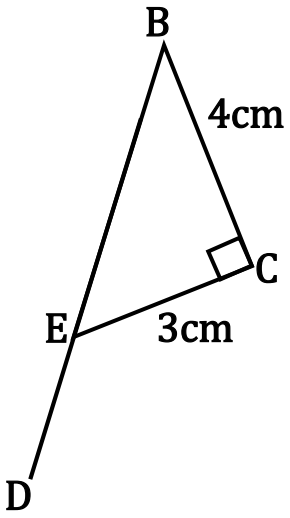
$$EF = \sqrt{EG^2 - FG^2}$$
$$= \sqrt{10^2 - 8^2}$$
$$= \sqrt{100 - 64}$$
$$= \sqrt{36}$$
$$= 6\text{cm}$$

$$\sin \hat{EGF} = \frac{6}{10}$$
$$= \frac{3}{5}$$

$$\therefore \sin \hat{EGH} = -\frac{3}{5}$$

2023.P1.Nov

In the diagram below, BCE is a right angled triangle in which BC = 4cm and CE = 3cm. BE is produced to D.



Find the value of cos CÊD.

Solution

$$\text{BE} = \sqrt{BC^2 + CE^2}$$

$$= \sqrt{4^2 + 3^2}$$

$$= \sqrt{16 + 9}$$

$$= \sqrt{25}$$

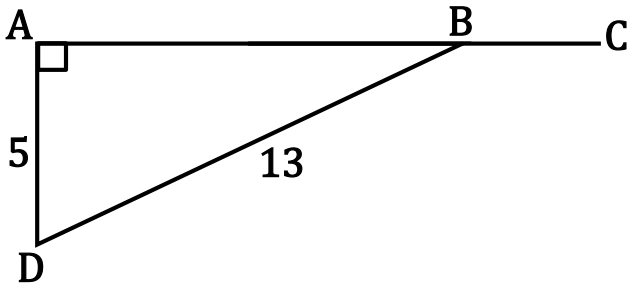
$$= 5\text{cm}$$

$$\cos C\hat{E}B = \frac{3}{5}$$

$$\therefore \cos C\hat{E}D = -\frac{3}{5}$$

2024.P1.GCE

In the diagram below, ABC is a straight line, AD = 5cm, BD = 13cm and angle BAD = 90°.



Find the value of tan $D\hat{B}C$.

Solution

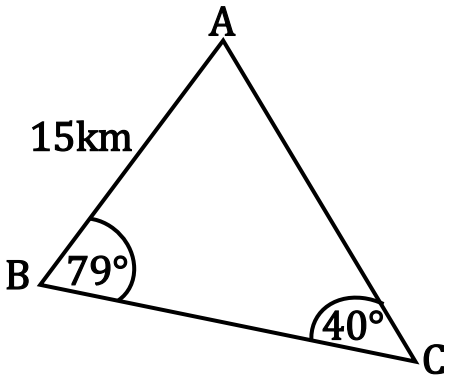
$$AB^2 = BD^2 - AD^2$$
$$AB^2 = BD^2 - AD^2$$
$$AB^2 = 13^2 - 5^2$$
$$AB^2 = 169 - 25$$
$$AB^2 = 144$$
$$AB = \sqrt{144}$$
$$AB = 12cm$$

$$\text{Tan } A\hat{B}D = \frac{5}{12}$$

$$\therefore \text{Tan } D\hat{B}C = -\frac{5}{12}$$

2018.P2.GCE

(a) Three village A, B and C are connected by straight paths as shown in the diagram below.



Given that AB = 15km, angle ABC = 79° and angle ACB = 40°, calculate the

(i) distance AC, [4]

(ii) area of triangle ABC, [2]

(iii) shortest distance from B to AC. [2]

(b) Solve the equation $\cos\theta = 0.937$ for $0° \leq \theta \leq 360°$. [2]

(c) Sketch the graph of $y = \sin\theta$ for $0° \leq \theta \leq 360°$. [2]

Solution

(a) (i) $\frac{b}{\sin B} = \frac{c}{\sin C}$

$\frac{AC}{\sin 79} = \frac{15}{\sin 40}$

$AC \sin 40 = 15 \sin 79$

$AC = \frac{15 \sin 79}{\sin 40}$

$AC = 22.907\,111\,975\,751\,2$

$AC = 22.9km$

(ii) $B\hat{A}C = 180 - 79 - 40$

$= 61°$

Area $= \frac{1}{2} bh \sin\theta$

$= \frac{1}{2} \times 15 \times 22.9 \sin 61$

$= 150.215\,934\,701\,191$

$= 150km^2$

(iii) SD $= \frac{2A}{b}$

$= \frac{2 \times 150}{22.9}$

$= 13.100\,436\,681\,222\,7$

$= 13.1$km

(b) $\cos\theta = 0.937$

$\alpha = \cos^{-1}(0.937)$

$= 20.4°$

1st Quadrant $\theta = \alpha$

$= 20.4°$

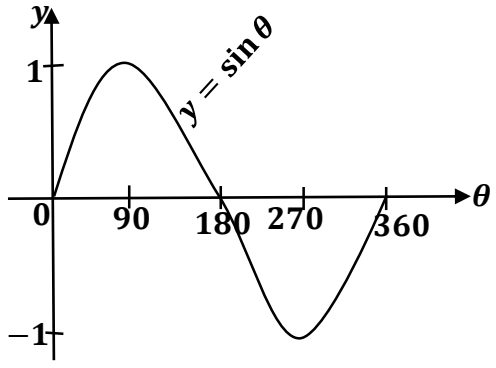
4th Quadrant $\theta = 360 - \alpha$

$= 360 - 20.4$

$= 339.6°$

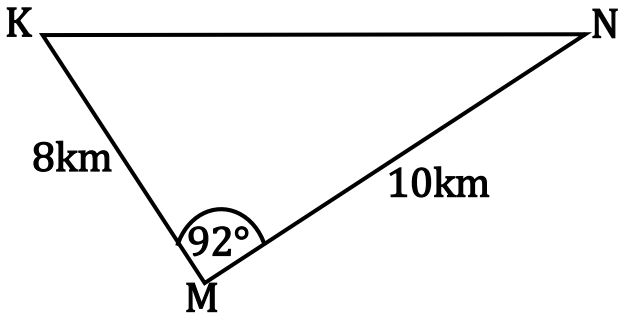
(c) $y = \sin\theta$

θ	0°	90°	180°	270°	360°
y	0	1	0	-1	0



2019.P2.Nov

(a) The diagram below shows a triangle KMN in which KM = 8km, MN = 10km and $K\widehat{M}N = 92°$.



Calculate

(i) KN, [5]

(ii) the area of triangle KMN, [2]

(iii) the shortest distance from M to KN. [2]

(b) Solve the equation $2\tan\theta = -3$ for $0° \leq \theta \leq 180°$. [1]

Solution

(a) (i) $m^2 = k^2 + n^2 - 2kn\cos M$

$m^2 = 10^2 + 8^2 - 2(10)(8)\cos 92$

$m^2 = 164 - 160\cos 92$

$m^2 = 169.583\,919\,472\,4$

$m = \sqrt{169.583\,919\,472\,4}$

$m = 13.022\,439\,075\,396$

$\therefore KN = 13km$

(ii) $A = \frac{1}{2}bh\sin\theta$

$= \frac{1}{2} \times 10 \times 8\sin 92$

$= 39.975\,633\,080\,763\,8$

$= 40km^2$

(iii) $SD = \frac{2A}{b}$

$= \frac{2\times 40}{13}$

$= \frac{80}{13}$

$= 6.153\,846$

$= 6.15km$

(b) $2\tan\theta = -3$

$\tan x = -\frac{3}{2}$

$\alpha = \tan^{-1}\left(\frac{3}{2}\right)$

$\alpha = 56.3°$

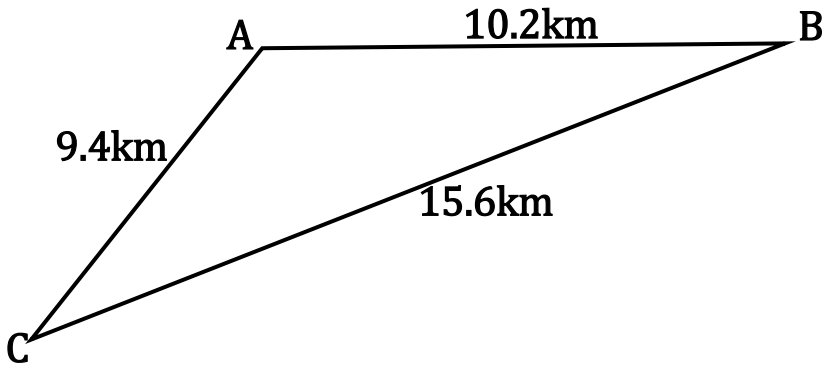
2nd Quadrant $\theta = 180° - \alpha$

$\theta = 180 - 56.3$

$\theta = 123.7°$

2020.P2.Nov

(a) Three villages A, B and C are joined by straight roads as shown in the diagram below.



Given that AB = 10.2km, BC = 15.6km and AC = 9.4km, calculate

(i) angle BAC to the nearest whole number, [5]

(ii) the area of triangle ABC, [2]

(iii) the shortest distance from A to BC. [2]

(b) Solve the equation $2\cos\theta = 1$ for $0° \leq \theta \leq 180°$. [1]

Solution

(a) (i) $\cos A = \frac{b^2+c^2-a^2}{2bc}$

$= \frac{9.4^2+10.2^2-15.6^2}{2(9.4)(10.2)}$

$= \frac{-50.96}{191.76}$

$A = \cos^{-1}\left(\frac{-50.96}{191.76}\right)$

$= 105.411\,454\,5$

$B\hat{A}C = 105°$

(ii) $A = \frac{1}{2}bh\sin\theta$

$= \frac{1}{2} \times 10.2 \times 9.4 \sin 105$

$= 46.306\,484\,11$

$= 46.3\text{km}^2$

(iii) $SD = \frac{2A}{b}$

$= \frac{2\times 46.3}{15.6}$

$= \frac{92.6}{15.6}$

$= 5.935\,897\,436$

$= 5.94\text{km}$

(b) $2\cos\theta = 1$

$\cos\theta = \frac{1}{2}$

$\alpha = \cos^{-1}\left(\frac{1}{2}\right)$

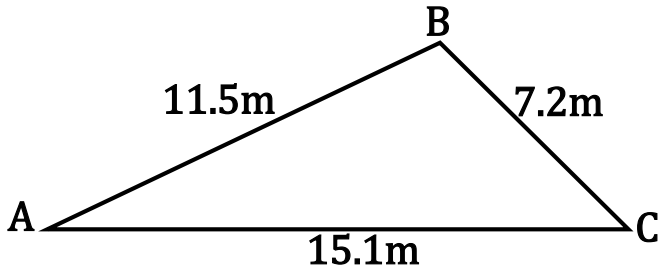
$\alpha = 60°$

1st Quadrant $\theta = \alpha$

$\theta = 60°$

2021.P1.Nov

(a) The diagram below shows triangle ABC in which AB = 11.5m, BC = 7.2m and AC = 15.1m.



Calculate

(i) angle ABC, [5]

(ii) the area of triangle ABC, [2]

(iii) the shortest distance from B to AC. [2]

(b) Solve the equation $15\tan\theta = 14$ for $180° \leq \theta \leq 270°$. [1]

Solution

(a) (iv) $\cos B = \frac{a^2+c^2-b^2}{2ac}$

$= \frac{7.2^2+11.5^2-15.1^2}{2(7.2)(11.5)}$

$= \frac{-43.92}{165.6}$

$B = \cos^{-1}\left(\frac{-43.92}{165.6}\right)$

$= 105.379\,870\,620\,706$

$A\widehat{B}C = 105.4°$

(v) $A = \frac{1}{2}bh\sin\theta$

$= \frac{1}{2} \times 7.2 \times 11.5 \sin 105.4$

$= 39.913\,549\,735\,292\,2$

$= 39.9m^2$

(vi) $SD = \frac{2A}{b}$

$= \frac{2\times39.9}{15.1}$

$= \frac{79.8}{15.1}$

$= 5.284\,768\,211\,920\,53$

$= 5.28m$

(b) $15\tan\theta = 14$

$\tan\theta = \frac{14}{15}$

$\alpha = \tan^{-1}\left(\frac{14}{15}\right)$

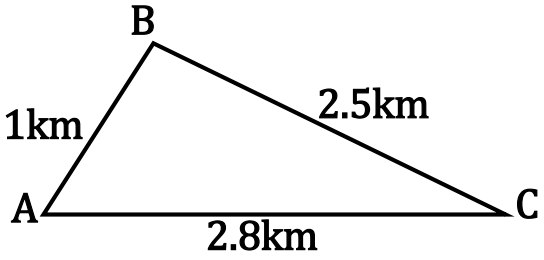
$\alpha = 43°$

3rd Quadrant $\theta = 180 + \alpha$

$\theta = 180 + 43$

2022.P2.Nov

(a) The diagram below shows a triangular piece of land ABC.



Given that AB = 1km, BC = 2.5km and AC = 2.8km, calculate the

(i) angle ABC, [5]

(ii) area of triangle ABC, [2]

(iii) shortest distance from B to AC. [2]

(b) Solve the equation $2\cos\theta = -1$ for $180° \leq \theta \leq 360°$. [1]

Solution

(a) (vii) $\cos B = \frac{a^2+c^2-b^2}{2ac}$

$= \frac{2.5^2+1^2-2.8^2}{2(2.5)(1)}$

$= \frac{-0.59}{5}$

$B = \cos^{-1}\left(\frac{-0.59}{5}\right)$

$= 96.776\,690\,91$

$A\hat{B}C = 96.8°$

(viii) $A = \frac{1}{2}bh\sin\theta$

$= \frac{1}{2} \times 2.5 \times 1 \sin 96.8$

$= 1.241206885$

$= 1.24\text{km}^2$

(ix) $SD = \frac{2A}{b}$

$= \frac{2\times 1.24}{2.8}$

$= \frac{2.28}{2.8}$

$= 0.885714285$

$= 0.89\text{km}$

(b) $2\cos\theta = -1$

$\cos\theta = -\frac{1}{2}$

$\alpha = \cos^{-1}\left(\frac{1}{2}\right)$

$\alpha = 60°$

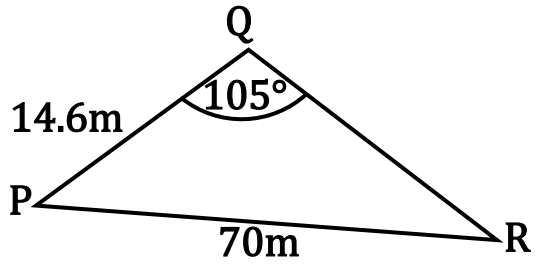
2nd Quadrant $\theta = 180 + \alpha$

$\theta = 180 + 60$

$= 240°$

2023.P2.Nov

(a) In the following triangle, PQ = 41.6m, PR = 70m and angle PQR = 105°.



Calculate

(i) angle QPR, [5]

(ii) the area of triangle PQR, [2]

(iii) the shortest distance from Q to PR. [2]

(b) Solve the equation $\tan\theta = 2.75$ for $180° \leq \theta \leq 270°$. [1]

Solution

(a) (i) $\frac{q}{\sin Q} = \frac{r}{\sin R}$

$\frac{70}{\sin 105} = \frac{41.6}{\sin R}$

$70 \sin R = 41.6 \sin 105$

$\sin R = \frac{41.6 \sin 105}{70}$

$\sin R = 0.574035919$

$\hat{R} = \sin^{-1}(0.574035919)$

$= 35°$

$\therefore Q\hat{P}R = 180 - 105 - 35$

$= 40°$

(ii) Area $= \frac{1}{2}bh \sin\theta$

$= \frac{1}{2} \times 41.6 \times 70 \sin 40$

$= 935.8987597$

$= 936m^2$

(iii) SD $= \frac{2A}{b}$

$= \frac{2 \times 936}{70}$

$= 26.74285714$

$= 26.7m$

(b) $\tan\theta = 2.75$

$\alpha = \tan^{-1}(2.75)$

$= 70°$

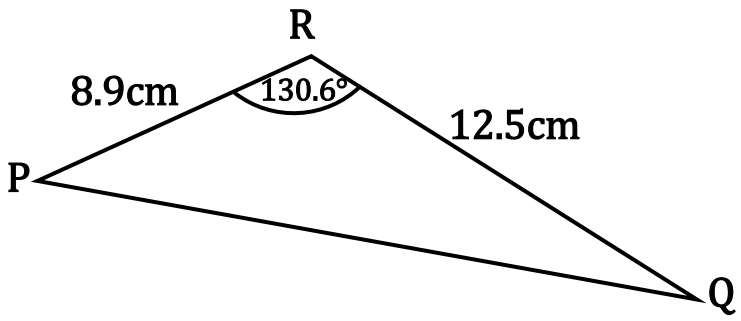
3rd Quadrant $\theta = 180 + \alpha$

$= 180 + 70$

$= 250°$

2024.P2.GCE

(a) The diagram below shows a triangle PQR in which PR = 8.9cm, QR = 12.5cm and $\text{P}\hat{\text{R}}\text{Q} = 130.6°$.



Calculate

(i) PQ, [5]

(ii) the area of triangle PQR, [2]

(iii) the shortest distance from R to PQ. [2]

(b) Solve the equation $2\sin x = 1$ for $90° \leq x \leq 270°$. [1]

Solution

(a) (i) $r^2 = p^2 + q^2 - 2pq\cos R$

$$r^2 = 12.5^2 + 8.9^2 - 2(12.5)(8.9)\cos 130.6$$

$$r^2 = 235.46 - 222.5\cos 130.6$$

$$r^2 = 380.257\,263\,341\,652$$

$$r = \sqrt{380.257\,263\,341\,652}$$

$$r = 19.500\,186\,238\,640\,2$$

$$\therefore PQ = 19.5cm$$

(ii) $\mathrm{A} = \frac{1}{2}bh\sin\theta$

$= \frac{1}{2} \times 12.5 \times 8.9 \sin 130.6$

$= 42.234\,466\,470\,502\,7$

$= 42.2\text{cm}^2$

(iii) $\mathrm{SD} = \frac{2A}{b}$

$= \frac{2 \times 42.2}{19.5}$

$= \frac{84.4}{19.5}$

$= 4.328\,205\,1$

$= 4.33\text{cm}$

(b) $2\sin x = 1$

$\sin x = \frac{1}{2}$

$\alpha = \sin^{-1}\left(\frac{1}{2}\right)$

$\alpha = 30°$

2nd Quadrant $\theta = 180° - \alpha$

$x = 180 - 30$

$x = 150°$

MENSURATION

Objectives

By the end of this topic, learners should

1 Define Mensuration.
2 Calculate the area of a sector.
3 Calculate the surface area of three-dimensional figures.
4 Calculate the volume of prisms.
5 Solve problems involving area and volume.

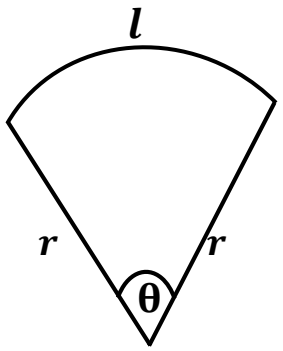
Definition

- Mensuration is the branch of mathematics that studies the measurement of geometric figures and their parameters like length, volume, shape, surface area, lateral surface area, etc.
- It is also defined as the act of measuring something. The world we live in is three-dimensional.
- In both primary and secondary school mathematics, the concept of measuring is crucial. Furthermore, measuring has a direct impact on our daily life.
- We learn to measure objects in both 3D and 2D shapes as we learn to measure them.
- Both standard and nonstandard units of measurement can be used to measure objects or quantities.

Area and Length of an arc of a sector

- A sector is any part of a circle bounded by two radii.
- An arc is a part of the circumference of a circle bounded by two radii.
- The area of a sector is proportional to the angle subtended at the centre of the circle.

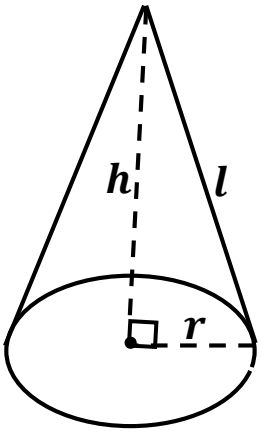
- Consider the diagram below which shows a sector



$\therefore A = \frac{\theta}{360} \times \pi r^2$ and $l = \frac{\theta}{360} \times 2\pi r$

Volume and Surface Area of a cone

- A cone is defined as a three-dimensional object that has a circular base and one vertex.
- To calculate the total surface area of a cone, we need to add the area of the curved surface and the area of the circular base.
- Volume is the amount of space that a three-dimensional object occupies.
- The diagram below shows a full cone with perpendicular height (h), radius (r) and a slant height (l).



$\therefore TSA = \pi r(r + l)$

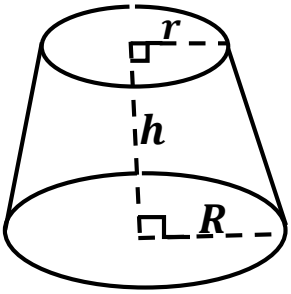
$Volume = \frac{1}{3} \times base\ area \times perpendicular\ height$

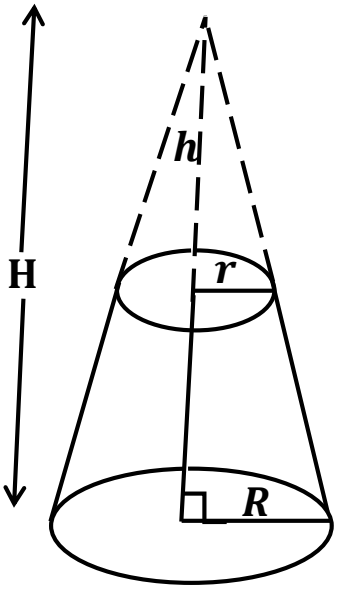
$V = \frac{1}{3}\pi r^2 h$

The volume of a frustum

- ✓ A frustum is that part of a cone or pyramid that remains if the top part has been cut off.
- ✓ The cut is always made parallel to the base of the cone or pyramid.
- ✓ However, a frustum with a triangular, square or rectangular cross-section is the frustum of a pyramid.
- ✓ A frustum with a circular cross-section is the frustum of a cone.

The frustum of a cone



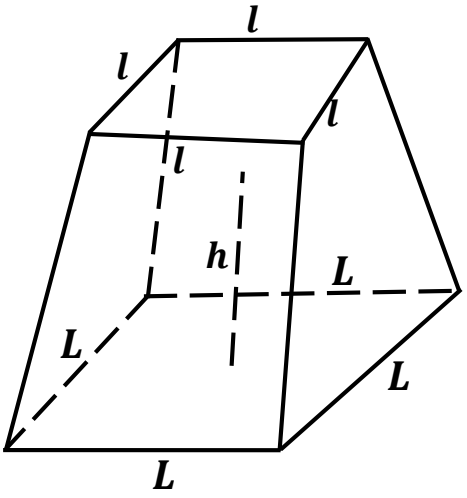


$$V = \frac{1}{3}\pi(R^2H - r^2h)$$

$$OR$$

$$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$$

The frustum of a square-based pyramid

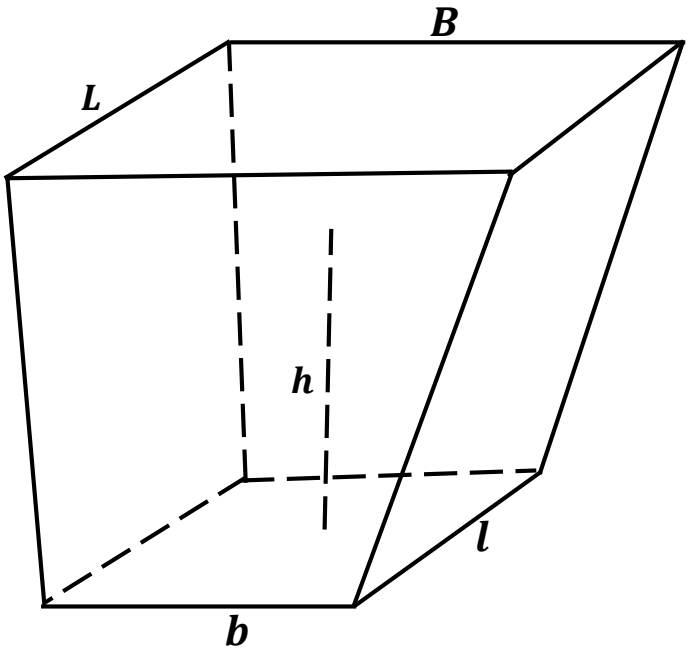


$$\therefore V = \frac{1}{3}(L^2H - l^2h)$$

$$OR$$

$$V = \frac{1}{3}\left(L^2 + l^2 + \sqrt{L^2l^2}\ \right)h$$

The frustum of a rectangular-based pyramid



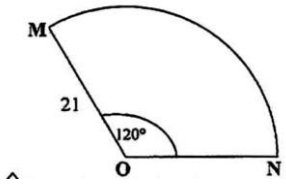
$$\therefore V = \frac{1}{3}(LBH - lbh)$$

$$OR$$

$$V = \frac{1}{3}\left(LB + lb + \sqrt{L \times B \times l \times b}\ \right)h$$

2016.P1.November

In the diagram below, MN is an arc of a circle whose Centre is O and radius 21cm.



Given that $M\hat{O}N = 120°$, calculate the area of the sector MON.
[Take π to be $\frac{22}{7}$]

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{120}{360} \times \frac{22}{7} \times 21^2$$

$$A = \frac{1}{3} \times 22 \times 3 \times 21$$

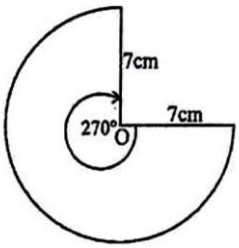
$$A = \frac{1 \times 22 \times 3 \times 21}{3}$$

$$A = 1 \times 22 \times 21$$

$$\therefore A = 462cm^2$$

2017.p1.November

The diagram below is a sector with centre O and radius 7cm. Angle at O is $270°$.



Calculate the area of the sector. $\left[\pi = \frac{22}{7}\right]$

Solution

$$\mathbf{A} = \frac{\theta}{360} \times \pi r^2$$

$$\mathbf{A} = \frac{270}{360} \times \frac{22}{7} \times 7^2$$

$$\mathbf{A} = \frac{3}{4} \times 22 \times 7$$

$$\mathbf{A} = \frac{3 \times 11 \times 7}{2}$$

$$\mathbf{A} = \frac{231}{2}$$

$$\mathbf{A} = 115.5\text{cm}^2$$

2017.p1.GCE

The curved surface area of a cone is 88cm². Given that the base radius is 4cm, calculate the slant height of the cone. $\left[\pi = \frac{22}{7},\ A = \pi r l\right]$

Solution

$$A = \pi r l$$

$$88 = \frac{22}{7} \times 4 \times l$$

$$88 \times 7 = 22 \times 4 \times l$$

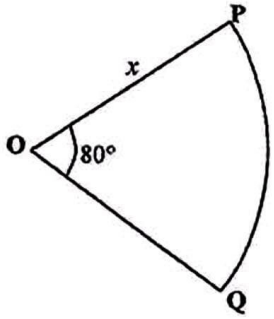
$$22 \times 4 \times 7 \times = 22 \times 4 \times l$$

$$l = \frac{22 \times 4 \times 7}{22 \times 4}$$

$$l = 7cm$$

2018.p1.November

In the diagram below, OPQ is a sector of a circle whose centre is O, radius x cm and angle POQ $=80°$.



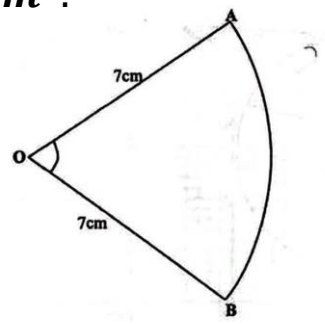
Given that the area of the sector OPQ is 308cm², calculate the value of x. $\left[\pi = \frac{22}{2}\right]$

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$
$$308 = \frac{80}{360} \times \frac{22}{7} \times x^2$$
$$308 = \frac{2}{9} \times \frac{22}{7} \times x^2$$
$$308 = \frac{2 \times 22 \times x^2}{9 \times 7}$$
$$2 \times 22 \times x^2 = 308 \times 9 \times 7$$
$$x^2 = \frac{308 \times 9 \times 7}{2 \times 22}$$
$$x^2 = \frac{2 \times 2 \times 7 \times 11 \times 9 \times 7}{2 \times 2 \times 11}$$
$$x^2 = 7 \times 9 \times 7$$
$$x^2 = 49 \times 9$$
$$x^2 = (7 \times 3)^2$$
$$x = 7 \times 3$$
$$x = 21cm$$

2018.p1.GCE

The diagram below shows a sector AOB of a circle with centre O and radius 7cm. the area of the sector is $25\frac{2}{3}cm^2$.



Calculate $A\hat{O}B$.

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$

$$25\frac{2}{3} = \frac{\theta}{360} \times \frac{22}{7} \times 7^2$$

$$\frac{77}{3} = \frac{\theta}{180} \times 11 \times 7$$

$$\frac{77}{3} = \frac{\theta \times 11 \times 7}{180}$$

$$3 \times \theta \times 11 \times 7 = 60 \times 3 \times 7 \times 11$$

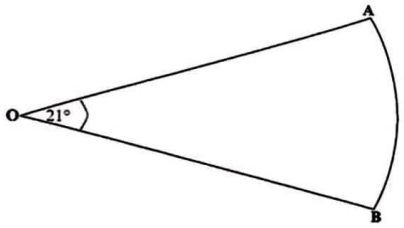
$$\theta = \frac{60 \times 3 \times 7 \times 11}{3 \times 11 \times 7}$$

$$\theta = 60°$$

$$\therefore A\hat{O}B = 60°$$

2019.p1.November

The diagram below shows a sector AOB. Arc AB subtends an angle of $21°$ at the centre O.



Given that the area of the sector is $14.85cm^2$, calculate the radius. $\left[\pi = \frac{22}{7}\right]$

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$

$$14.85 = \frac{21}{360} \times \frac{22}{7} \times r^2$$

$$\frac{297}{20} = \frac{1}{60} \times 11 \times r^2$$

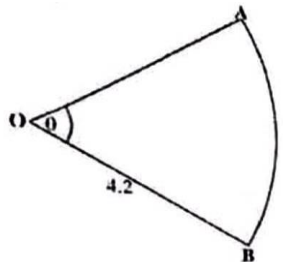
$$\frac{297}{20} = \frac{1 \times 11 \times r^2}{60}$$

$$20 \times 1 \times 11 \times r^2 = 297 \times 60$$

$$r^2 = \frac{27 \times 11 \times 20 \times 3}{20 \times 1 \times 11}$$
$$r^2 = 27 \times 3$$
$$r^2 = 3 \times 9 \times 3$$
$$r^2 = 9 \times 9$$
$$r^2 = (3 \times 3)^2$$
$$r = 3 \times 3$$
$$r = 9cm$$

2019.p1.GCE

The diagram below shows a sector of a circle with centre O and radius 4.2cm. Angle AOB $= \theta$.



Given that the area of the sector AOB is 9.24cm², find the value of θ. $\left[\theta = \frac{22}{7}\right]$

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$

$$9.24 = \frac{\theta}{360} \times \frac{22}{7} \times 4.2^2$$

$$\frac{231}{25} = \frac{\theta}{180} \times \frac{11}{7} \times \frac{21}{5} \times \frac{21}{5}$$

$$\frac{231}{25} = \frac{\theta \times 11 \times 7 \times 3 \times 3 \times 7}{3 \times 3 \times 2 \times 5 \times 2 \times 7 \times 5 \times 5}$$

$$\frac{7 \times 11 \times 3}{5 \times 5} = \frac{\theta \times 11 \times 7}{2 \times 5 \times 2 \times 5 \times 5}$$

$$\theta \times 11 \times 7 \times 5 \times 5 = 7 \times 11 \times 3 \times 2 \times 5 \times 2 \times 5 \times 5$$

$$\theta = \frac{7 \times 11 \times 3 \times 2 \times 5 \times 2 \times 5 \times 5}{11 \times 7 \times 5 \times 5}$$

$$\theta = 3 \times 2 \times 5 \times 2$$

$$\therefore \theta = 60°$$

2020.P1.Nov

Calculate the radius of a semicircle whose area is 77cm². $\left[\text{Take } \pi \text{ to be } \frac{22}{7}\right]$

Solution

$$\text{Area} = \left(\frac{1}{2}\right)\pi r^2$$

$$77 = \left(\frac{1}{2}\right)\pi r^2$$

$$77 = \left(\frac{1}{2}\right) \times \left(\frac{22}{7}\right) \times r^2$$

$$154 = \left(\frac{22}{7}\right) \times r^2$$

$$1078 = 22r^2$$

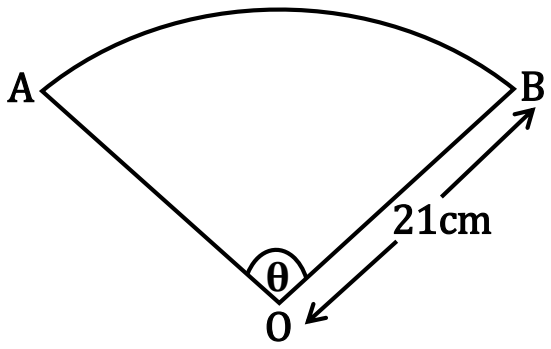
$$49 = r^2$$

$$r = \sqrt{49}$$

$$r = 7\text{cm}$$

2021.P1.Nov

The diagram below shows a sector AOB with centre O and radius 21cm. The angle subtended at the centre is θ and the area of the sector is 462cm^2.



Calculate the value of θ. $\left[\pi = \frac{22}{7}\right]$

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$

$$462 = \frac{\theta}{360} \times \frac{22}{7} \times 21^2$$

$$462 = \frac{\theta}{180} \times \frac{11}{1} \times 3 \times 21$$

$$462 = \frac{77\theta}{20}$$

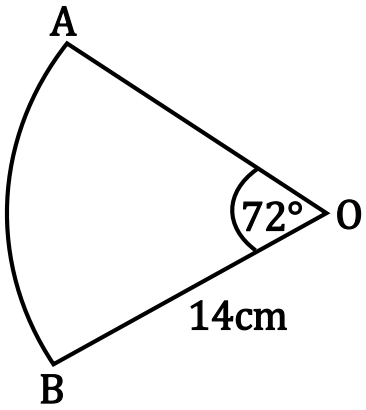
$$77\theta = 462 \times 20$$

$$\theta = \frac{462 \times 20}{77}$$

$$\theta = 120°$$

2022.P1.Nov

In the diagram below, AOB is a sector of a circle, centre O. Angle AOB = 72° and the radius is 14cm.



Calculate the area of the sector. $\left[\pi = \frac{22}{7}\right]$

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{72}{360} \times \frac{22}{7} \times 14^2$$

$$A = \frac{1}{5} \times 22 \times 28$$

$$A = \frac{616}{5}$$

$$A = 123\frac{1}{5}$$

$$A = 123.2cm^2$$

2023.P1.Nov

The curved surface area of a cone with radius 4.2cm is 138.6cm^2. Find its slant height. $\left[\pi = \frac{22}{7}\right]$

Solution

$$\text{CSA} = \pi r l$$

$$138.6 = \frac{22}{7} \times 4.2 \times l$$

$$\frac{693}{5} = \frac{22}{7} \times \frac{21}{5} \times l$$

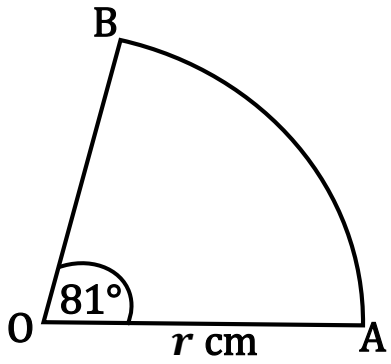
$$\frac{693}{5} = \frac{66l}{5}$$

$$l = \frac{693}{66}$$

$$l = 10.5cm$$

2024.P1.GCE

The diagram below shows a sector AOB whose area is 34.65cm^2. The angle subtended at the centre is 81° and the radius is r cm. $\left(\pi = \frac{22}{7}\right)$



Find the radius (r) of the sector.

Solution

$$A = \frac{\theta}{360} \times \pi r^2$$

$$34.65 = \frac{81}{360} \times \frac{22}{7} \times r^2$$

$$\frac{693}{20} = \frac{9}{20} \times \frac{11r^2}{7}$$

$$\frac{693}{20} = \frac{99r^2}{140}$$

$$20 \times 99r^2 = 140 \times 693$$

$$r^2 = \frac{140 \times 693}{20 \times 99}$$

$$r^2 = 49$$

$$r = \pm\sqrt{49}$$

$$r = 7cm$$

2016.p2.October

(a) The cross section of a rectangular tank measures 1.2m by 0.9m. if it contains fuel to a depth of 10m, find the number of litres of fuel in the tank. ($1m^3 = 1\,000$ litres). [3]

(b) A cone has a perpendicular height of 12cm and slant height of 13cm. Calculate its total surface area. [Take $\pi = 3.142$]. [3]

Solution

(a) First we determine the volume of the tank:

$$V = l \times w \times h$$
$$= 1.2 \times 0.9 \times 10$$
$$= 10.8m^3$$

Now we convert units:

$$10.8m^3 \times \frac{1\,000l}{1m^3} = 10\,800l$$

$\therefore$ **The number of litres of fuel in the tank is 10 800 litres.**

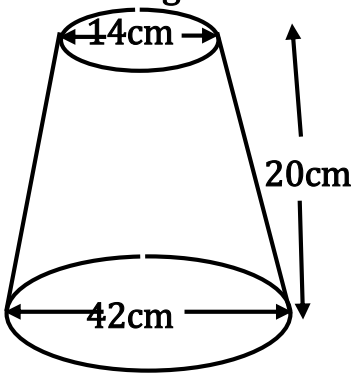
(b) First we need to determine the radius for the circular base of the cone.

$$r^2 = l^2 - h^2$$
$$r^2 = 13^2 - 12^2$$
$$r^2 = 25$$
$$r = \sqrt{25}$$
$$r = 5cm$$

$$\therefore TSA = \pi r(r + l)$$
$$= 3.142 \times 5(5 + 13)$$
$$= 15.71(18)$$
$$= 282.78$$
$$= 283cm^2$$

2017.p2.November

The figure below is a frustum of a cone. The base diameter and top diameter are 42cm and 14cm respectively, while the height is 20cm. (Take π as 3.142)



Calculate its volume. [6]

Solution

First formula:

$V = \frac{1}{3}\pi(R^2H - r^2h)$

$r : R = h : H$

$7 : 21 = h : h + 20$

$1 : 3 = h : h + 20$

$\frac{1}{3} = \frac{h}{h+20}$

$3h = h + 20$

$3h - h = 20$

$2h = 20$

$h = 10cm$

$H = 10 + 20$

$H = 30cm$

$\therefore V = \frac{1}{3}\pi(R^2H - r^2h)$

$= \frac{1}{3} \times 3.142(21^2 \times 30 - 7^2 \times 10)$

$= \frac{3.142}{3}(13\,230 - 490)$

$= \frac{3.142}{3}(12\,740)$

$= \frac{40029.08}{3}$

$= 13\,343.026$

$= 13\,300cm^3$

Second formula:

$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$

$= \frac{1}{3} \times 3.142(21^2 + 7^2 + 21 \times 7)20$

$= \frac{3.142}{3}(490 + 147)20$

$= \frac{3.142}{3}(637)20$

$= \frac{3.142}{3}(12\,740)$

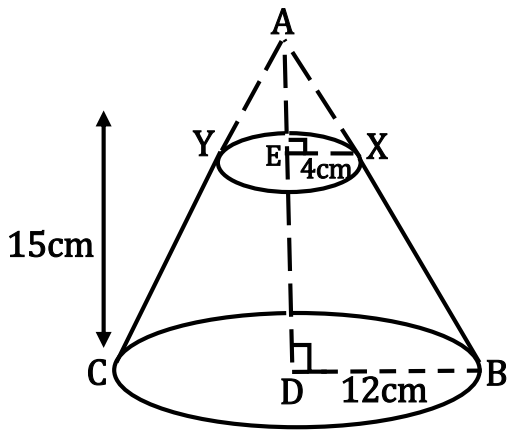
$= \frac{40\,029.08}{3}$

$= 13\,343.02667$

$= 13\,300cm^3$

2017.p2.GCE

The figure below is a cone ABC from which BCXY remained after the small cone AXY was cut off. [Take π as 3.142]



Given that EX = 4cm, DB = 12cm and DE = 15cm, calculate

(i) The height AE, of the smaller cone AXY. [2]

(ii) The volume of XBCY, the shape that remained. [4]

Solution

(i) $r : R = h : H$

$4 : 12 = h : h + 15$

$1 : 3 = h : h + 15$

$\frac{1}{3} = \frac{h}{h+15}$

$3h = h + 15$

$3h - h = 15$

$2h = 15$

$h = 7.5$

$\therefore The\ height\ AE = 7.5cm$

(ii) $V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$

$= \frac{1}{3} \times 3.142(12^2 + 4^2 + 12 \times 4)15$

$= \frac{3.142}{3}(160 + 48)15$

$= \frac{3.142}{3}(208)15$

$= \frac{3.142}{3}(3\ 120)$

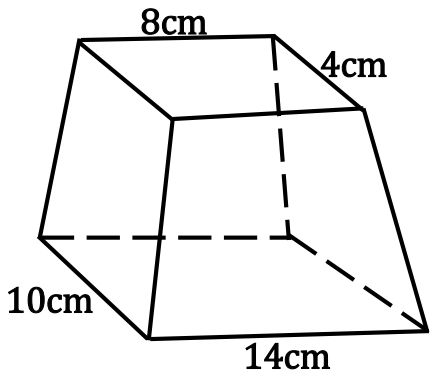
$= \frac{9\ 803.04}{3}$

$= 3\ 267.68$

$= 3\ 270cm^3$

2018.p2.November

The diagram below is a frustum of a rectangular pyramid with a base 14cm long and 10cm wide. The top of the frustum is 8cm long and 4cm wide.



Given that the height of the frustum is 11.4cm, calculate its volume. [6]

Solution

$$V = \frac{1}{3}(LB + lb + \sqrt{L \times B \times l \times b}\,)h$$

$$= \tfrac{1}{3}(10 \times 14 + 4 \times 8 + \sqrt{10 \times 14 \times 4 \times 8})11.4$$

$$= \frac{1}{3}(140 + 32 + \sqrt{4\,480})11.4$$

$$= \frac{1}{3}(172 + 8\sqrt{70})11.4$$

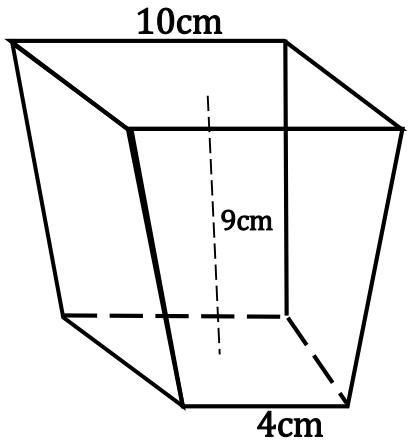
$$= \frac{11.4(172 + 8\sqrt{70})}{3}$$

$$= 907.944648066359$$

$$= 908cm^3$$

2018.p2.GCE

The diagram below shows a bin in the form of a frustum with square ends of sides 4cm and 10cm respectively. The height of the bin is 9cm.



Find the volume of the bin. [6]

Solution

$$V = \frac{1}{3}\left(L^2 + l^2 + \sqrt{L^2 l^2}\ \right)h$$

$$= \frac{1}{3}\left(10^2 + 4^2 + \sqrt{10^2 \times 4^2}\right)9$$

$$= \frac{1}{3}\left(116 + \sqrt{1\,600}\right)9$$

$$= \frac{1}{3}(116 + 40)9$$

$$= \frac{1}{3}(156)9$$

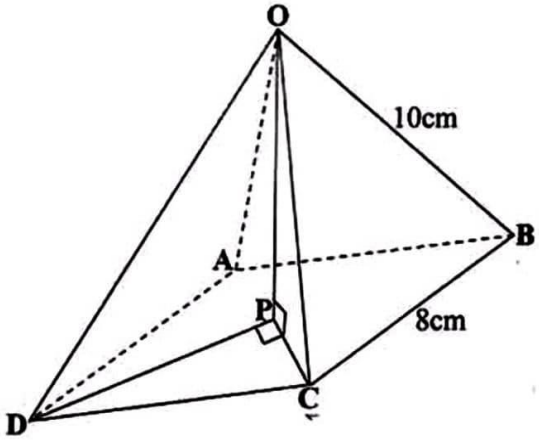
$$= \frac{1}{3}(1\,404)$$

$$= \frac{1\,404}{3}$$

$$= 468cm^3$$

2019.p2.November

The figure below shows a right pyramid with a vertex O and a square base ABCD of side 8cm. $\mathbf{C\hat{P}D = O\hat{P}C = 90°}$.



Given that OA = OB = OC = OD = 10cm, calculate

(i) the length of OP, [4]

(ii) the angle between the edge OC and the base PC. [2]

Solution

(i) $\mathbf{AC^2 = 8^2 + 8^2}$

$\mathbf{AC^2 = 64 + 64}$

$\mathbf{AC = \sqrt{128}}$

$\mathbf{AC = 11.3cm}$

$\mathbf{OC^2 = OP^2 + PC^2}$

$\mathbf{10^2 = OP^2 + 5.65^2}$

$\mathbf{100 = OP^2 + 31.9225}$

$\mathbf{OP^2 = 100 - 31.9225}$

$\mathbf{OP = \sqrt{68.0775}}$

$\mathbf{OP = 8.25cm}$

(ii) SHO-CAH-TOA

$$\cos P\hat{C}O = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

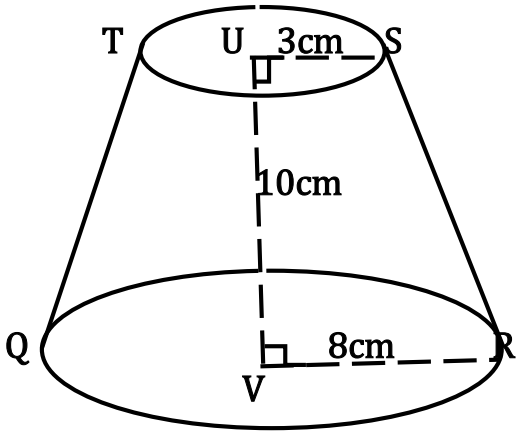
$$\cos P\hat{C}O = \frac{5.65}{10}$$

$$P\hat{C}O = \cos^{-1}\left(\frac{5.65}{10}\right)$$

$$P\hat{C}O = 56.6°$$

2019.p2.GCE

The diagram below shows a frustum TQRS of a cone. [Take π as 3.142]



Given that US = 3cm, UV = 10cm and RV = 8cm, calculate its volume. [6]

Solution

$$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$$

$$= \frac{1}{3} \times 3.142(8^2 + 3^2 + 8 \times 3)10$$

$$= \frac{3.142}{3}(73 + 24)10$$

$$= \frac{3.142}{3}(97)10$$

$$= \frac{3.142}{3}(970)$$

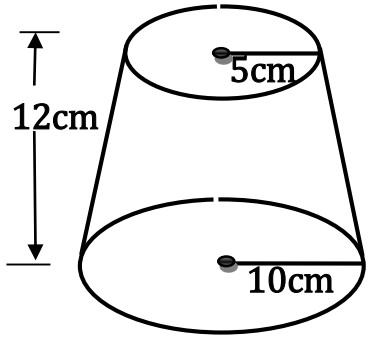
$$= \frac{3047.74}{3}$$

$$= 1\,015.913$$

$$= 1\,020 cm^3$$

2020.p2.November

The figure below is a frustum of a cone. The base radius and top radius are 10cm and 5cm respectively, while the height is 12cm. (Take π as 3.142)



Calculate its volume. [6]

Solution

$$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$$

$$= \frac{1}{3} \times 3.142(10^2 + 5^2 + 10 \times 5)12$$

$$= \frac{3.142}{3}(125 + 50)12$$

$$= \frac{3.142}{3}(175)12$$

$$= \frac{3.142}{3}(2\ 100)$$

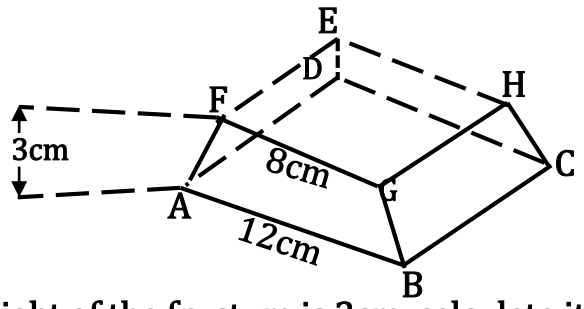
$$= \frac{6\ 598.2}{3}$$

$$= 2\ 199.4$$

$$= 2\ 200cm^3$$

2020.p2.GCE

The figure ABCDEFGH is a frustum of a pyramid. EFGH and ABCD are square of sides 8cm and 12cm respectively.



Given that the height of the frustum is 3cm, calculate its volume. [6]

Solution

$$V = \frac{1}{3}\left(L^2 + l^2 + \sqrt{L^2 l^2}\ \right)h$$

$$= \frac{1}{3}\left(12^2 + 8^2 + \sqrt{12^2 \times 8^2}\right)3$$

$$= \frac{1}{3}\left(208 + \sqrt{9\,216}\right)3$$

$$= \frac{1}{3}(208 + 96)3$$

$$= \frac{1}{3}(304)3$$

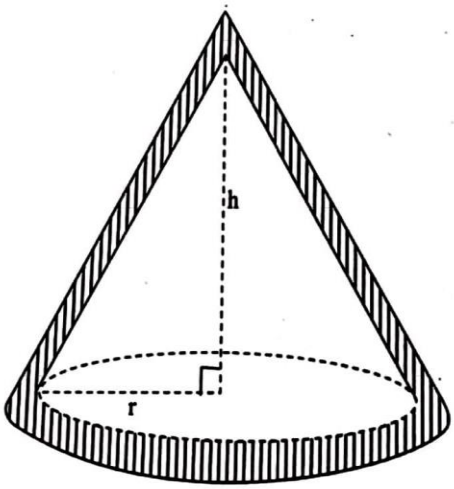
$$= \frac{1}{3}(912)$$

$$= \frac{912}{3}$$

$$= 304cm^3$$

2021.p2.November

The diagram below shows a cone. The shaded part is the thickness of the cone. The internal volume is 34 650cm³. The internal base area is 3 850cm². [Take $\pi = 3.142$]



(i) Calculate the internal height (h) and radius (r). [4]

(ii) Given that the cone is 0.7cm thick, calculate the external volume of the cone. [2]

Solution

(i) $\text{Volume} = \frac{1}{3} \times \text{base area} \times h$

$34\,650 = \frac{1}{3} \times 3\,850 \times h$

$34\,650 = \frac{3\,850h}{3}$

$3\,850h = 34\,650 \times 3$

$3\,850h = 103\,950$

$h = \frac{103\,950}{3\,850}$

$h = 27cm$

$\text{Base Area} = \pi r^2$

$3\,850 = 3.142r^2$

$r = \sqrt{\frac{3\,850}{3.142}}$

$r = 35cm$

(ii) $V = \frac{1}{3}\pi r^2 h$

$= \frac{1}{3} \times 3.142 \times 35.7^2 \times 27.7$

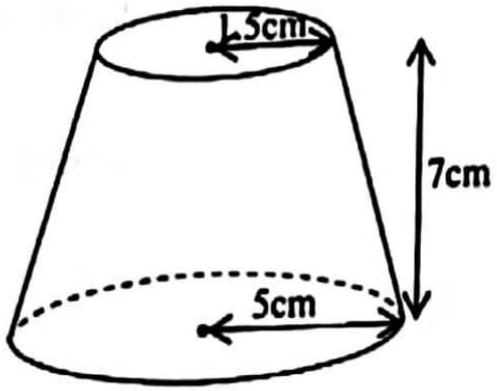
$= \frac{110\,923.197\,966}{3}$

$= 36\,974.399\,322$

$= 37\,000cm^3$

2021.p2.GCE

The following diagram shows a frustum of a cone. The radii of the circles at the top and bottom are 1.5cm and 5cm respectively. Its height is 7cm.



Calculate the volume of the frustum. (Take π as 3.142) [6]

Solution

$$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$$

$$= \frac{1}{3} \times 3.142(5^2 + 1.5^2 + 5 \times 1.5)7$$

$$= \frac{3.142}{3}(27.25 + 7.5)7$$

$$= \frac{3.142}{3}(34.75)7$$

$$= \frac{3.142}{3}(243.25)$$

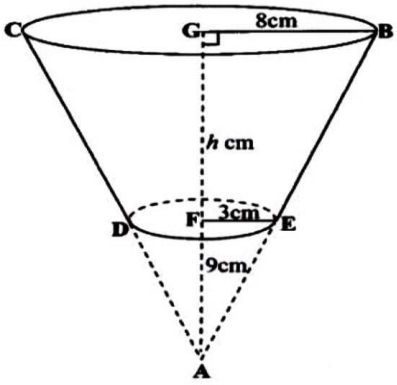
$$= \frac{764.291\,5}{3}$$

$$= 254.763\,83$$

$$= 255cm^3$$

2022.p2.November

The diagram shows a bin BCDE which was made by cutting off the smaller cone ADE from the cone ABC. [Take $\pi = 3.142$]



Given that AF = 9cm, BG = 8cm, EF = 3cm and GF = h cm, calculate the

(i) value of h, [2]

(ii) volume of the bin BCDE. [4]

Solution

(i) $r : R = h : H$

$3 : 8 = 9 : 9 + h$

$\frac{3}{8} = \frac{9}{9+h}$

$3(9+h) = 9 \times 8$

$27 + 3h = 72$

$3h = 72 - 27$

$3h = 45$

$h = \frac{45}{3}$

$h = 15cm$

(ii) $V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$

$= \frac{1}{3} \times 3.142(8^2 + 3^2 + 8 \times 3)15$

$= \frac{3.142}{3}(73 + 24)15$

$= \frac{3.142}{3}(97)15$

$= \frac{3.142}{3}(1\,455)$

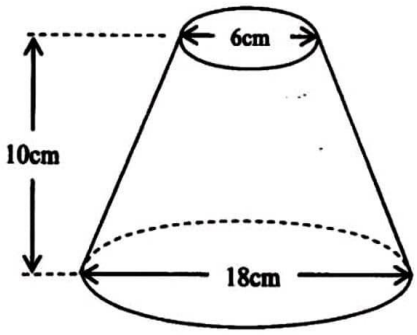
$= \frac{4\,571.61}{3}$

$= 1\,523.87$

$= 1\,520cm^3$

2022.p2.GCE

The following diagram shows the frustum of a cone. The top and bottom diameters are 6cm and 18cm respectively. [Take π as 3.142]



Given that its perpendicular height is 10cm, calculate the volume of the frustum. [6]

Solution

$$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$$

$$= \frac{1}{3} \times 3.142(9^2 + 3^2 + 9 \times 3)10$$

$$= \frac{3.142}{3}(90 + 27)10$$

$$= \frac{3.142}{3}(117)10$$

$$= \frac{3.142}{3}(1\,170)$$

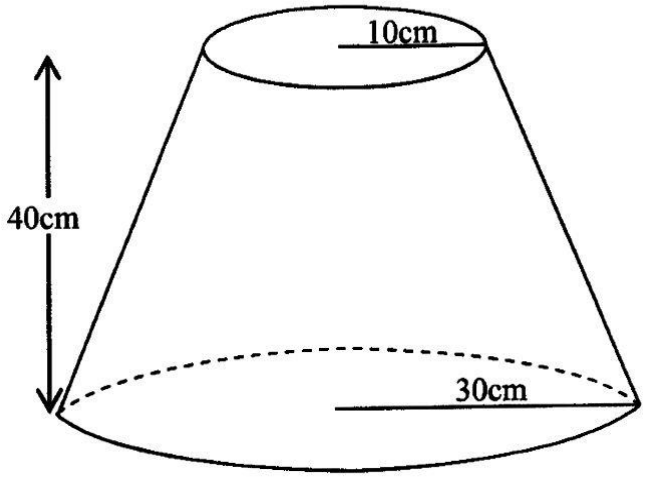
$$= \frac{3\,676.14}{3}$$

$$= 1\,225.38$$

$$= 1\,230 cm^3$$

2023.p2.GCE

The diagram below shows the frustum of a cone. The perpendicular height is 40cm. The top and bottom radii are 10cm and 30cm respectively. [Take π as 3.142]



Calculate the volume of the frustum. [6]

Solution

$$V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$$

$$= \frac{1}{3} \times 3.142(30^2 + 10^2 + 30 \times 10)40$$

$$= \frac{3.142}{3}(1000 + 300)40$$

$$= \frac{3.142}{3}(1300)40$$

$$= \frac{3.142}{3}(52000)$$

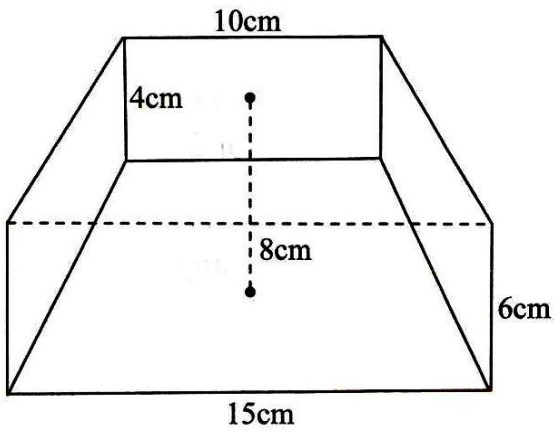
$$= \frac{163384}{3}$$

$$= 54461.33333$$

$$= 54\,500cm^3$$

2023.p2.Nov.

The following diagram is a frustum of a rectangular pyramid with a base 15cm long and 6cm wide. The top of the frustum is 10cm long and 4cm wide. Its perpendicular height is 8cm.



(i) Find the height of the pyramid. [2]
(ii) Calculate the volume of the frustum. [4]

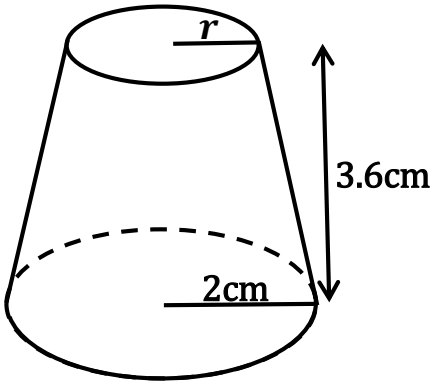
Solution

(i)
$$b : B = h : H$$
$$10 : 15 = h : h + 8$$
$$2 : 3 = h : h + 8$$
$$\frac{2}{3} = \frac{h}{h+8}$$
$$3h = 2(h + 8)$$
$$3h = 2h + 16$$
$$3h - 2h = 16$$
$$h = 16cm$$
$$H = 8 + 16$$
$$H = 24cm$$

(ii)
$$V = \frac{1}{3}\left(LB + lb + \sqrt{L \times B \times l \times b}\right)h$$
$$= \frac{1}{3}\left(6 \times 15 + 4 \times 10 + \sqrt{6 \times 15 \times 4 \times 10}\right)8$$
$$= \frac{1}{3}\left(90 + 40 + \sqrt{3600}\right)8$$
$$= \frac{1}{3}(130 + 60)8$$
$$= \frac{1}{3}(190)8$$
$$= \frac{1}{3}(1520)$$
$$= \frac{1520}{3}$$
$$= 506.6666667$$
$$= 507cm^3$$

2024.P2.GCE

The figure below is a frustum of a cone. The base radius is 2cm, top radius is r cm and the height is 3.6cm. [Take π as 3.142]



Calculate the

(i) value of r if the height of the cone from which the frustum was made was 6cm, [2]

(ii) volume of the frustum. [4]

Solution

(i) $r : R = h : H$

$r : 2 = 2.4 : 6$

$\frac{r}{2} = \frac{2.4}{6}$

$6r = 4.8$

$r = 0.8cm$

(ii) $V = \frac{1}{3}\pi(R^2 + r^2 + Rr)h$

$= \frac{1}{3} \times 3.142(2^2 + 0.8^2 + 2 \times 0.8)3.6$

$= \frac{3.142}{3}(4.64 + 1.6)3.6$

$= \frac{3.142}{3}(6.24)3.6$

$= \frac{3.142}{3}(22.464)$

$= \frac{70.581\,888}{3}$

$= 23.527\,296$

$= 23.5cm^3$

PROBABILITY

Objectives

By the end of this topic, learners should

1. Define Probability.
2. Compute probabilities using the laws of probability.
3. Calculate probabilities using tree diagrams and grids.
4. Calculate probabilities of mutually exclusive events and compound events.
5. Find probabilities of independent events.
6. Apply probabilities to solve real-life problems.

Definition

- Probability is a branch of mathematics that deals with the study of chance events and their likelihood of occurrence.
- It is a measure of the likelihood or uncertainty of an event occurring.

Laws of Probability

- The probability of something happening is the likelihood of that thing happening.
- A probability is always expressed as a number from 0 to 1.
- If we conduct an experiment once, this is called a trial.
- A result of an experiment is an outcome of the experiment.
- An event is one particular outcome in which we are interested.
- The total number of possible outcomes of a trial form a sample space.

- If A is one possible outcome of a trial, then the probability of A is given by $P(A)\ \frac{\text{Number of outcomes favourable to A}}{\text{total number of possible outcomes in the trial}}$.
- If A and B are the only possible outcomes of a trial, then $P(A) + P(B) = 1$.
- If in a trial the probability of an event A is P(A) and the trial is carried out in times, then the expected number of events is by $n \times P(A)$.

Tree diagrams and grids

- Compound events are two or more events that together form one trial of an experiment.
- Tree diagrams are used to solve problems involving compound events.
- A probability grid is a table that shows all the possible outcomes of a trial that consists of two events.
- Events A and B are mutually exclusive if they cannot happen at the same time.
- For any two events A and B that are mutually exclusive, $P(A \text{ or } B) = P(A) + P(B)$. So $P(A \cup B) = P(A) + P(B)$.
- For any two events A and B that are not mutually exclusive, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Events are independent if the occurrence of one does not influence the occurrence of the other in any way.
- For any two events A and B that are independent, $P(A \text{ and } B) = P(A) \times P(B)$.
- A discrete sample space contains a finite number of outcomes.
- A continuous sample space contains an infinite number of outcomes.

2018.P1.Oct

A box contains 5 green, 3 yellow and 6 red beads of the same type. Find the probability of picking a yellow bead at random from the box.

Solution

$$\text{Total} = 5 + 3 + 6$$
$$= 14$$
$$P(Y) = \frac{3}{14}$$

2019.P1.Nov

A number is chosen at random from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Find the probability that it is a perfect square.

Solution

$$\text{Total} = 10$$
Perfect square numbers: {1, 4, 9}
$$P(PS) = \frac{3}{10}$$

2020.P1.Nov

The probability of Masamushi failing an examination is $\frac{3}{10}$. What is the probability that she will pass that particular examination?

Solution

$$P(P) + P(F) = 1$$
$$P(P) + \frac{3}{10} = 1$$
$$P(P) = 1 - \frac{3}{10}$$
$$P(P) = \frac{7}{10}$$

2021.P1.Nov

The probability of getting a black button from a box is $\frac{2}{9}$. Find the probability of getting a button which is not black.

Solution

$$\mathbf{P(B) + P(B') = 1}$$
$$\frac{2}{9} + \mathbf{P(B') = 1}$$
$$\mathbf{P(B') = 1 - \frac{2}{9}}$$
$$\mathbf{P(B') = \frac{7}{9}}$$

2022.P1.Nov

The probability that a boy will be late for school on any particular day is x. Find, in terms of x, the probability that he will not be late for school.

Solution

$$\mathbf{P(late) + P(not\ late) = 1}$$
$$\boldsymbol{x} + \mathbf{P(not\ late) = 1}$$
$$\mathbf{P(not\ late) = 1 - } \boldsymbol{x}$$

2023.P1.Nov

A letter is chosen at random from the letters of the word "EXCELLENT". What is the probability that the letter "E" is chosen?

Solution

$$\mathbf{P(E) = \frac{3}{9}}$$
$$\mathbf{= \frac{1}{3}}$$

2024.P1.GCE

A fair die is rolled once. Find the probability of 4 showing up.

Solution

Since the die is fair, each of the 6 faces has an equal chance of showing up.

There is only 1 face with the number 4.

So, the probability of 4 showing up is:

$$P(4) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{1}{6} \text{ or } 0.17 \text{ (rounded to two decimal places).}$$

2018.P2.GCE

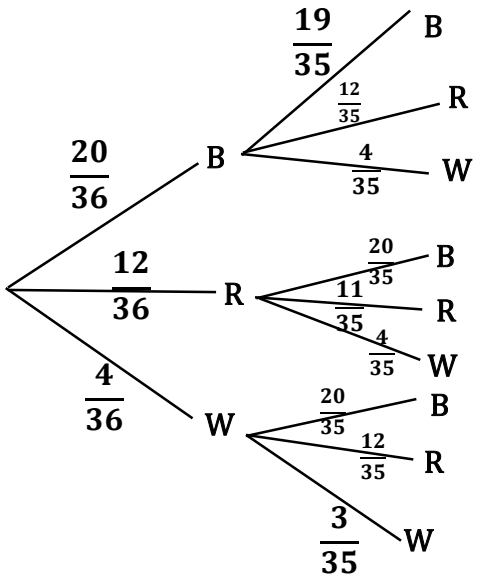
A box contains identical buttons of different colours. There are 20 black, 12 red and 4 white buttons in the box. Two buttons are picked at random one after another and not replaced in the box.

(i) Draw a tree diagram to show all the possible outcomes. [3]

(ii) What is the probability that both buttons are white? [2]

Solution

(i)



(ii) $P(WW) = \frac{4}{36} \times \frac{3}{35}$

$= \frac{12}{1\,260}$

$= \frac{1}{105}$

2019.P2.Nov

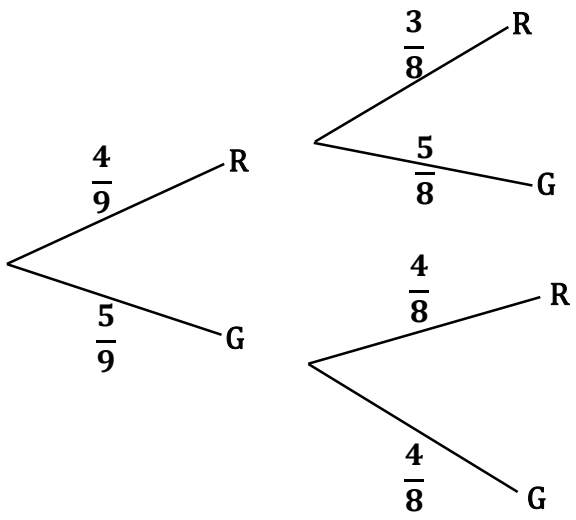
A box contains 4 red pens and 5 green pens. A pen is picked at random from the box without replacement and a second pen is then picked.

(i) Draw a tree diagram to illustrate the outcomes. [3]

(ii) What is the probability of one red pen and one green pen? [2]

Solution

(i)



POSSIBLE OUTCOMES

$P(RR) = \frac{4}{9} \times \frac{3}{8} = \frac{1}{6}$

$P(RG) = \frac{4}{9} \times \frac{5}{8} = \frac{15}{18}$

$P(GR) = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$

$P(GG) = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$

(ii) P(RG) or P(GR)

$$\frac{5}{18} + \frac{5}{18}$$

$$\frac{5}{9}$$

2020.P2.Nov

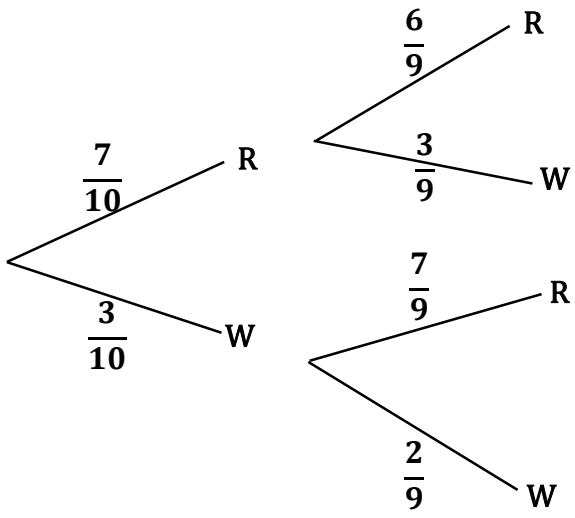
A bag contains 7 red and 3 white identical balls. Two balls are taken from the bag at random one after the other without replacement.

(i) Draw a tree diagram to show all the possible outcomes. [2]

(ii) Find the probability of taking at least one white ball. [3]

Solution

(i)



POSSIBLE OUTCOMES

$P(RR) = \frac{7}{10} \times \frac{6}{9} = \frac{7}{15}$

$P(RW) = \frac{7}{10} \times \frac{3}{9} = \frac{7}{30}$

$P(WR) = \frac{3}{10} \times \frac{7}{9} = \frac{7}{30}$

$P(WW) = \frac{3}{10} \times \frac{2}{9} = \frac{1}{15}$

(ii) P(RW) or P(WR) or P(WW)

$$\frac{7}{30} + \frac{7}{30} + \frac{1}{15}$$

$$\frac{7+7+2}{30}$$

$$\frac{16}{30}$$

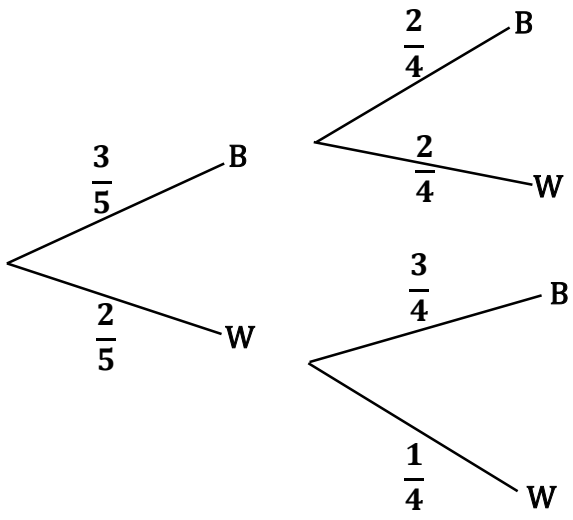
$$\frac{8}{15}$$

2021.P2.Nov

A box contains 3 black and 2 white marbles of the same size. A marble is taken out at random from the box and not replaced. A second marble is then drawn. Calculate the probability that both marbles are

(i) white, [2]
(ii) of different colours. [3]

Solution



POSSIBLE OUTCOMES

$P(BB) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$

$P(BW) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$

$P(WB) = \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$

$P(WW) = \frac{2}{5} \times \frac{1}{4} = \frac{1}{10}$

(i) $P(WW) = \frac{1}{10}$

(ii) $P(BW) \text{ or } P(WB) = \frac{3}{10} + \frac{3}{10}$

$= \frac{6}{10}$

$= \frac{3}{5}$

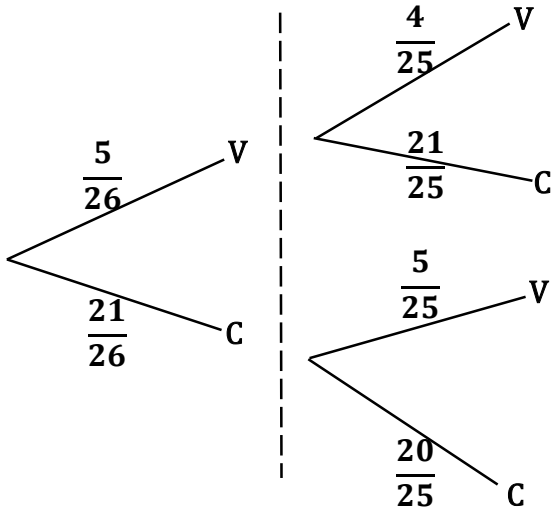
2022.P2.Nov

A box contains identical cards and on each card a letter of the English alphabet is printed. A card is selected at random from the box and not replaced, and a second card is then selected. Find the probability that the two cards selected

(i) have vowels printed, [2]

(ii) one has a vowel and the other a consonant printed. [3]

Solution



POSSIBLE OUTCOMES

$P(VV) = \frac{5}{26} \times \frac{4}{25} = \frac{2}{65}$

$P(VC) = \frac{5}{26} \times \frac{21}{25} = \frac{21}{130}$

$P(CV) = \frac{21}{26} \times \frac{5}{25} = \frac{21}{130}$

$P(CC) = \frac{21}{26} \times \frac{20}{25} = \frac{42}{65}$

(i) $P(VV) = \frac{2}{65}$

(ii) $P(VC) \text{ or } P(CV) = \frac{21}{130} + \frac{21}{130}$

$= \frac{42}{130}$

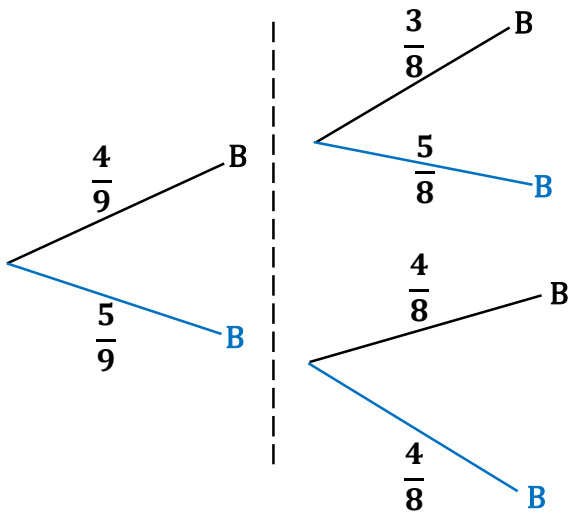
$= \frac{21}{65}$

2023.P2.Nov

A box contains 4 black and 5 blue identical pens. Two pens are picked from the box at random one after the other without replacement. Find the probability that the pens picked are

(i) both blue, [2]
(ii) of the same colour. [3]

Solution



POSSIBLE OUTCOMES

$P(BB) = \frac{4}{9} \times \frac{3}{8} = \frac{1}{6}$

$P(BB) = \frac{4}{9} \times \frac{5}{8} = \frac{5}{18}$

$P(BB) = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$

$P(BB) = \frac{5}{9} \times \frac{4}{8} = \frac{5}{18}$

(i) $P(BB) = \frac{5}{18}$

(ii) $P(BB) \text{ or } P(BB) = \frac{1}{6} + \frac{5}{18}$

$= \frac{4}{9}$

2024.P2.GCE

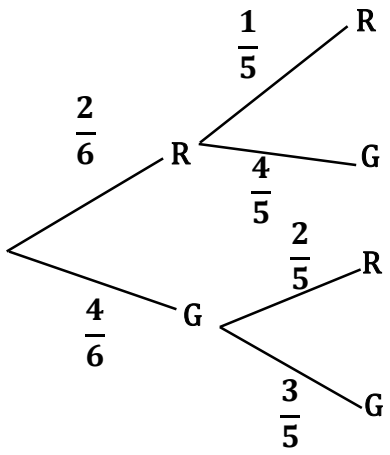
A box contains 6 identical beads, 2 of which are red and the rest are green. Two beads are selected at random from the box, one after the other without replacement.

(i) Draw a tree diagram to illustrate this information. [2]

(ii) Calculate the probability that the beads selected are of different colours. [3]

Solution

(i)



(ii) P(RG) or P(GR)

$$\left(\frac{2}{6}\times\frac{4}{5}\right)+\left(\frac{4}{6}\times\frac{2}{5}\right)$$

$$\frac{4}{15}+\frac{4}{15}$$

$$\frac{8}{15}$$

STATISTICS

Objectives

By the end of this topic, learners should

1. Define statistics.
2. Construct cumulative frequency tables using grouped and ungrouped data.
3. Draw cumulative frequency curves.
4. Draw relative cumulative frequency curves.
5. Calculate the range, interquartile range and semi-interquartile range.
6. Calculate the percentiles.
7. Calculate variance and standard deviation for ungrouped and grouped data.

Definition

- Statistics is the science of collecting, organizing, analyzing, interpreting and presenting data.
- It involves the use of mathematical and computational methods to summarize and describe data, as well as to draw conclusions and make predictions based on data.
- Statistics is one of the most useful branches of mathematics in real life.
- Statistics are used in many careers, including nursing, farming weather-reporting, life and short-term insurance, education and many more.

Cumulative frequency tables

- A cumulative frequency table is a table that has three columns, one each for data values, frequencies and cumulative frequencies.
- The frequency of a data value is the number of times that the data values appears in a data set.
- The cumulative frequency of a data value is the sum of all the frequencies up to and including the frequency of that data value.
- Ungrouped data are data vales that are used individually.
- Grouped data are data values that are grouped into class intervals.
- A class interval is an interval with a lower and upper value within which data is grouped.
- A cumulative frequency curve is a curve obtained by plotting a graph of the cumulative frequencies against the number of data values.
- A relative cumulative frequency curve is similar to a cumulative frequency curve, but instead of plotting cumulative frequencies, we plot relative cumulative frequencies.
- A relative cumulative frequency is a cumulative frequency divided by the total frequency.

Measures of dispersion

- Measures of dispersion give us information about how spread out together the data values in a data set are.
- There are different measures of dispersion namely:
  (i) Range
  (ii) Quartiles
  (iii) Interquartile range
  (iv) Semi-interquartile range
  (v) Percentiles

- The range of a data set is the difference between the highest and the lowest data values in a data set.
- Range = highest data value – lowest data value.
- Quartiles are measures of dispersion that divide an ordered data set into four equal quarters namely:
  - (i) The middle quarter (Q_2) which divides the data set into two equal halves. It is the same as the median.
  - (ii) The lower quartile (Q_1) which divides the bottom half of the data set into two equal halves.
  - (iii) The upper quartile (Q_3) which divides the top half of the data set into two equal halves.
- The interquartile range (IQR) is the difference between the upper quartile (Q_3) and the lower quartile (Q_1).
- IQR = $Q_3 - Q_1$.
- The semi-interquartile range (SIQR) is the half of the interquartile range (IQR).
- SIQR $= \frac{1}{2}(Q_3 - Q_1)$
  $= \frac{Q_3 - Q_1}{2}$
- Percentiles are measures of dispersion that divide an ordered data into 100 equal hundredths.
- n^{th} percentile $= \frac{n}{100} \times$ total frequency and the final result is always obtained on a graph.

Variance and Standard deviation

- Variance is a measure of the average distance between each number in a data set and a mean.
- Variance $(\sigma) = \frac{\sum(x-\bar{x})^2}{n}$ for ungrouped data where x is each score, $\bar{x}$ is the mean and n is the total frequency.
- For grouped data, Variance $(\sigma) = \frac{\sum f(x-\bar{x})^2}{\sum f}$ where f is the frequency of each class interval, x is the midpoint of each class interval, $\bar{x}$ is the mean and $\sum f$ is the total frequency.
- Standard deviation is the square root of variance.
- Standard deviation $= \sqrt{\frac{\sum(x-\bar{x})^2}{n}} = \sqrt{\left\{\frac{\sum x^2}{n} - (\bar{x})^2\right\}}$ for ungrouped data where x is each score, $\bar{x}$ is the mean and n is the total frequency.
- For grouped data,

  standard deviation $= \sqrt{\frac{\sum f(x-\bar{x})^2}{\sum f}} = \sqrt{\left\{\frac{\sum fx^2}{\sum f} - (\bar{x})^2\right\}}$ where f is the frequency of each class interval, x is the midpoint of each class.
- Mean $(\bar{x}) = \frac{\sum fx}{\sum f}$.

2018.P2.GCE

A farmer planted 60 fruits. In a certain month, the number of fruits per tree was recorded and results were shown in the table below.

Fruits per tree	2	3	4	5	6	7	8
Frequency	1	5	4	6	10	16	18

(a) Calculate the standard deviation.

(b) Answer this part of the question on a sheet of graph paper.

(i) Using the table above, copy and complete the relative cumulative frequency table below.

Fruits per tree	2	3	4	5	6	7	8
Frequency	1	6	10	16	26	42	60
Relative cumulative frequency	0.02	0.1	0.17	0.27			

(ii) Using a scale of 1cm to represent 1 unit on the x-axis for $0 \leq x \leq 8$ and 2cm to represent 0.1 unit on the y-axis for $0 \leq y \leq 1$, draw a smooth relative cumulative frequency curve.

(iii) Showing your method clearly, use your graph to estimate the 70th percentile.

Solution

(a)

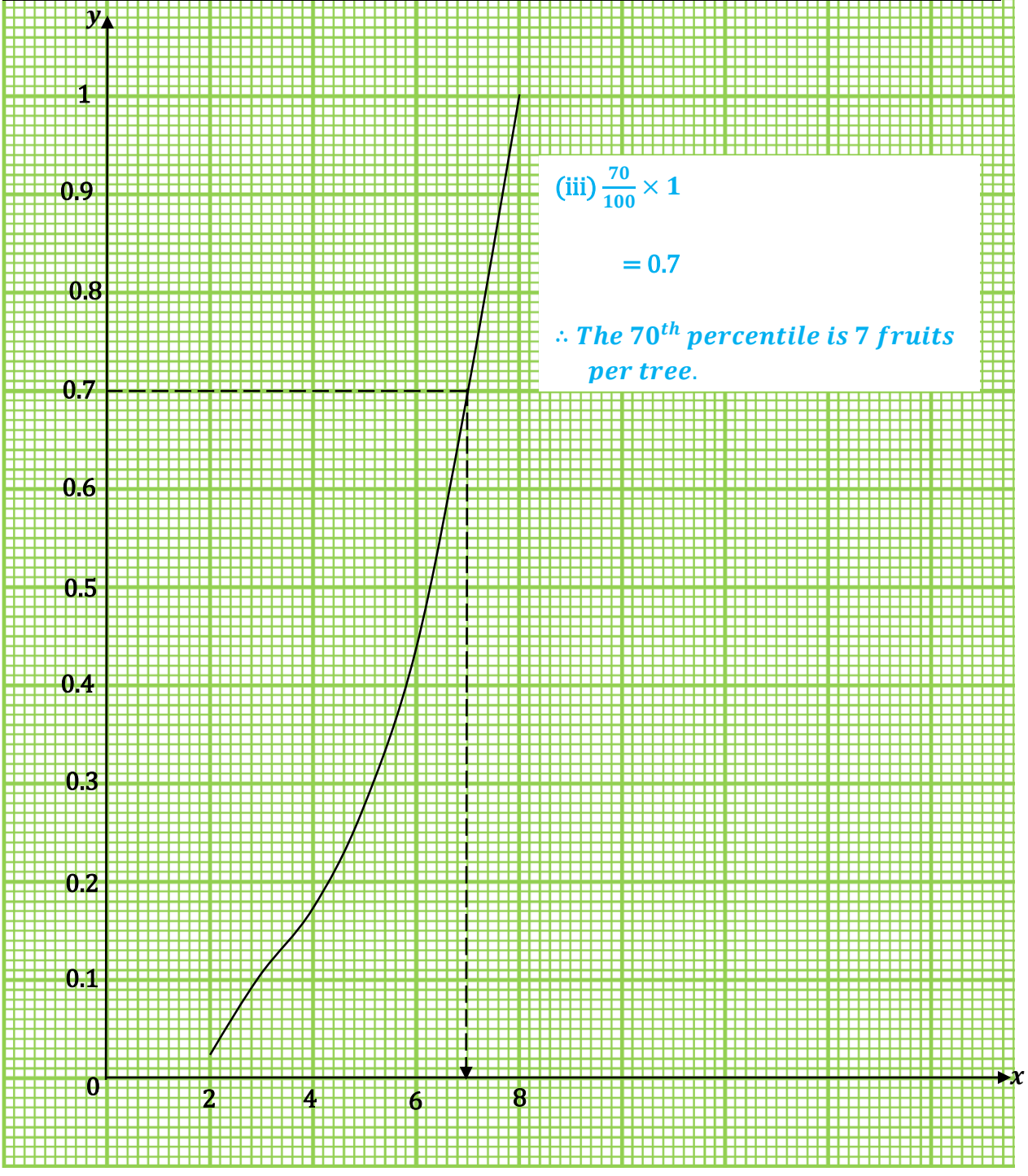
	f	x	fx	fx^2
	1	2	2	4
	5	3	15	45
	4	4	16	64
	6	5	30	150
	10	6	60	360
	16	7	112	784
	18	8	144	1 152
$\sum$	60		379	2 559

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f}$$
$$= \frac{379}{60}$$
$$= 6.32$$

$$\text{SD} = \sqrt{\sum \frac{fx^2}{\sum f} - (\bar{x})^2}$$
$$= \sqrt{\frac{2\,559}{60} - (6.32)^2}$$
$$= \sqrt{2.707\,6}$$
$$= 1.645\,478\,653\,766\,13$$
$$= 1.65$$

(b)

Fruits per tree	2	3	4	5	6	7	8
Cumulative Frequency	1	6	10	16	26	42	60
Relative cumulative frequency	0.02	0.1	0.17	0.27	0.43	0.7	1



(iii) $\frac{70}{100} \times 1$

$= 0.7$

$\therefore$ *The* 70th *percentile is 7 fruits per tree.*

2019.P2.Nov

The table below shows the expenditure of 90 farmers in a particular farming season.

Amount (K)	$0 < x \leq 100$	$100 < x \leq 200$	$200 < x \leq 300$	$300 < x \leq 400$	$400 < x \leq 500$	$500 < x \leq 600$	$600 < x \leq 700$	$700 < x \leq 800$
No. of farmers	5	16	17	17	14	12	7	2

(a) Calculate the standard deviation.

(b) Answer this part of the question on a sheet of graph paper.

(i) Using the table above, copy and copy and complete the cumulative frequency table below.

Amount (K)	≤ 0	≤ 100	≤ 200	≤ 300	≤ 400	≤ 500	≤ 600	≤ 700	≤ 800
Frequency	0	5	21	38	55	69			

(ii) Using a scale of 2cm to represent 100 units on the horizontal axis and 2cm to represent 10 units on the vertical axis, draw a smooth cumulative curve.

(iii) Showing your method clearly use your graph to estimate the interquartile range.

Solution

(a)

	f	x	fx	fx^2
	5	50	250	12 500
	16	150	2 400	360 000
	17	250	4 250	1 062 500
	17	350	5 950	2 082 500
	14	450	6 300	2 835 000
	12	550	6 600	3 630 000
	7	650	4 550	2 957 500
	2	750	1 500	1 125 000
$\sum$	90		31 800	14 065 000

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f}$$

$$= \frac{31\,800}{90}$$

$$= 353.33333333$$

$$\text{S.D} = \sqrt{\frac{\sum fx^2}{\sum f} - (\bar{x})^2}$$

$$= \sqrt{\frac{14\,065\,000}{90} - (353.33333333)^2}$$

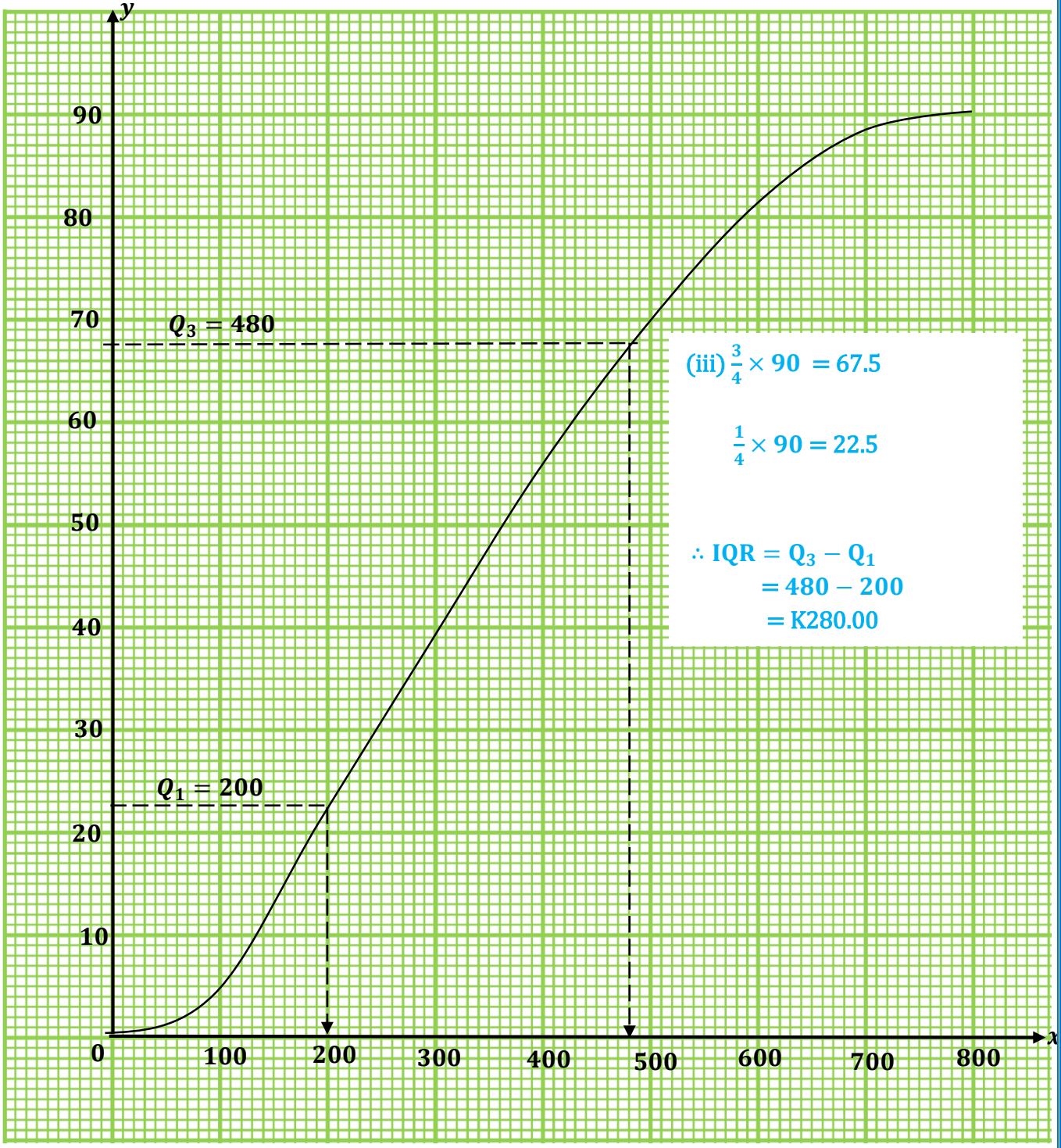
$$= \sqrt{31\,433.333\,333\,333\,3}$$

$$= 177.294\,481\,959\,629$$

$$= 177.3 \text{ or } 177$$

(b) (i)

Amount (K)	≤ 0	≤ 100	≤ 200	≤ 300	≤ 400	≤ 500	≤ 600	≤ 700	≤ 800
Frequency	0	5	21	38	55	69	81	88	90



(iii) $\frac{3}{4} \times 90 = 67.5$

$\frac{1}{4} \times 90 = 22.5$

$\therefore IQR = Q_3 - Q_1$
$= 480 - 200$
$= K280.00$

2020.P2.Nov

The table below shows the distribution of lengths of plots in a certain locality.

Length of plot (m)	$1 \leq x \leq 5$	$6 \leq x \leq 10$	$11 \leq x \leq 15$	$16 \leq x \leq 20$	$21 \leq x \leq 25$	$26 \leq x \leq 30$	$31 \leq x \leq 35$
Frequency	0	2	11	44	31	8	4

(a) Calculate the standard deviation.

(b) Answer this part of the question on a sheet of graph paper.

(i) Using the table above, copy and copy and complete the cumulative frequency table below.

Length of plot (m)	≤ 5	≤ 10	≤ 15	≤ 20	≤ 25	≤ 30	≤ 35
Frequency	0	2	13				100

(ii) Using a scale of 2cm to represent 5 units on the horizontal axis and 2cm to represent 10 units on the vertical axis, draw a smooth cumulative curve.

(iii) Showing your method clearly use your graph to estimate the interquartile range.

Solution

(a)

f	x	fx	fx^2
0	3	0	0
2	8	16	128
11	13	143	1 859
44	18	792	14 256
31	23	713	16 399
8	28	224	6 272
4	33	132	4 356
100		2 020	43 270

$$\text{Mean } (\bar{x}) = \frac{\sum fx}{\sum f}$$

$$= \frac{2\,020}{100}$$

$$= 20.2$$

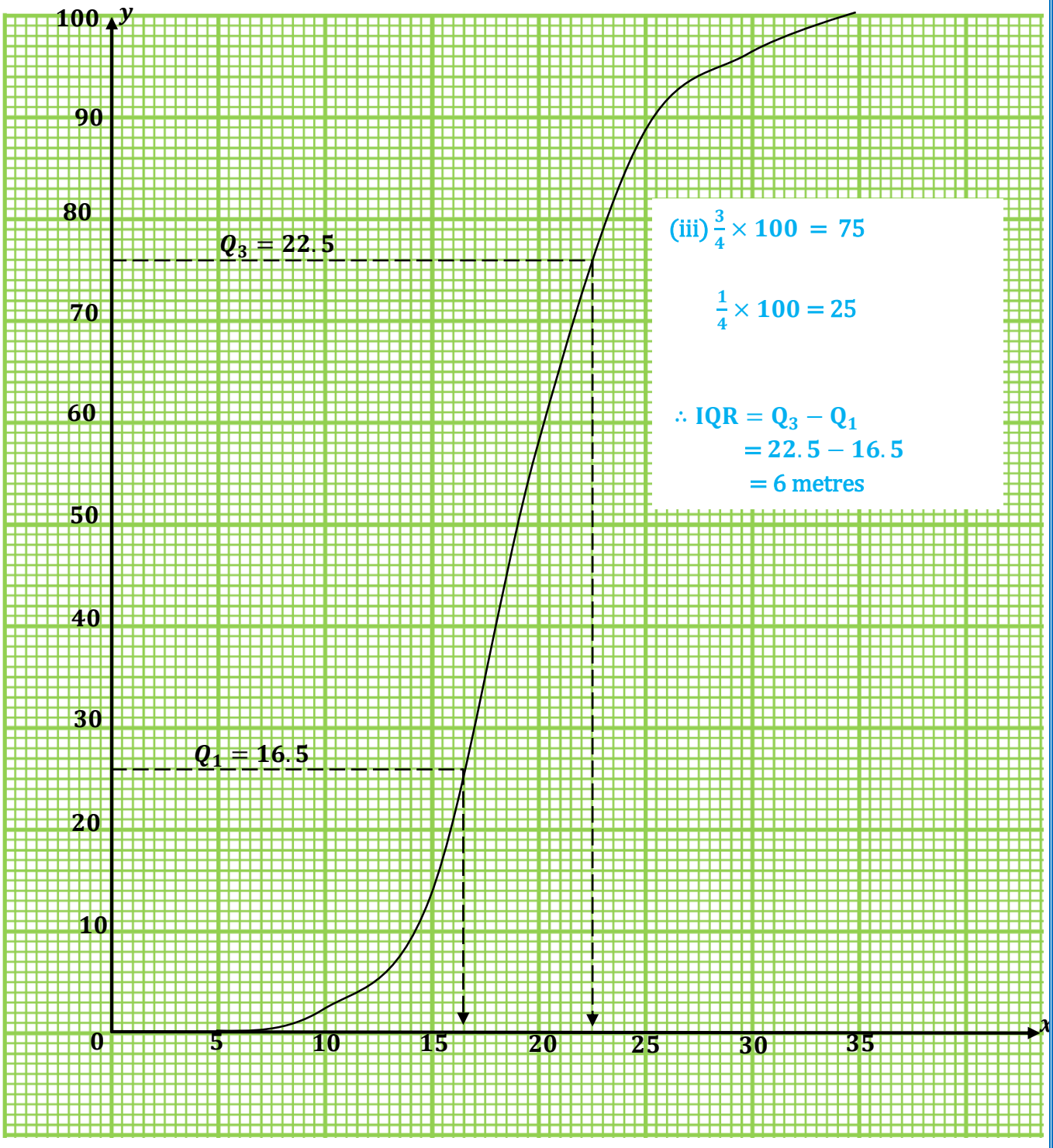
$$\text{S.D} = \sqrt{\frac{\sum fx^2}{\sum f} - (\bar{x})^2}$$

$$= \sqrt{\frac{43\,270}{100} - (20.2)^2}$$

$$= \sqrt{24.66}$$
$$= 4.965\,883\,607\,174\,055$$
$$= 4.97$$

(b) (i)

Length of plot (m)	≤ 5	≤ 10	≤ 15	≤ 20	≤ 25	≤ 30	≤ 35
Frequency	0	2	13	57	88	96	100



(iii) $\frac{3}{4} \times 100 = 75$

$\frac{1}{4} \times 100 = 25$

$\therefore IQR = Q_3 - Q_1$
$= 22.5 - 16.5$
$= 6$ metres